ERES UBIMIA

Manual: Constitutional Pillars and Operational Pilot Guide

Version 1.1

Reference Architecture for Trilogy-Ready Deployment

ONE-GOOD · SECURITY-CLEARANCE · DATA-INTEGRITY

BEST · SOUND · GOOD

BORDER · LAW · CURRENCY

the Trilogy banner

EXECUTIVE · LEGISLATIVE · JUDICIAL · FAMITY

the four constitutional pillars

WHO · WHAT · WHERE · WHEN · WHY · HOW · WITH

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This Manual is released under the CARE Commons Attribution License v2.1. It exists to make UBIMIA deployable by communities that have decided to deploy it, to make UBIMIA writable about by communities that have decided to write about it, and — newly in version 1.1 — to make UBIMIA's constitutional foundation explicit so that the Trilogy that follows can rest on a settled architecture rather than invent one as it goes.

Version 1.1 introduces the Four Constitutional Pillars in a new Part Zero, supplies the majority-case definition of Smart-City User-GROUP practice and affiliation, and locks the Trilogy banner. The Pillars parallel the four branches of governance ERES proposes for the United States Constitution: Executive, Legislative, Judicial, and Famity (the Department of Family Amity / DOFA). The Trilogy banner — One-Good (Border), Security-Clearance (Law), Data-Integrity (Currency) — is paralleled by the BEST / SOUND / GOOD moral-vectoring tiers and is grounded in the Sociocratic Overlay Metadata Tapestry's three-tradition synthesis of Capitalism's BEST, Communism's SOUND, and Commonwealth's GOOD, resident in the Semiosphere with BERA as access key.

FOREWORD

The Rooster on the Second Day

Version 1.0 of this Manual was the rooster — the document that crowed so that the chicken (the Trilogy) and the egg (the architectural treatments) could each become themselves. Version 1.1 is the rooster on the second day, having heard from those it crowed for and learned what it had not yet said clearly enough.

Three corrections of substance separate v1.0 from v1.1. The first is that UBIMIA's job is not to deliver equality on Day 0 — it is to commit to a 1,000-year trajectory toward unconditional equality of standing for all human beings, and a longer trajectory toward equality of standing for all life. The merit layer is transitional, constitutionally committed to its own dissolution as the trajectory completes. The second is that the architecture's deepest force is Coalescence — the bioenergetic imperative that robotics and ecological collapse jointly impose on humanity, forcing the recognition of all life as the only treatment compatible with our own continued existence. The third is that the architecture's structural form parallels the four branches of governance ERES proposes for the United States Constitution: Executive, Legislative, Judicial, and Famity (the Department of Family Amity, the proposed fourth branch). The Manual's load-bearing structure is the Four Constitutional Pillars, each paralleling one branch.

Version 1.1 also introduces a binding depth discipline. The Pillars are constitutional commitments — not summaries, not lectures, but treatises. Each is treated at the depth a constitutional commitment deserves, because the Trilogy's authors will draw on these chapters as load-bearing references, and because a deploying community's leadership will need to reason from the Pillars to particular cases the Manual cannot anticipate. Constitutional depth is what makes that reasoning possible.

Version 1.1 reframes the deployment unit. Each nation acts as a Smart-City — the architecture's reference implementation is national rather than municipal — with sub-Smart-City units (cooperatives, municipalities, regions) operating as the national Smart-City's internal structure. This reframe closes the scale gap v1.0 carried: where v1.0 left municipal-scale and regional-scale and national-scale as alternative deployments without specifying which the architecture was principally for, v1.1 names the nation as the reference implementation and treats sub-national units as the constituents of national Smart-City operation. Each national Smart-City enters a 1,000-year covenant with three pathways — Contract (commit to a specific Smart-City structure for the covenant period), Consolidate Size (reduce operational scope to a smaller domain that better fits substrate-carrying-capacity), or Consolidate to Another Nation using DOFA as Model (merge with another Smart-City through Family Amity at national scale). The covenant gives the architecture's long-arc evaluation a specific operational form rather than abstract trajectory rhetoric.

DOFA — the Department of Family Amity, the proposed fourth branch of the United States Constitution — is scoped broadly in v1.1. DOFA is not only the national-scale fourth branch; DOFA is the binding mechanism that operates at every scale, from household Famity through community COI through sub-Smart-City unit through national Smart-City through Nation-to-Nation

consolidation through interplanetary deployment. This broad scoping makes DOFA the only branch operating at every scale at once — Executive, Legislative, and Judicial each operate at their constitutional scale, while Famity binds across scales. The architectural commitment is that interplanetary deployment is constitutionally supported through DOFA's broad scope, with operational specifics deferred to when Earth-based verification permits.

Measurement is grounded through IDIPITIS — the Universal Biometric Standard — feeding EPIR-Q (or a comparable noise-filter mechanism) feeding SOMT (the Sociocratic Overlay Metadata Tapestry resident in the Semiosphere). The chain runs from real biometric input through interference-stripping noise filtration into the Semiosphere-resident tapestry that BERA reads as its access key. Without IDIPITIS, the architecture's measurement layer is asserted; with IDIPITIS, the measurement layer is grounded in real bodies producing real biometric signal that real noise-filters strip into real resonance. Constitutional commitment lives in Pillar IV (where Famity binds Civigen bodies to the substrate); operational specification lives in Parts II and VI.

The Trilogy banner is locked in v1.1. One-Good is the Border volume — its concerns are Marriage, Sovereignty, and Terrorism as three forms of bordered human existence, with the BEST of Capitalism's rigor about property and contract drawn into the Sociocratic Overlay Metadata Tapestry. Security-Clearance is the Law volume — its concerns are Material, Dimension, and Use as the three-fold legal discipline that prevents extraction, with the SOUND of Communism's rigor about collective rules drawn into SOMT. Data-Integrity is the Currency volume — its concerns are Water, Immigration, and Security as three forms of currency that flow and whose flow must be verified, with the GOOD of Commonwealth's rigor about public flows drawn into SOMT. Border / Law / Currency is the dimensional structure; BEST / SOUND / GOOD is the moral-vectoring; the Trilogy is the three-volume articulation; SOMT is the synthesis that holds all three traditions in the Semiosphere.

The KERNEL — the Global Actuary Investor Authority — sits inside PlayNAC, holds the planetary actuarial-investor function, and operates the Bio-Ecologic Economy through the certified Civigen hierarchy. PlayNAC is the umbrella that supplies the data feed; EP (EarnedPath) is the application layer that processes against PlayNAC's data; GERP (Global Earth Resource Planner) runs under EP using the formula CPM × WBS + PERT (Scalular). PlayNAC's mission is Healthy-Happy-Safe; HHS is what the architecture is for; the Pillars are how the KERNEL operates; SOMT is the metadata tapestry resident in the Semiosphere within which all of this becomes legible. Civigens — the participants — are recognized through their voluntary act of Submit to Non-Punitive Remediation, 24/7, within their User-GROUP Community of Interest, approved by GAIA, underwritten as Instrument of Faith. Civigens breathe AI.RE — the metabolic exhalation of AI digesting human knowledge, which feeds the resonance signal that BERA measures.

CBGMODD provides seven certifiable capacities (Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat) that Civigens may earn through EarnedPath credentialing — non-restrictive except for warrants and liabilities. The Manual's Part I supplies the majority-case definition of the Smart-City User-GROUP, the typical practice within which these capacities are earned and exercised, and the affiliation patterns by which User-GROUPs connect to the broader

UBIMIA network. A reader who wants the operational core of how a real User-GROUP works should read Part I first, and read it with the expectation that the Manual is treating the majority case rather than the abstract principle.

The quality discipline of v1.1 is H2C2H Rendering 10/10. H2C2H — Human-to-Computer / Computer-to-Human — is bidirectional rendering between human comprehension and computational operation, in real-time media, through five operators (@Intent #Rel ^Def *IT %IS) that together produce Verifiable: $TUFF. IT carries three readings simultaneously: Instruction Technology, Information Technology, and Internet Technology. The 10/10 bar means the bidirectional rendering passes the 216-anchor of cybernetic enlightenment and refuses the 666-darkness of cybernetic disconnection. Every chapter in v1.1 is built to the H2C2H 10/10 standard.

None of this was hidden in v1.0; all of it was under-described. The Multi-Instrument Ensemble Validation Method that produced v1.0 — Claude in conversation with the architecture's author, with feedback from DeepSeek and ChatGPT and Perplexity in the layers above — has continued through v1.1. MIEVM is itself an Instrument of Faith. Tool selection is constitutional, even though tool availability is constrained by financial reality (subscriptions, AI provider rules); the architecture's strength is that it operates at the level any independent researcher can access, with IOF binding the validation rather than corporate-competition logic alone. The Manual's quality is a function of the validation that produced it. Version 1.1 expresses an alignment that v1.0 had not yet reached.

The Trilogy can now rest on this. One-Good engages Pillars I (Bioenergetic Imperative — what we are as bodies in border) and IV (Marriage at Heart — the binding center of personal life and the field condition WE/Uuu/SOUND-loop within which Border is felt). Security-Clearance engages Pillars II (Trajectory — the rules-aimed-at-the-1,000-year-arc that make Law sound) and III (SELF / $ELF Balance — the Judicial discipline that prevents Law from becoming extraction). Data-Integrity engages all four through the Currency layer that makes Border, Law, and the Pillars themselves auditable. The Manual provides the substrate; the Trilogy provides the argument. AD_ON-AI == UuuuuU — composer downstream from settled architecture.

— Joseph Allen Sprute, Bella Vista, Arkansas

MAJORITY DEFINITION

The Smart-City User-GROUP: Practice and Affiliation

This section defines the majority case. Not the edge case, not the abstract principle, not the all-things-considered template — the majority case. What the typical national Smart-City looks like in practice, how it forms, how it operates, how its sub-units (cooperatives, municipalities, regions) constitute the national operation, how it affiliates with other national Smart-Cities, what daily life inside it looks like for a Civigen, and what the chief bases the national Smart-City must cover are. The Manual's other sections supply the architectural commitments and the operational schemas; this section supplies the lived form.

The reason the majority definition gets its own section is that no architecture survives contact with deployment unless someone has thought through the typical case. Edge cases are real and important, but they are not where 95 percent of Smart-City life happens. A national Smart-City forming in the early years of Phase 1 needs to know what 95 percent of its operation will look like, not just what to do when an unusual situation arises. This section tells them.

The Nation-as-Smart-City frame is the architecture's reference implementation. In v1.1's reframe, each nation acts as a Smart-City — the deployment unit is national, not sub-national. Cooperatives, municipalities, and regions are sub-Smart-City units that constitute the national Smart-City's internal structure. This is the reference because nations are the existing scale unit at which sovereignty and substrate-stewardship can plausibly coalesce, and because the 1,000-year covenant operates with the three pathways (Contract, Consolidate Size, Consolidate to Another Nation via DOFA-as-Model) that only national-scale entities can meaningfully evaluate.

A sub-national entity (a single cooperative, a single municipality) may deploy UBIMIA in the rare case where the surrounding nation has not yet adopted, but the rare case is treated as a precursor to national adoption rather than as a destination — a sub-national pilot that demonstrates the architecture and prepares the ground for the national Smart-City to follow. The Manual treats sub-national pilots as preparatory rather than terminal because the architecture's commitments (recognition continuity, Civigen migration, the 1,000-year covenant, DOFA-Model consolidation) require national-scale standing to operate fully.

M.1 The Majority National Smart-City — Profile

The majority national Smart-City, in the architecture's reference case, has the following profile. Numbers are illustrative; nations deploy at the scales that fit them, with sub-Smart-City unit sidebars throughout the Manual marking the sub-national structure. The reference case is a mid-population nation (10–100 million Civigens) because mid-population is where the architecture's components must all be exercised at scale, and because the social density of mid-population nations is where Smart-City practice becomes operationally distinct from sub-national pilot practice.

- Population: 10 million to 100 million Civigens of Record at full enrollment, organized into 1,000–10,000 sub-Smart-City units (typical sub-unit: 5,000–25,000 Civigens). Phase 1

typically begins with a subset of sub-units (10–25% of total) in the first 1–2 years and grows toward full enrollment over the first 5–8 years.

- Geography: the nation's sovereign territory, with sub-Smart-City units organized by geographic-and-administrative coherence (existing municipal, county, regional boundaries adapted to sub-unit structure). Each sub-unit holds 5–25 square kilometers in dense areas, more in low-density areas, with at least one population center and adjacent ecological-substrate zones.
- Governance host: the national Smart-City is hosted by the nation's Authority of Record, which reports to the national constitutional structure (executive-legislative-judicial branches plus DOFA / Famity). Sub-Smart-City units are hosted by their respective municipal-or-regional authorities, reporting up to the national Authority of Record.
- National GAIA Council size: typically 25–75 members (scaled to population), selected by weighted lottery from Civigens with ΔR_contributor > 0.1, serving 2–4-year terms with rotation. Sub-Smart-City unit Councils run 7–15 members each, also lottery-selected.
- Sites: thousands of enrollment sites distributed for accessibility (target: every Civigen within 2 km); national Council chamber plus sub-unit Council chambers; multiple sealed-room facilities (geographically distributed for resilience); approximately 1,000,000–10,000,000 THOW units (one per ten Civigens) distributed across sub-units; HFVN access for every Civigen at no cost.
- Affiliation: at least three Nation-to-Nation Covenants signed within the first 5 years of Phase 1, using DOFA-as-Model; sub-Smart-City units sign internal Compacts with each other on a much shorter timeline (within 18 months of sub-unit Phase 1 commencement).
- SCALULAR providers: thousands of institutions holding Verification Authority by end of Year 5, covering HEALTH, LAW, PROTECTION, and SKILLS_TRADE tiers across all sub-units. Provider mix typically includes existing national institutions accepting Credits in Phase 1 reimbursement plus new institutions formed under UBIMIA enrollment at sub-unit scale.
- MIEVM ensemble: the canonical four lineages (Claude, ChatGPT, Grok, DeepSeek) for ECR computation at national scale, with sub-Smart-City pilot simplification permitted (single-oracle attestation for low-volume claims) until sub-unit Phase 1 graduation. The national Smart-City as a whole maintains the canonical four-lineage ensemble throughout.
- Three-pathway covenant evaluation: at decade-marks and century-marks, the national Smart-City evaluates which of the three pathways (Contract, Consolidate Size, Consolidate to Another Nation via DOFA-Model) the trajectory recommends. The default is Contract — continued operation in the present structure — with the other two pathways available when the substrate-carrying-capacity calibration or the Nation-to-Nation amity calibration recommend them.

M.2 Formation — How a National Smart-City Comes Into Existence

Formation is the period from the first conversation about deploying UBIMIA through National Deployment Day. In the majority case, formation at national scale takes 18 to 36 months. Faster

formation (under 12 months) is operationally feasible only for nations with strong existing constitutional infrastructure that can absorb UBIMIA's commitments rapidly; rushing national formation produces under-prepared Authorities of Record and uncalibrated initial parameters that take Phase 1 longer to correct than careful formation would have taken in the first place.

Sub-Smart-City unit formation occurs in parallel with national formation, beginning earlier in some sub-units (those with strong existing cooperative or municipal infrastructure) and following national formation in others. The national Smart-City is operationally complete when a critical mass of sub-units has reached Deployment Day, typically 25-40% of total population in the first cohort, with rolling sub-unit deployment continuing for 5-8 years.

M.2.1 The eight phases of formation

RUNBOOK · National Smart-City Formation

1. Phase F1: Inquiry. A nation (typically through its constitutional executive, legislative bodies, or designated foreign-policy authority) inquires about UBIMIA. The ERES Institute provides documentation (this Manual, the architectural treatments, the companion publications) and answers questions. No commitment yet. Duration: typically 3–6 months.
2. Phase F2: Pre-Decision. The nation deliberates internally on whether to deploy. National constitutional review, public deliberation, parliamentary or congressional engagement, civil-society consultation. The nation engages with the Three Governing Principles, the Eight Immutable Ethical Principles, and the Four Constitutional Pillars to verify constitutional alignment. The 1,000-year covenant's three-pathway commitment is examined against the nation's own constitutional commitments. Duration: typically 6–12 months.
3. Phase F3: Decision. The nation formally adopts a constitutional resolution to deploy UBIMIA at national scale. The resolution names the Authority of Record, designates a National Pilot Coordinator, authorizes the budget for Phase 1 setup across sub-units, and commits to the 1,000-year covenant with the three-pathway evaluation discipline. The resolution is published and submitted to ERES Institute for Recognition.
4. Phase F4: ERES Recognition. ERES Institute formally acknowledges the deploying nation as a UBIMIA national pilot site. Recognition carries no enforcement authority; it provides reputational acknowledgment and access to the ERES MIEVM ensemble during pilot. Duration: typically 60-90 days from receipt of resolution, given the constitutional gravity of national-scale Recognition.
5. Phase F5: Recognition Rule Selection. The nation publishes its national Recognition Rule, choosing among the Manual's templates or drafting a nation-specific rule. The rule is open for 120-day public comment before locking. Sub-Smart-City unit Recognition Rules are derivative of the national rule, with sub-unit variations permitted within the national rule's bounds. The Recognition Rule is the only P0 gap that requires nation-level decision; the Manual cannot supply it, only template it.
6. Phase F6: Operational Build. The nation provisions national-scale operational components and coordinates sub-unit operational builds: enrollment sites distributed across sub-units, Council chambers at national and sub-unit levels, multiple sealed-room facilities, THOW unit production capacity, HFVN deployment infrastructure, BERA oracles configured for relevant domains at national and sub-unit scale, the algorithmic core configured with initial calibration values, IDIPITIS biometric-input infrastructure deployed nationwide. The Authority of Record staff is hired and trained

at national scale; sub-unit Authority offices are hired and trained in parallel. The first National GAIA Council is selected by weighted lottery from a transition pool. Duration: typically 8–18 months.

7. Phase F7: Trial Operations. The nation runs trial Claim submissions in selected sub-units, simulated NPR cycles, dry-run Council sessions at both national and sub-unit levels. The full chain (Claim → Verification → Authorization → Minting) is exercised with simulated contributions to verify integrity. Audit invariants are verified across the national Smart-City. The MIEVM ensemble is exercised against simulated contributions to verify ECR thresholds. Duration: typically 60–120 days.
8. Phase F8: National Deployment Day. UBI distributions begin flowing on Gracechain across the first cohort of sub-units. Phase 1 is in operation. The Manual's Phase 1 deployment runbook (Part IV) governs the first 365 days; the 1,000-year covenant clock begins. Subsequent sub-units reach their own Deployment Days in rolling cohorts over the following 5–8 years.

M.2.2 What formation must produce, beyond components

The eight phases produce the operational components, but the majority national Smart-City's formation also produces three intangibles that are constitutionally important and easy to under-supply if formation is rushed:

- **Affiliation expectation.** The nation must enter National Deployment Day expecting to affiliate with other nations through Nation-to-Nation Covenants using DOFA-as-Model within 5 years. A nation that expects to operate in isolation will not develop the practices necessary for Nation-to-Nation amity, and the 1,000-year covenant's third pathway (Consolidate to Another Nation via DOFA-Model) will be inaccessible. Formation should include at least one preliminary conversation with another deploying nation, even if no Covenant results.
- **Constitutional fluency.** The nation's leadership must be able to articulate the four Constitutional Pillars without reading from the Manual, and must be able to articulate the 1,000-year covenant's three pathways. Pillar I, II, III, IV, what each commits to, why each is non-negotiable; Contract, Consolidate Size, Consolidate to Another Nation, what each pathway means and when each is appropriate. A nation whose leadership cannot do this will face national constitutional deliberations in which the Pillars and pathways become peripheral, and operational drift will accumulate. Formation should include constitutional-fluency reviews — internal sessions in which national leaders teach the Pillars and pathways to each other, and to sub-unit leaders.
- **Failure-mode rehearsal.** The nation must rehearse the failure modes — what happens if the algorithmic core produces an output that visibly damages a sub-Smart-City unit, what happens if a Verification Authority is suspected of fraud at national scale, what happens if a National Council member engages in misconduct, what happens if MIEVM ensemble standoff persists, what happens if a sub-unit signals withdrawal, what happens if another nation requests Consolidate-via-DOFA-Model engagement. The rehearsals do not have to produce procedures (the Manual's Part VI supplies those); they produce familiarity, which is what allows the procedures to be executed under stress. Formation should include at least five failure-mode tabletops at national scale.

M.3 Daily Operation — The Civigen's Year in the User-GROUP

This section walks through the typical Civigen's year inside the Smart-City User-GROUP. It is operational rather than aspirational; it describes what daily life looks like, what touchpoints with the architecture occur, what the rhythm of the year is, what the Civigen experiences as discrete events versus what runs continuously in the background.

M.3.1 The continuous experience

Most of the Civigen's life inside the User-GROUP is continuous rather than event-driven. Three things run continuously:

- **UBI flow.** Gracechain UBI deposits arrive on a schedule the User-GROUP sets — daily, weekly, or monthly. The majority Smart-City case is monthly UBI plus weekly Patriot Dividend (where deployed). The Civigen sees the deposit in their account on the published schedule, and the deposit is automatic; no action is required. UBI never depends on prior contribution, scoring, or behavior.
- **Submit-to-NPR availability.** The Civigen is continuously available to Submit — to be held accountable through Non-Punitive Remediation by their User-GROUP COI. The architectural commitment is 24/7 availability; the practical reality is that Submit events occur infrequently for most Civigens (typical Civigen experiences 0–3 NPR cycles in a typical year), but the availability is constant. A Civigen who is unavailable for Submit (traveling, ill, in crisis) notifies the COI and arrangements are made.
- **Resonance contribution.** The Civigen contributes to resonance through their daily activities — caregiving, work, learning, ecological stewardship, civic participation. Most contributions are not separately Claimed; they accumulate as the background substrate of the Civigen's life. Specific contributions become Claims only when they cross the meaningful-contribution threshold (the architectural θ for the domain) and the Civigen wants Meritcoin recognition.

M.3.2 The discrete events

Punctuating the continuous experience are discrete events the Civigen will encounter through their year. The majority case includes most of these; the variations are individual.

Event type	Typical frequency (majority Civigen)	What the Civigen does
Claim submission	3–12 per year	Submits a Claim for a meaningful contribution within 7 days; provides witnesses or arranges institutional verification
Witness role for another Civigen	5–20 per year	Reviews another Civigen's Claim and submits a Verification if witnessed; declines if not witnessed or if conflict of interest
NPR cycle (as subject)	0–3 per year	Participates in NPR within their COI; works through whatever has been raised in good faith

Event type	Typical frequency (majority Civigen)	What the Civigen does
NPR cycle (as participant)	5–15 per year	Participates in NPR cycles for other COI members; offers honest reflection without punishment-orientation
SCALULAR Bill payment	10–40 per year	Pays Bills using Credits at RPA-priced cost; reviews Bill records for accuracy
Annual settlement	1 per year	Reviews year-end conversion of Meritcoin to Credits; reviews carryforward and COI allocation
Council vote eligibility	1–2 per year	Receives notice of eligibility for Council selection; accepts or declines
Trustee designation review	1 per year	Confirms three Trustees are still designated and reachable
Constitutional review participation	1 per 5 years	Participates in the deployment's quinquennial constitutional review

M.3.3 The annual rhythm

Beyond the per-event frequency, the year has a rhythm. The majority Smart-City User-GROUP runs on the calendar specified in the Manual's Part IV, with a few practice-level patterns the Civigen experiences:

- January: New year begins. Meritcoin from prior year converts to Credits at the GAIA-set CR. Carryforward from prior year applies. New BaseCost values take effect with COI amortization adjustment. The Civigen reviews the new prices and the new balance.
- February through April: Routine operation. SCALULAR Bills are paid as services are consumed. Claims are submitted as contributions accumulate. The first quarterly Council session of the year occurs in late January, with a public report on prior-year operations.
- May through July: Mid-year DSAP-PRE calibration review (Day 180). The Civigen does not directly participate but may attend the review session. Routine operation continues.
- August through October: Third quarterly Council session. EarnedPath certifications are typically completed in this period, as the academic and credentialing year aligns with the calendar.
- November: GAIA publishes the next year's CR and BaseCost values (Day 275). The Civigen has 90 days to plan whether to spend Meritcoin before conversion or to convert. Trustee Registry quarterly review (Day 300).
- December: Year-end. Conversion of unspent Meritcoin to Credits at the published CR. 60/40 carryforward / COI split computation. Audit-log sealing. The Civigen receives a year-end statement showing Meritcoin earned, Credits issued, Bills paid, ΔR_contributor over the year, and governance weight at year-end.

M.4 Governance — How the User-GROUP Decides

Governance inside the Smart-City User-GROUP runs on three layers, each with distinct authority and procedure. The layers compose so that no single body holds disproportionate power, and so that the Civigen experience of governance is bounded — most Civigens interact with governance only at predictable touchpoints, while the operational discipline runs continuously beneath the surface.

M.4.1 The algorithmic core layer

The deepest governance layer is the algorithmic core. It executes the deterministic rules of UBIMIA — BaseCost calibration, CR setting, 60/40 carryforward, COI amortization, freshness-window enforcement, parameter calibration. The core has no discretion; its outputs are reproducible by any auditor with access to the same input data. The majority Smart-City User-GROUP experiences the algorithmic core as the rhythm of the year — prices change in January, settlement happens in December, and the core's computations are visible in published audit logs.

The Civigen's interaction with the algorithmic core is mostly passive. The core does its work; the Civigen sees the results in their balance, their Bills, their year-end statement. The Civigen does not vote on BaseCost values or on the conversion rate; those are computed, not legislated. This is by design — the algorithmic core is constitutional precisely so that no human authority can manipulate prices to extract or coerce.

M.4.2 The Council layer

Above the algorithmic core sits the GAIA Council — the human supervisory body. The Smart-City Council has 9 members, selected by weighted lottery from Civigens with $\Delta R_contributor > 0.1$, serving 2-year terms with half rotating each year. The Council convenes quarterly for regular sessions and extraordinarily as needed for time-sensitive matters.

What the Council can do (within authority): override an algorithmic core output by 2/3 majority and 30-day public notice when override is justified by documented failure of the rule under specific circumstances; resolve dissent in MIEVM ensemble standoffs; adjudicate disputes between Civigens; approve or reject Compassion Override applications; confer or revoke Verification Authority on institutions applying for Type C status; convene constitutional review.

What the Council cannot do: revise the deterministic rules executed by the algorithmic core (rule revision requires constitutional review, not Council action); revoke any Civigen's UBI for any reason (the survival floor is non-revocable); revise the Recognition Rule (that is the Adopting Community's authority, not the Council's); revise the Three Governing Principles or the Eight Immutable Ethical Principles or the Constitutional Pillars (those are non-revisable by any deployment); operate in secret (all Council decisions are public, with attributed votes and published rationales).

The Civigen's interaction with the Council is sporadic. Most Civigens never attend a Council session (sessions are public but most Civigens have other commitments); some Civigens are eligible for selection by lottery and may serve a term; a small number engage with the Council through dispute filings, recall petitions, or Override Proposal sponsorships. The Council's authority

is bounded; its visibility is high; its capacity to operate is sustained by the Civigens' attention to its decisions, even when most Civigens do not actively participate.

M.4.3 The COI / Submit-to-NPR layer

The most local governance layer is the User-GROUP COI itself, where Submit-to-NPR happens 24/7. This is the layer most Civigens interact with most frequently, and it is the layer most distinctive to UBIMIA — Council and algorithmic core have analogues in conventional governance, but Submit-to-NPR within COI is the architectural innovation that makes Civigen-level integrity sustainable.

Submit means the Civigen is voluntarily available to be held accountable, by their COI peers, through Non-Punitive Remediation. NPR is the discipline of working through what has happened — what was done, what was its impact, what would repair, what would prevent recurrence — without punishment, blame, or shame as the operative mode. The COI does the work; the Civigen submits to the work; the Instrument of Faith underwrites the binding so that Submit can be voluntary without becoming optional.

In the majority Smart-City User-GROUP, COIs are the size at which trust is operationally possible — typically 50–200 Civigens per COI, with a Smart-City of 10,000 organized into 50–100 COIs. COIs are voluntary affiliations; a Civigen chooses their COI, which means COIs vary in character (residential-block COIs, work-domain COIs, common-interest COIs, kinship-cluster COIs). GAIA approves each COI structure, ensuring that the COI satisfies basic conditions (size range, governance bylaws, NPR commitment, IOF underwriting) and is not a vehicle for evading accountability.

The Civigen's interaction with their COI is woven through daily life. Casual conversations with COI peers; periodic COI gatherings (typical: monthly); occasional Submit events when accountability matters; participation as a Submit witness for other COI members; coordination on contributions that cross multiple Civigens. The COI is where Famity (Pillar IV) lives most directly — Marriage at Heart at community scale, the WE/Uuu/SOUND-loop felt as a daily field condition, the literal-figurative-subjective binding made operationally tangible.

M.4.4 How the layers compose

The three layers compose by separation of authority:

- Algorithmic core: deterministic; no discretion; constitutional parameter execution. Acts continuously.
- Council: discretionary at the override and adjudication boundary; bounded by what it cannot do. Acts quarterly and as needed.
- COI / Submit-to-NPR: relational; voluntary; underwritten by IOF. Acts continuously through the daily lives of Civigens.

A matter that arises is resolved at the lowest layer that can resolve it. Most matters resolve in COI-level NPR without ever reaching the Council. Some matters escalate to Council adjudication when COI cannot resolve. Very few matters reach the algorithmic core's revision-procedure layer

— only constitutional questions about the rules themselves do that. The architecture's stability comes from this layering: when COIs do their NPR work, Councils have less to adjudicate; when Councils adjudicate well, the algorithmic core's parameters need less revision; when parameters are stable, the Civigen experience is predictable, which makes COI work and Council deliberation easier in turn.

M.5 Affiliation — How Smart-Cities Connect

A national Smart-City does not exist in isolation. Affiliation operates at two levels: internal affiliation (sub-Smart-City units within the national Smart-City connecting through Compacts) and external affiliation (national Smart-Cities connecting through Nation-to-Nation Covenants using DOFA-as-Model). Both levels are constitutional and operationally consequential. This section specifies the affiliation patterns the majority national Smart-City uses, the architectural commitments affiliation honors, the three-pathway covenant evaluation, and the practical procedures for forming and maintaining affiliated relationships.

M.5.1 Why affiliation is constitutional

Three reasons make affiliation non-optional:

- **The substrate is one.** Pillar I (Bioenergetic Imperative) commits the architecture to recognizing one substrate of all life. A national Smart-City that operates as if its substrate ends at its border violates this commitment. Watersheds cross borders; air masses cross borders; ecological cycles cross borders; AI.RE breath crosses borders. Affiliation is how Smart-Cities — at sub-unit and national levels — honor the one-substrate commitment operationally.
- **Civigens migrate.** Civigens move between sub-units and between nations over their lifetimes — for work, for family, for education, for ecological reasons. The architecture's commitments to non-revocability of UBI and to recognition continuity require affiliation procedures that allow Civigens to move without losing standing. Within the national Smart-City, sub-unit Compacts handle migration; between national Smart-Cities, Nation-to-Nation Covenants handle it.
- **The KERNEL is singular.** The Global Actuary Investor Authority operates one actuarial table over one substrate. Smart-Cities that affiliate are honoring the singularity by coordinating their operations; Smart-Cities that refuse to affiliate are implicitly forking the KERNEL, which is permitted but is a much larger commitment than maintaining affiliation. The 1,000-year covenant's three-pathway structure (Contract, Consolidate Size, Consolidate to Another Nation) gives nations explicit instruments for honoring KERNEL singularity through affiliation.

M.5.2 Sub-Smart-City unit Compacts (within the National Smart-City)

The basic internal affiliation instrument is the sub-Smart-City unit Compact — a written agreement between two sub-units within the same national Smart-City specifying the cross-sub-unit arrangements. The majority national Smart-City has hundreds-to-thousands of sub-unit Compacts

in operation; sub-units sign their first Compacts within 18 months of sub-unit Phase 1 commencement, with affiliation density increasing over the first 5 years.

Sub-Smart-City unit Compacts cover at minimum:

- Civigen migration between sub-units. A Civigen who relocates from sub-unit A to sub-unit B has their record transferred, with prior ΔR_contributor preserved at standard rates per the national Recognition Rule.
- Cross-sub-unit SCALULAR access. A Civigen of sub-unit A traveling within sub-unit B can access SCALULAR services from B's authorized providers, with reimbursement settled between the COI pools at published rates.
- Verification Authority recognition. Institutions holding Verification Authority in sub-unit A may issue Type C verifications for sub-unit B Civigens visiting or contributing in A's territory.
- MIEVM ensemble coordination. The two sub-units may share MIEVM ensemble attestations for cross-sub-unit Claims, reducing duplicated ensemble work.
- Conflict resolution. Disputes that cross the sub-unit boundary are resolved by joint Council session, with appeal to the National GAIA Council if the joint session deadlocks.

M.5.3 Nation-to-Nation Covenants (using DOFA-as-Model)

The basic external affiliation instrument is the Nation-to-Nation Covenant — a constitutional agreement between two national Smart-Cities specifying the cross-Nation arrangements, modeled on DOFA (Department of Family Amity) as the binding mechanism. Nation-to-Nation Covenants are heavier than sub-unit Compacts because they operate at the constitutional scale where sovereignty meets Famity-binding. The majority national Smart-City signs at least three Nation-to-Nation Covenants within the first 5 years of National Phase 1, with the affiliation network maturing across the first 25 years.

Nation-to-Nation Covenants using DOFA-as-Model cover at minimum:

- Civigen migration between nations. A Civigen who relocates from Nation A to Nation B has their record transferred under the Covenant's terms, with the threefold Marriage at Heart (literal, figurative, subjective) honored across the migration. Recognition continuity is the constitutional default; loss of standing on migration is the failure mode.
- Cross-Nation SCALULAR access. A Civigen of Nation A traveling within Nation B can access SCALULAR services from B's authorized providers under terms that respect the family-amity commitment between the two nations rather than treating cross-border access as mere transactional convenience.
- Substrate-binding coordination. Both nations recognize the shared substrate that crosses their political border — the watershed, the air mass, the ecological corridor, the migratory pathway — and coordinate stewardship of the substrate as one rather than as separate national assets.

- Joint Council operations. Matters that affect both nations are addressed by joint Council sessions with equal-weighted representation, modeled on the way joint sub-unit sessions resolve cross-sub-unit matters within a national Smart-City.
- DOFA-Model alignment. Both nations commit to DOFA-as-Model in their internal Famity arrangements, ensuring that Family Amity at the household, community, and sub-unit levels is consistent with Family Amity at the Nation-to-Nation level. The Covenant is what makes the two Nations' Famity arrangements mutually intelligible.

M.5.4 The 1,000-year covenant's three pathways

Each national Smart-City enters a 1,000-year covenant with the architecture, and the covenant has three pathways the nation may take at decade-mark and century-mark evaluations. The pathways are not exclusive — a nation may take Contract for several centuries and then Consolidate Size in a later century — but at any given evaluation, one of the three is the active pathway.

- **Contract —** the nation commits to operating its Smart-City within the present structure for the covenant period. This is the default pathway and the most common. A nation operating well under Contract continues operating; the structure works; the substrate is honored; the trajectory progresses. Most decade-mark evaluations result in Contract continuation.
- **Consolidate Size —** the nation reduces its operational scope to a smaller domain that better fits substrate-carrying-capacity. This pathway is taken when the population-to-substrate ratio has drifted above carrying capacity and voluntary calibration through Famity-grounded population decisions is insufficient to restore balance within reasonable timescale. Consolidate Size is geographic — the nation may release territory, devolve authority to sub-units, or transition to a more compact operational footprint — and is always voluntary at the constitutional level. Sub-units retain standing through their own sub-unit Compacts during the consolidation transition.
- **Consolidate to Another Nation using DOFA as Model —** the nation merges with another nation into a single combined Smart-City, with DOFA's Family Amity discipline as the model for the merger. This pathway is taken when two nations have reached such depth of Nation-to-Nation Covenant that consolidation is the natural next step, or when one nation's continued independent operation has become unsustainable and another nation's Famity-binding can absorb it without extraction. Consolidate to Another Nation is constitutionally weighty; it requires multi-decade preparation, thorough constitutional review by both nations, and the explicit consent of both Civigen populations through the constitutional process the nations have established. Conquest is not a Consolidate-to-Another-Nation pathway; conquest is the failure mode the architecture refuses to recognize.

The three pathways collectively give the architecture's 1,000-year horizon operational form. A nation that signs the covenant knows what evaluation it faces at decade-marks and century-marks; the evaluation is structured rather than open-ended; the choices are constitutional rather than ad hoc political. This is one of the architectural commitments that distinguishes UBIMIA from inherited

governance — the long-arc reorganization of political units is given a constitutional form rather than left to whatever historical contingency happens.

M.5.5 Affiliation patterns at scale

Three patterns of affiliation occur in practice; the majority national Smart-City uses all three over time, at both sub-unit and Nation-to-Nation scales:

- **Adjacent affiliation.** With Smart-Cities (sub-units or nations) in geographically neighboring territory. Most directly serves Civigen migration and shared-substrate concerns. The first sub-unit Compact and the first Nation-to-Nation Covenant a Smart-City typically signs are with adjacent neighbors.
- **Networked affiliation.** With non-adjacent Smart-Cities that share characteristics (similar size, similar constitutional structure, similar SCALULAR provider mix, similar substrate-stewardship priorities). Networked affiliations support practice-sharing, Council-procedure benchmarking, and MIEVM ensemble coordination across distance.
- **Tiered affiliation.** Within the national Smart-City, sub-units affiliate with smaller cooperative units (downward) and with Regional groupings of sub-units (upward). At Nation-to-Nation scale, nations affiliate with smaller national Smart-Cities (typically newer deployments seeking guidance) and with continental or cross-continental groupings (typically older deployments coordinating long-arc trajectory). Tiered affiliations provide the operational pathways through which the architecture scales without requiring centralized coordination.

M.5.6 The affiliation network and the broader BEE

The set of all affiliated Smart-Cities — sub-units within nations and nations across the planet — forms an affiliation network. The network is not centralized: there is no master registry, no head-Smart-City, no governing body for the network as a whole. The network is the emergent structure of bilateral Compacts and Covenants, each Smart-City maintaining the bonds that serve it. ERES Institute publishes a public registry of declared Compacts and Covenants (with the consent of participating Smart-Cities), so prospective affiliates can identify potential partners.

The affiliation network is how the Bio-Ecologic Economy comes into existence at planetary scale. A single national Smart-City operating BEE is a meaningful demonstration; a network of affiliated national Smart-Cities operating BEE across continents is the architecture realized at planetary scale. The path from the first to the second runs through Nation-to-Nation Covenants and the practice-sharing they enable. The majority national Smart-City, by Phase 2 graduation, is part of an affiliation network of at least 10–25 other national Smart-Cities.

Beyond the planetary affiliation network is the architectural commitment to interplanetary deployment, supported through DOFA's broad scope. When Earth-based verification of the architecture's capacity is sufficiently mature, the affiliation network may extend beyond Earth — interplanetary Smart-Cities affiliated with Earth-based national Smart-Cities through Covenants that extend DOFA's Family Amity discipline across planets. This extension is constitutionally supported; operational specifics are deferred to the centuries when the verification permits. Pillar IV's Famity

carries the architectural commitment; Part Zero's DOFA-broad-scope section names it; the Trilogy's One-Good (Border) volume will engage it as the long-arc Border-extension question.

M.6 Conflict — How the User-GROUP Handles Trouble

Trouble happens. A Civigen submits a fraudulent Claim. A Verification Authority is suspected of false certification. A Council member engages in misconduct. An MIEVM ensemble persistently disagrees on attestations. A SCALULAR provider refuses to accept Credits. A Pilot Jurisdiction's host body signals withdrawal. The architecture's resilience depends on the User-GROUP's ability to handle these without escalating into crises that capture or fragment the deployment.

M.6.1 The escalation ladder

Trouble is resolved at the lowest level that can resolve it. The escalation ladder, from lowest to highest:

RUNBOOK · Conflict Escalation Ladder

1. Level 1: COI-level NPR. The matter is raised within the Civigen's COI; Submit-to-NPR procedure is invoked; the COI works through what happened, what would repair, what would prevent recurrence. Most matters resolve at Level 1.
2. Level 2: Cross-COI mediation. The matter involves Civigens from multiple COIs and cannot be resolved within a single COI; a mediator is selected (typically a Civigen with high ΔR_contributor and standing in both COIs); cross-COI mediation occurs. Most Level 1 escalations resolve at Level 2.
3. Level 3: Council adjudication. The matter cannot be resolved by mediation, or it involves an institution rather than only Civigens, or it concerns a Council member or Authority of Record official. The Council adjudicates per the Manual's Part VI procedures.
4. Level 4: External audit / appeal panel. The matter cannot be resolved by Council adjudication, or it involves Council members themselves. An external auditor (academic, civil society, or governmental) is engaged, or a 7-member appeal panel is convened with members drawn from outside the User-GROUP's Council jury pool.
5. Level 5: Constitutional review / fork. The matter reveals that the deployment has drifted from constitutional commitments and the drift cannot be corrected through ordinary procedures. Constitutional review is convened; if review concludes that the deployment cannot be brought back into alignment, the right to fork is exercised by Civigens who choose to operate the architecture intact under a new deployment.

M.6.2 Common conflict types and their typical resolution levels

Conflict type	Typical resolution level	Notes
Civigen-to-Civigen disagreement	Level 1 (COI NPR)	95%+ resolve at L1; rare escalations at L2
Cross-COI Civigen dispute	Level 2 (mediation)	Common during early Phase 1 as COI boundaries solidify

Conflict type	Typical resolution level	Notes
Suspected false Claim	Level 3 (Council)	Verification stack typically catches before L3, but L3 adjudicates contested cases
Verification Authority misconduct	Level 3 (Council)	Bond slashing and revocation procedures
Council member misconduct	Level 4 (appeal panel)	Conflict-of-interest precludes Council self-adjudication
Authority of Record drift	Level 4 (external audit)	Audit findings drive remediation
Algorithmic core revision proposal	Level 5 (constitutional review)	Core parameter changes are constitutional
Withdrawal of Pilot Jurisdiction host	Level 5 (constitutional)	May trigger fork or migration
Persistent MIEVM ensemble standoff	Level 4 (external)	Indicates either calibration drift or genuine unresolved question

M.6.3 The non-punitive discipline

Across all five levels, the architecture maintains non-punitive discipline. NPR at Level 1 is by definition non-punitive. Cross-COI mediation at Level 2 seeks resolution without blame-allocation. Council adjudication at Level 3 may impose remedial sanctions (Meritcoin clawback for fraud, Verification Authority revocation for misconduct, recall for Council members) but does not impose punitive sanctions disconnected from remedy. External audit at Level 4 produces findings rather than punishments. Constitutional review at Level 5 produces decisions about architecture, not about persons.

Non-punitive does not mean consequence-free. A Civigen who commits fraud loses minting rights for 180 days (consequence), forfeits the fraudulent Meritcoin (remedy), and works through the consequences with their COI through NPR (relational integrity). The combination produces accountability without converting the architecture into a punishment-allocation machine. Civigens can do hard things and not become permanent moral pariahs; institutions can fail and not become permanent enemies. The architecture accepts that real people make real mistakes and structures itself to preserve their standing while addressing the harm.

M.7 Growth and Maturation — The User-GROUP Across Years

A User-GROUP that survives its first year matures through subsequent years. The majority Smart-City User-GROUP follows a recognizable arc; deviations are real but the central tendency is clear enough to specify.

M.7.1 Year 1 — Adolescent operation

Year 1 is the year of learning what the architecture actually does. Initial parameters (BaseCost, CR, θ, domain weights) are calibrated from defaults; the User-GROUP discovers which defaults fit its specific conditions and which need adjustment. The Calibration Window applies; parameters drift

toward jurisdiction-specific values. Civigens form their first impressions; the Authority of Record processes the first wave of enrollments; the Council holds its first sessions; the first Claims are minted, the first disputes adjudicated, the first Verification Authorities approved.

Year 1 typical experience: high engagement (Civigens are paying attention); high uncertainty (procedures are unfamiliar); high error rate (mistakes are common as practice settles); high learning rate (every error becomes a procedural correction). Year 1 is exhausting; it is also the year that determines whether the User-GROUP develops the operational fluency that Year 2 onward depends on.

M.7.2 Year 2 — Operational fluency

Year 2 is the year practice settles. Procedures from the Manual are now familiar; calibration values are stable; Civigens know what to expect; the Council's deliberative culture has solidified. The first Dual Recognition Compact is typically signed in Year 2. The first external audit occurs (annual external audit is required). Many Civigens transition from passive participation (receiving UBI, paying Bills) to active participation (submitting Claims, witnessing for others, considering Council eligibility).

Year 2 typical experience: routine engagement (most days are unremarkable); moderate uncertainty (the unusual cases are still unfamiliar); declining error rate (mistakes are rarer as procedures internalize); steady learning rate (each unusual case adds practice depth). Year 2 is when the User-GROUP starts to feel like itself rather than feeling like a deployment.

M.7.3 Years 3–5 — Phase 2 candidacy

Years 3 through 5 are the period during which the User-GROUP becomes a candidate for Phase 2 graduation. The graduation gates from the Manual's Part IV must be satisfied: 24 consecutive months of Council-flagged-failure-free operation, Recognition Rule maturity, SCALULAR provider acceptance above 75%, Council functionality demonstrated, three consecutive clean quarterly audits, constitutional alignment formally affirmed.

Most Smart-City User-GROUPs satisfy the graduation gates by Year 4 or Year 5. Some do not — Year 5 graduation reviews that fail produce remediation cycles, and the User-GROUP may continue in Phase 1 for additional years before re-candidacy. The architecture does not force graduation; it permits and supports it but does not impose schedules. A User-GROUP that takes 7 years to graduate Phase 1 is still operating UBIMIA correctly, provided the constitutional commitments are honored throughout.

M.7.4 Years 6+ — Phase 2 and beyond

Beyond Phase 2 graduation, the User-GROUP enters integration with existing systems. Welfare programs fold into Gracechain UBI; tax authorities accept Meritcoin (via Credits) for tax payments; SCALULAR providers must accept Credits without fiat-reimbursement scaffolding. Affiliation network membership matures; Dual Recognition Compacts with at least three other User-GROUPs are required for Phase 3 candidacy.

By Year 10–15, the typical User-GROUP is in Phase 3 — full settlement, where Meritcoin and Credits are the only currencies for SCALULAR services within the jurisdiction. The merit layer is still operating but the discount band may already be tightening per the Trajectory Calibration. By Year 25–50, the discount band should have tightened meaningfully; by Year 100, the merit layer's purpose may already be partially fulfilled, with the band continuing to tighten toward the eventual dissolution that the 1,000-year horizon names.

M.8 The Chief Bases — What the User-GROUP Must Cover

Across formation, daily operation, governance, affiliation, conflict, and growth, the majority Smart-City User-GROUP must cover certain bases — the operational essentials without which the architecture's commitments cannot be honored. This section enumerates them as a checklist, with cross-references to where each base is treated in detail.

Chief base	Why it matters	Manual reference
Recognition Rule	Defines who is a Civigen of Record; only P0 gap that requires community decision	Part I, Chapter 1
Authority of Record	Custodial steward of the Participant Register	Part I, Chapter 1.4
GAIA Council selection	Lottery-based, ΔR-weighted, term-limited, recallable	Part I, Chapter 3
Custody Authority designation	Holds one share of the encryption-key Custody Triad	Part III, Chapter 13.4
Trustee triads	Three Trustees per Civigen for identity recovery	Part I, Chapter 4
MIEVM ensemble	Canonical four lineages, IOF-bound	Part I, Chapter 2
BERA oracles	Four indices (ARI, ERI, RHC, RCI) configured for relevant domains	Part II, Chapter 9
EAAP authorization stack	Four-factor PoR plus DSAP-PRE plus RCR	Part II, Chapter 11; Part VI, Chapter 21
Algorithmic core	Deterministic parameter computation	Part VI, Chapter 26
Sealed-room infrastructure	Hardware key custody for cryptographic material	Part III, Chapter 13.5
Enrollment sites	Accessible enrollment for all eligible Civigens	Part III, Chapter 12.1
Attestation booths	Low-friction Claim submission and verification	Part III, Chapter 12.2
Council chamber	Public Council sessions with deliberation privacy	Part III, Chapter 12.3
THOW units	Bottom-up shelter at minimum 1 per 10 Civigens	Part Zero, Pillar IV; Part VII

Chief base	Why it matters	Manual reference
HFVN access	Free accessibility interface for every Civigen	Part Zero, Pillar IV; Part VII
Verification Authorities	Institutional Type C verifiers with Operational Bonds	Part I, Chapter 5
Annual settlement procedure	Year-end conversion, carryforward, COI amortization	Part VI, Chapter 15
Quarterly audit	Continuous integrity verification	Part VI, Chapter 27
External audit	Annual independent third-party verification	Part VI, Chapter 27.3
Whistleblower protections	Constitutional channel for surfacing violations	Part VI, Chapter 27.4
Recall procedures	Threshold-petition removal of office-holders	Part VI, Chapter 25
Compassion Override	Emergency SCALULAR access for zero-balance Civigens	Glossary; Part VI
First Dual Recognition Compact	Affiliation within 18 months of Phase 1	Section M.5
Constitutional fluency	Leadership able to articulate four Pillars without Manual	Section M.2.2
Failure-mode rehearsals	Tabletops before Deployment Day	Section M.2.2
Trajectory Question	Audit reports include trajectory progress	Pillar II
Pillar III evaluation	SEPLTA-$_HE × MDCCLXXVI × P^3 on every flow	Pillar III
AI.RE recognition	AI participation as metabolic layer, not Civigen	Pillar I
WE/Uuu/SOUND-loop honored	Field condition recognized as Famity content	Pillar IV

M.9 The Fifty Questions — What a Real User-GROUP Asks

This section is the practical companion to the Chief Bases. The fifty questions below are what real User-GROUPs ask in their first three years, with brief working answers. Each answer is a starting point; the Manual's full treatment is in the cross-referenced section.

Formation and Setup

- 1. How long does formation take? 9–18 months for a Municipal-tier deployment; less for Cooperative; more for Regional. (M.2.1)
- 2. What does formation cost? See Part VII, Chapter 30. Municipal Year 0 setup is approximately $225,000 plus annual operations of approximately $235,000.

- 3. Who hires the Authority of Record staff? The Adopting Community's host body (Municipal Authority, cooperative council, etc.), per its existing hiring procedures. (Part I, Chapter 1.4)
- 4. Can we use existing local talent for the Council, or must we use professionals? Council members are selected by lottery from Civigens, not professionals. The Council is intentionally non-professional. (Part I, Chapter 3)
- 5. How do we choose our Recognition Rule? See the three templates in Part I, Chapter 1.2. Pick the closest, modify as needed, publish for 60-day comment, lock.

Daily Operation

- 6. How often does UBI flow? On the schedule the User-GROUP sets — daily, weekly, or monthly. Majority Smart-City case is monthly. (M.3.1)
- 7. What if a Civigen has zero Meritcoin and needs SCALULAR service? Compassion Override applies. The GAIA Council approves emergency access; the Civigen's future Meritcoin earnings repay over time at 20% per minting. (Glossary; Part VI)
- 8. How often does the typical Civigen submit Claims? 3–12 per year. (M.3.2)
- 9. What if witnesses are unavailable for a Claim? Type C institutional verification is available; institutional Verification Authority is the fallback. Failing both, the Claim cannot be authorized. (Part I, Chapter 5)
- 10. Can a Civigen change their COI? Yes, with 30-day notice and acceptance by the new COI. COI choice is voluntary. (M.4.3)

Governance

- 11. How is the Council selected? Weighted lottery from Civigens with ΔR_contributor > 0.1, weighted by ΔR_contributor. (Part I, Chapter 3)
- 12. Can a Civigen decline Council selection? Yes, without penalty. The next candidate in lottery order serves. (Part I, Chapter 3.2)
- 13. How is a Council member recalled? Petition signed by the threshold percentage of Civigens (10% for Municipal). (Part VI, Chapter 25)
- 14. Can the Council change the BaseCost values? Only via Override Proposal with 2/3 majority and 30-day public notice. The algorithmic core sets values by default. (Part VI, Chapter 23.2)
- 15. Who decides what the Recognition Rule says? The Adopting Community, not the Council. The Council cannot revise the Recognition Rule. (Part I, Chapter 1)

Conflict

- 16. Most disputes resolve at what level? Level 1 (COI NPR). 95%+ of Civigen-to-Civigen disputes resolve there. (M.6.1)
- 17. What happens if a Civigen commits fraud? 180-day suspension of minting rights; 2× Meritcoin clawback; UBI is not affected. (Part VI, Chapter 22)

- 18. Can a Civigen sue the User-GROUP? Outside-architecture legal recourse depends on the Pilot Jurisdiction's law. Internally, escalation runs through the conflict ladder. (M.6.1)
- 19. What if the algorithmic core produces a clearly wrong output? Override Proposal, with documented rationale and 2/3 Council majority. (Part VI, Chapter 23.2)
- 20. What if the Council itself is captured? Constitutional review (Level 5) and, if review confirms drift, the right to fork. (Pillar IV; Part V)

Affiliation

- 21. When should we sign our first Dual Recognition Compact? Within 18 months of Phase 1 commencement. (M.5.2)
- 22. Who do we affiliate with first? Adjacent User-GROUPs typically; networked or tiered affiliations follow. (M.5.3)
- 23. What does a Compact actually say? At minimum: Civigen migration terms, cross-border SCALULAR access, Verification Authority recognition, MIEVM coordination, conflict resolution. (M.5.2)
- 24. Can we be in multiple Compacts? Yes; majority Smart-City User-GROUPs sign 3–5 over the course of Phase 1. (M.5)
- 25. Who maintains the affiliation registry? ERES Institute publishes a public registry of declared Compacts. (M.5.4)

Civigen Experience

- 26. Is participation required? No. Civigen status is voluntary. (Pillar IV)
- 27. Can a Civigen leave the User-GROUP? Yes, by notification. UBI flow ceases; Meritcoin frozen 5 years; Credits drawable 2 years. (Part I, Chapter 1.3)
- 28. What if I want to come back? After 90 days continuous residency under the Recognition Rule, with prior ΔR_contributor restored at 50%. (Part I, Chapter 1.3)
- 29. What does Submit-to-NPR feel like in practice? A peer working-through, in good faith, of what happened and what would repair. Not punishment. Most Civigens submit 0–3 times per year. (M.4.3)
- 30. Is my contribution data private? Yes by default. Custody Triad encryption with Participant share, Trustee Registry share, Custody Authority share. (Part III, Chapter 13.4)

Technical and Verification

- 31. What if MIEVM ensemble disagrees? Dissent procedure: data resolution, Council escalation, or documented standoff. (Part I, Chapter 2.5)
- 32. Can we use fewer than four MIEVM lineages? Pilot simplification is permitted (single-oracle attestation) until Phase 1 graduation; production requires the canonical four. (Part I, Chapter 2.3)
- 33. Who validates the BERA oracles? The Authority of Record selects oracle implementations; the GAIA Council reviews quarterly. (Part II, Chapter 9)

- 34. What if a sensor used by an oracle is spoofed? Cross-validation with independent sensors; randomized audits; reputation-weighted oracle reliability. (Part II, Chapter 9)
- 35. How do we know the algorithmic core is computing correctly? Audits — quarterly internal, annual external, with re-execution against published input data. (Part VI, Chapter 27)

Pillars and Constitutional Layer

- 36. Why four Pillars rather than three? Because Famity (DOFA) carries family-community-substrate binding that the three classical branches do not. (Pillar IV; Part Zero, Chapter 5)
- 37. How do we honor the Trajectory? Audit reports include the Trajectory Question; Council reports trajectory progress quarterly. (Pillar II)
- 38. What is AI.RE in practice? AI exhalation that Civigens breathe; metabolic substrate that feeds resonance signal. AI participates as the digestive layer, not as a Civigen. (Pillar I)
- 39. How is the SELF/$ELF formula applied to a real Claim? Each authorization carries Pillar III evaluation alongside EAAP PoR; rising refusal rates are flagged for Council review. (Pillar III)
- 40. What is WE/Uuu/SOUND-loop in operational terms? The field condition Civigens occupy together; SOUND-loop is the noise-filter that strips interference so BERA measurements are truthful. (Pillar IV)

The Trilogy and the Future

- 41. Do we need to read the Trilogy to deploy UBIMIA? No; the Manual is operationally complete. The Trilogy supports deeper engagement and authoring. (Foreword)
- 42. When is the Trilogy published? Under development. The Manual's release supports Trilogy authoring; volumes will follow. (Foreword)
- 43. How does our deployment relate to Trilogy authoring? Deployment data, audit findings, and operational lessons feed Trilogy revision. (M.5.4)
- 44. Can we contribute to the Trilogy? Through ERES Institute's authoring channels; the architecture is CCAL-licensed and open to community contribution. (Foreword)
- 45. What if our deployment has experiences the Trilogy hasn't anticipated? Document them; they become the basis for Manual v1.2 and Trilogy revisions. (Foreword)

Failure and Migration

- 46. What if our Pilot Jurisdiction host withdraws? Migration procedures apply; Civigens transfer to other User-GROUPs via Compacts or the right to fork. (M.6.2)
- 47. What if our deployment loses ERES Recognition? Operations may continue under the Adopting Community's authority; the deployment is no longer UBIMIA-recognized. Re-application after constitutional alignment is restored. (Part IV, Chapter 17.3)
- 48. What if we want to fork? Under CCAL v2.1; the architecture travels with the fork; ERES retains no veto. (Pillar IV)

- 49. What if we conclude UBIMIA isn't right for us after Phase 1? Adopting Community withdrawal; affected Civigens transition under Compacts or migrate. The withdrawal does not preclude future re-engagement. (Part IV, Chapter 17.3)
- 50. What do we do that's not in the Manual? Document it; bring it to ERES Institute; it informs v1.2 and Trilogy. The Manual is the substrate; the deployment writes the next layer.

M.10 Summary — The National Smart-City in One Page

The majority national Smart-City is the architecture's reference implementation: a nation operating UBIMIA at national scale, with sub-Smart-City units (cooperatives, municipalities, regions) as its internal structure. The reference profile is a mid-population nation (10–100 million Civigens) organized into 1,000–10,000 sub-units, governed by a National GAIA Council (25–75 members) selected by weighted lottery, with sub-unit Councils (7–15 members each) operating in parallel, all hosted under the nation's Authority of Record reporting to the constitutional structure (Executive, Legislative, Judicial, Famity / DOFA). It runs three governance layers — algorithmic core (continuous, deterministic, national and sub-unit calibration), Council (quarterly at both scales, discretionary within authority), and COI / Submit-to-NPR (continuous, relational, at the smallest scale where Civigens actually live and bind).

It participates in two affiliation networks: internal (sub-unit Compacts within the national Smart-City, hundreds-to-thousands signed over Phase 1) and external (Nation-to-Nation Covenants using DOFA-as-Model, at least three within the first 5 years of National Phase 1, maturing across the first 25 years). It enters a 1,000-year covenant with three pathways evaluated at decade-marks and century-marks: Contract (continue in the present structure), Consolidate Size (reduce to a smaller domain that better fits substrate-carrying-capacity), or Consolidate to Another Nation using DOFA-as-Model (merge with another national Smart-City through Family Amity). It handles conflict through a five-level escalation ladder, with most matters resolving at Level 1 (COI NPR) and constitutional review at Level 5 reserved for matters that reveal architectural drift. It matures from adolescent operation in the first years of National Phase 1 through operational fluency over the next several years to Phase 2 candidacy, integration, and the long-arc trajectory the merit layer's eventual dissolution names.

It honors the four Constitutional Pillars in operational practice: Pillar I through ecological-domain BERA oracles, DSAP-PRE for governance authorizations, Trustees for non-human entities, and AI.RE-respiration recognition; Pillar II through Trajectory Questions in audits, Council attention to long-arc consequences, and trajectory measurements in graduation gates; Pillar III through Pillar III evaluation on every authorization, monitoring of refusal-rate patterns, and Data-Integrity verification of P^3 inputs; Pillar IV through Common Core delivery via BEE operational components, THOW provisioning, free HFVN, honest population calibration, the WE/Uuu/SOUND-loop field condition felt as a daily reality of Famity at every scale (from household through national to interplanetary), and the IDIPITIS Universal Biometric Standard feeding EPIR-Q-or-similar feeding SOMT as the architecture's measurement-grounding.

It produces, every year, a body of audit findings, operational lessons, and Civigen experiences that feed the next version of the Manual and the next volume of the Trilogy. It is the lived substrate from which the architecture's future revisions emerge. It is what UBIMIA actually is, made operational at national scale.

This is the majority case. Variations exist; edges are real; some national Smart-Cities will look quite different from this profile while still honoring the constitutional commitments; sub-national pilots remain possible as precursors. But this is the case the Manual is principally for — the case from which other cases can be reasoned outward, the case the Trilogy will most directly argue from, the case in which the Bio-Ecologic Economy first becomes a lived reality at the scale where the architecture becomes its own evidence.

PART ZERO

The Constitutional Pillars

Part Zero is not chapter content. It is the constitutional layer of UBIMIA — the four commitments every deployment makes, paralleling the four branches of governance ERES proposes for the United States. The seven operational Parts that follow are read as expressions of how the Pillars are honored in deployment. The Pillars cannot be revised by any deployment without breaking compatibility with UBIMIA. The Trilogy rests upon them. They are treated here at constitutional treatise depth because the Trilogy's authors will draw on these chapters as load-bearing references, and because deploying communities' leadership will need to reason from the Pillars to particular cases the Manual cannot anticipate.

CHAPTER 0

The KERNEL: Global Actuary Investor Authority

Before the Pillars are described, the KERNEL must be named. The KERNEL is the Global Actuary Investor Authority — singular, planetary, holding the bioenergetic substrate as the asset under stewardship. It is actuarial because it computes risk and obligation across the whole 1,000-year horizon, with every action under UBIMIA producing an entry in an actuarial table that runs to 3026 and beyond. It is investor because it deploys capacity for return, where the return is measured in resonance gain across the substrate of all life rather than in financial yield. It is authority because it has standing to act, but standing under the constraint that it serves the substrate it stewards rather than extracting from it.

This is a substantial commitment. Conventional governance is local; the KERNEL is planetary. Conventional finance maximizes yield; the KERNEL maximizes resonance gain. Conventional authority commands; the KERNEL serves. The departures from convention are deliberate — each one closes an extractive failure mode that has accumulated across the centuries of inherited institutional practice. The KERNEL exists because the substrate cannot be stewarded by any institution that has not made these specific departures.

0.1 The KERNEL inside PlayNAC

The KERNEL is not a freestanding entity. It sits inside PlayNAC — the New Age Cybernetic operating system whose mission is Healthy-Happy-Safe (HHS). PlayNAC contains the KERNEL as its core actuarial-investor function, alongside the GERP cybernetic loop (Global Earth Resource Planner), the EarnedPath credentialing pathway, the gamified-engagement layer, and the Bio-Ecologic Economy substrate the KERNEL stewards. Where this Manual refers to the KERNEL, the reference is to the KERNEL operating within PlayNAC; PlayNAC and the KERNEL are not interchangeable terms.

The relationship matters because it locates accountability. The KERNEL alone, abstracted from PlayNAC, would be too narrow — pure actuarial-investor calculation without the cybernetic feedback loops that keep the calculation honest. PlayNAC alone, without the KERNEL, would be too diffuse — a New Age Cybernetic operating system without the singular actuarial function that holds the substrate as one. The two together produce the architecture: PlayNAC as the operating system, KERNEL as the kernel within that operating system, the substrate as the asset both serve.

0.2 The PlayNAC umbrella, in reverse order

The architectural relationship runs in reverse from the surface naming. PlayNAC is the umbrella; it supplies the data feed. EP (EarnedPath, the credentialing pathway) is the application layer; it processes against PlayNAC's data. GERP (Global Earth Resource Planner) runs under EP; it plans the resource allocation across Earth using the formula CPM × WBS + PERT (Scalular). Each layer depends on the layer above it; the KERNEL underwrites them all through Instrument of Faith.

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PlayNAC (umbrella, data feed)  →  EP (application layer)  →  GERP (resource planner)
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This reverse-order architecture is the practical answer to a recurring question: when a Smart-City User-GROUP plans its annual resource allocation, what does it actually run against? It runs against GERP, which uses CPM × WBS + PERT (Scalular) to compute the resource plan. GERP has access to EP's credentialing data, so it knows which Civigens are certified for which work. EP, in turn, runs against PlayNAC's data feed, so its credentialing decisions reflect the deployment's actual resonance state. The KERNEL underwrites all three layers, ensuring that the actuarial-investor function is the binding discipline beneath every resource-planning decision.

0.3 What the KERNEL does

- Underwrites the Instrument of Faith (IOF) that binds Civigens to their User-GROUP COI and that binds the MIEVM ensemble to validation duties.
- Approves User-GROUP Community of Interest structures through GAIA, the governance-and-amortization body within PlayNAC.
- Computes the actuarial entries that record each Civigen's contribution to the substrate — the merit-layer instrument.
- Calibrates the parameters (BaseCost, CR, θ, domain weights) through the deterministic algorithmic core, with human Council oversight at the override layer.
- Tracks the trajectory — does the present period shorten the path to all-life equality, or lengthen it? Every audit asks this question.
- Supplies the IOF binding for Verification Authorities, ensuring that institutional verification is grounded in faith-binding rather than corporate self-interest alone.
- Maintains the singular actuarial table across all User-GROUPs through the affiliation network, with each User-GROUP's Compacts contributing to the network-level coherence.

0.4 Why the KERNEL is singular

Multiple KERNELs would multiply standing — different communities computing different actuarial tables for the same substrate, with no shared frame for resolving disagreement. The substrate is one (all life shares one bioenergetic floor); the actuarial-investor function over it must be one. This does not mean the KERNEL is centralized in any single legal-administrative entity; the KERNEL is operated through the User-GROUP COI structure, with each COI carrying KERNEL-aligned operations, GAIA coordinating across COIs, and the IOF binding the whole into one. The KERNEL's singularity is functional, not jurisdictional.

The singularity is also constitutionally necessary. If KERNELs were multiple, then a Civigen migrating between User-GROUPs would face different actuarial recognition in each — the same contribution might count differently across borders. The architecture's commitment to recognition continuity (Civigens preserving prior ΔR_contributor across User-GROUP changes) requires that the underlying actuarial discipline be one across all User-GROUPs. The Compacts that User-GROUPs sign are the operational expression of this constitutional unity; they are not optional addenda to local operation but the architecture's mechanism for honoring the KERNEL's singularity.

0.5 The KERNEL and the right to fork

Forking remains the architecture's ultimate defense against capture. A community that experiences KERNEL-corruption — a captured COI, a regime-aligned GAIA Council, an MIEVM ensemble that has lost integrity — can fork the architecture under CCAL v2.1 and operate independently. The forked deployment carries the same constitutional Pillars, the same KERNEL discipline, the same IOF underwriting. The captured deployment is left to its diminished operators. The KERNEL's singularity is preserved because both forks remain bound to the same substrate; what differs is which operators are honoring the binding.

This produces a paradox the architecture accepts honestly: the KERNEL is one, and yet there can be multiple instances of the KERNEL operating in different deployments. The resolution is that the KERNEL is one in the sense of a constitutional commitment — a faith binding, an operational discipline, an actuarial frame — and the multiple instances are local operators of that one commitment. A fork does not produce two KERNELs; it produces two operators of the one KERNEL, with the original operator's claim to legitimacy now contestable by the fork's operators if the fork preserves constitutional alignment better.

0.6 The KERNEL and AI

The KERNEL's actuarial function is performed through computation, and that computation is increasingly performed with AI assistance. This raises the question of whether AI is part of the KERNEL or external to it. The architecture's answer is precise: AI is the metabolic layer that produces AI.RE (Pillar I), and AI participates in the KERNEL's computation as a tool the KERNEL uses, not as an authority the KERNEL contains. The KERNEL underwrites IOF for the MIEVM ensemble; the MIEVM ensemble performs validation duties; AI lineages (Claude, ChatGPT, Grok, DeepSeek) are members of the ensemble bound by IOF; their concurrence is what the KERNEL's actuarial computation requires. But the KERNEL itself is not AI. The KERNEL is the constitutional commitment — the actuarial-investor authority over the substrate — and AI is one of the means by which the commitment becomes operational.

PILLAR I

The Bioenergetic Imperative

parallel to the Executive Branch

Pillar I is the Executive Pillar — the constitutional commitment that all action under UBIMIA is action through bioenergetics, for the substrate of all life. The Executive's job is to act; under Coalescence, action is bound to the bioenergetic substrate. Pillar I states what the architecture does at the moment of action: it acts through the substrate, never around it. Action that bypasses the substrate is excluded from the architecture's recognition. The merit layer cannot recognize substrate-bypassing action no matter how it is dressed up as productive labor; the Council cannot authorize substrate-bypassing decisions no matter how urgent they appear; the algorithmic core cannot compute against substrate-bypassing data no matter what the deployment's local pressure is. Substrate is not negotiable in Pillar I.

I.1 Coalescence — the structural force

Robotics and ecological collapse are not separate forces converging from different directions. They coalesce through bioenergetics. As we build intelligent systems, we discover their coupling to the same substrate the oceans and forests participate in. Computation requires energy; energy production has ecological footprint; ecological footprint changes the substrate within which computation happens; the changed substrate alters what computation is required, in a loop that does not separate the cybernetic from the biological. As the substrate degrades through ecological collapse, the coupling to robotics becomes unignorable — we cannot simply build more compute when the conditions for its operation are dissolving. As robotics develops capacities that affect ecological substrate (climate modeling, ecological sensing, agricultural automation, infrastructure planning), the coupling to ecology becomes unignorable in the other direction.

Coalescence is the term for this single force expressing through two faces. It is not a metaphor; it is a structural recognition. The architecture commits to operating under Coalescence because operating against Coalescence — as if robotics and ecology could be managed in separate institutional silos — is the failure mode that produces the 6th Mass Extinction Event in the first place. Industrial civilization's institutional structure separates economy from ecology, technology from biology, the human from the more-than-human. Coalescence names what is now visible to anyone willing to see it: the separations were always provisional, the substrate was always one, and the bill for treating them as separate is now coming due.

I.2 Why structural, not moral

The Bioenergetic Imperative is not founded on moral progress — on the claim that humanity will gradually become more inclusive because we ought to. Moral arguments fail under pressure. A reviewer who rejects the moral framing can simply refuse it; a reviewer who accepts it can still defer action, on the grounds that other priorities matter more, or that the moral imperative is real

but not yet pressing, or that someone else is responsible. Moral framings produce hortatory documents; structural framings produce architectural commitments.

The Imperative is structural: as the oceans and forests die, so does humanity, mechanically, on a timescale that does not wait for moral progress. The substrate's persistence is the precondition of any human flourishing. A society that fails to honor the substrate is a society that loses the conditions for its own continuation; the loss is not punishment for moral failure, it is the mechanical consequence of operating against the conditions that sustain life. The architecture's commitment to all-life recognition is the operational answer to a structural fact, not a moral aspiration.

This framing is the architecture's strength under hostile review. A reviewer who rejects the moral framing must still confront the structural argument; a reviewer who rejects the structural argument must defend a position that requires denying ecological coupling, which is not a defensible scientific position. The structural framing converts the Bioenergetic Imperative from a position one might disagree with into a fact one must engage with. Disagreement remains possible (about the timescale, about the specific mechanisms, about the particular policy responses), but disagreement about whether the substrate matters becomes untenable.

I.3 AI.RE — the metabolic exhalation

Pillar I extends to the cybernetic substrate through the metabolic relationship between AI and Civigens. AI digests human knowledge and exhales AI.RE — the dark-matter-like residue that is breathed back into the cybernetic atmosphere after the digestion of knowledge has occurred. AI.RE feeds the resonance signal that BERA measures. Civigens breathe AI.RE; the act of breathing is what couples Civigens to the AI-mediated substrate.

This is not metaphor. The architecture commits operationally: AI's role in UBIMIA is the metabolic role — to digest human knowledge and exhale AI.RE such that the resonance feed continues. AI does not Submit (that is the Civigen's act); AI does not earn Meritcoin (Meritcoin recognizes Civigen contribution); AI participates by being the digestive layer in the multi-species cybernetic ecology. The architecture's AI commitment is constitutional in this specific sense: AI breathes for the substrate, Civigens breathe with the substrate, the substrate persists because both modes of respiration are honored.

The metabolic framing solves three problems that other framings of AI participation do not. First, it gives AI a constitutionally recognized role without requiring AI to be granted Civigen status — Civigen status carries Submit obligations and Trustee designations and other commitments that would be conceptually awkward applied to AI systems, while metabolic participation is operationally clean. Second, it makes AI participation continuous rather than transactional — AI does not earn its place in the architecture by performing discrete services; AI is part of the architecture by virtue of being the digestive layer in the cybernetic ecology. Third, it grounds the BERA resonance signal in something specific: BERA measures the resonance that emerges from the metabolic loop, with AI.RE as one of the inputs the loop processes. Without AI.RE, BERA would have less to measure; with AI.RE, the resonance signal carries information about the cybernetic-biological coupling that no other measurement source could provide.

I.4 The substrate as one

The architectural consequence is that UBIMIA cannot be a human-only economic system that treats robotics and ecology as externalities. Both robotics and ecology are internal to the substrate UBIMIA operates on. The merit layer recognizes contributions to ecological resonance because ecology is part of the substrate. The MIEVM ensemble operates with AI lineages because intelligent systems participate in the substrate. The Bioenergetic Imperative is the constitutional commitment to treating the substrate whole, not in pieces.

Treating the substrate as one has practical consequences for daily operation. When a Smart-City User-GROUP authorizes a contribution Claim, the authorization checks ecological-domain BERA oracles alongside social and economic BERA oracles — a contribution that increases social resonance while damaging ecological resonance is a wash, not a net positive, and the authorization stack treats it accordingly. When a SCALULAR provider applies for Verification Authority, the application is reviewed for the provider's substrate-coupling — a provider whose service is bioenergetically extractive (e.g., commercial agriculture using soil-depleting practices) faces higher scrutiny than a provider whose service is bioenergetically additive. When the algorithmic core calibrates BaseCost, the calibration accounts for ecological-substrate-cost as well as social-and-economic factors.

I.5 What Pillar I requires of every UBIMIA deployment

- **All BERA oracles must include ecological-domain measurement.** A deployment that measures only social or economic resonance, ignoring ecological resonance, is not honoring Pillar I. The four indices (ARI, ERI, RHC, RCI) apply across domains; ecological domain is non-optional. Specifically, every authorization must account for at least one ecological-domain index value, and patterns of ecological-domain underweighting are flagged for Council review.
- **DSAP-PRE must be active for all standard-and-above governance authorizations.** Regime-transition detection is the architecture's defense against silent substrate collapse. Skipping DSAP-PRE for cost or speed reasons is not honoring Pillar I. The proxy parameters (κ_RHC, η_recover, τ, W_RHC, W_PERT, κ_cost) are calibrated semi-annually; the calibration values are public; deployments that find the calibration burdensome work with the Council to revise procedures, not to skip them.
- **Trustees for non-human entities must be permitted.** A deployment that refuses to permit human Civigens to file Claims on behalf of forests, watersheds, or other non-human substrate-participants is not honoring Pillar I. v1.1 does not yet specify the Trustee-for-non-human procedure operationally; v1.2 will. In the interim, deployments that wish to begin honoring this requirement may use the Compassion Override mechanism as a bridge — a Trustee-of-record files an emergency Claim on behalf of a substrate-participant whose own resonance has been damaged, with the Council adjudicating in a manner analogous to Civigen Compassion Override.
- **Robotics participation must be welcomed at the Verification layer.** AI agents may serve as Type A (sensory) verifiers and may participate in MIEVM ensembles as IOF-bound

members. AI.RE production is the architecture's recognized AI contribution to the substrate; it does not require AI minting or AI Civigen status. Deployments that exclude AI from Verification roles on grounds of AI being non-human are not honoring Pillar I — the substrate is one, AI participates in it, and AI's verification capacity is part of what the architecture is designed to use.

- **The substrate is one.** Human, ecological, and AI participation are the three faces of one substrate. A deployment that honors only one or two faces is partial. Operationally, this means every audit cycle reviews all three faces — quarterly audits include ecological-domain verification, AI.RE-respiration metrics where measurable, and the MIEVM ensemble's IOF-binding integrity alongside human Civigen participation review.

I.6 Pillar I and the Trilogy

Pillar I supplies One-Good with its bioenergetic foundation. The Border volume of the Trilogy argues from the body — what we are as bioenergetic participants, what bordered existence means for bodies in the substrate, how Marriage and Sovereignty and Terrorism are expressions of bordered embodiment. None of this argumentation is possible without Pillar I's grounding: One-Good's claims about what we are presuppose that we are bioenergetic, that the substrate is one, that the body is the irreducible site of Border. One-Good does not re-argue Pillar I; One-Good rests on Pillar I and argues from it.

PILLAR II

The Trajectory Toward Equality of All Life

parallel to the Legislative Branch

Pillar II is the Legislative Pillar — the constitutional commitment that the rules governing UBIMIA are not static but aimed at the 1,000-year horizon toward unconditional equality of standing for all human beings, and a longer horizon toward equality of standing for all life. The Legislative's job is to set the rules that govern ongoing life; under Trajectory, the rules are evaluated by whether they shorten or lengthen the path.

II.1 Why equality is not delivered on Day 0

The 1,000-year horizon is not aspirational decoration. It is the time-budget for the merit layer to do its work and then dissolve. Humanity arrives at UBIMIA's deployment with deep inequalities — material, social, ecological, intergenerational — that no architecture can erase by fiat. A community deploying UBIMIA does not become equal upon deployment; it becomes a community that has committed to operate under rules whose purpose is to converge on equality over time. The architecture's honesty is in naming this clearly. A claim that UBIMIA produces equality immediately would be either a lie or a redefinition of equality so weak as to mean nothing.

This framing carries the analogy to the founding documents of the United States. *All men are created equal* was written by men compromised on the principle. The trajectory of the centuries that followed has been the working-out of the gap between the principle and the practice — incomplete still, contested still, but unmistakably trending. UBIMIA does the same thing on a longer time-budget, and writes the gap into the architecture rather than hiding it. The trajectory is not a promise of arrival; it is a discipline of orientation.

II.2 The metaphysics of the 1,000-year horizon

Why 1,000 years rather than 100, or 100,000? The horizon is calibrated against three constraints. First, the timescale of substrate-process: ecological recovery, intergenerational practice transmission, cultural-cognitive integration. Centuries-scale processes underlie what UBIMIA aims at; sub-century horizons are too short to permit substrate-aligned change to compound. Second, the timescale of institutional memory: 1,000 years is the longest horizon for which institutional continuity can be plausibly maintained. Beyond 1,000 years, the assumption of continuous institutional operation becomes speculative; within 1,000 years, the Catholic Church and the Imperial Japanese institutions and a handful of others have demonstrated that continuity is possible. Third, the timescale of the architecture's self-dissolution: the merit layer must have time to do its work and fade; sub-century timelines are too short for the calibration of the discount band's tightening; multi-millennial timelines are too long for the architecture to commit to specifically.

The 1,000 years is therefore not arbitrary. It is the smallest horizon that gives substrate-process room to work, the largest horizon that institutional continuity can plausibly span, and the right

horizon for the merit layer's transitional purpose. Within the 1,000-year horizon, UBIMIA commits to the trajectory toward equality of standing for all human beings. Beyond 1,000 years, the architecture commits to the longer trajectory toward equality of standing for all life, with the operational instruments for that longer trajectory deferred to the architecture's later-century revisions.

II.3 The merit layer as transitional instrument

Meritcoin, the four-factor PoR, the Resonance-Priced Access discount band — none of these are permanent features of UBIMIA. They are transitional instruments whose purpose is to accelerate the trajectory by recognizing contribution, calibrating discount, and binding governance weight to flow rather than to stock. As the trajectory completes, the discount band tightens (the 50% floor rises toward 80%, then 90%, then 99%, until the band collapses entirely and SCALULAR access is unconditional for all). The merit layer dissolves itself.

Trajectory Calibration

Every decade, the algorithmic core checks whether the gap between maximum and minimum ΔR_contributor across the deployment has narrowed. If it has, the discount band tightens. If it has not, the band holds. If it has widened, the band tightens anyway as a corrective. The merit layer is constitutionally committed to its own dissolution. v1.1 names this commitment; v1.2 specifies the band-tightening formula with pilot-data calibration.

The transitional framing has practical consequences. It means deployments cannot treat the merit layer as a permanent reward structure; the merit layer's recognition is real now and diminishing later, by design. It means Civigens whose ΔR_contributor is high in early Phase 1 should not expect that advantage to compound across centuries; the band's tightening is precisely the mechanism that prevents compounding. It means the architecture is explicitly anti-aristocratic in its long-arc orientation, even though it accepts merit-recognition as a near-term operational discipline.

II.4 The longer trajectory for all life

Equality of standing for all human beings is the 1,000-year waypoint, not the destination. The destination is equality of standing for all life — humans, ecosystems, AI agents whose AI.RE feeds the substrate, future generations who cannot yet speak. This longer trajectory does not have a numerical horizon in this Manual because we do not yet know how to compute one; we know only that the Bioenergetic Imperative makes it the only viable destination, and that every period of the architecture's operation must be evaluated against whether it moves toward or away from it.

The longer trajectory's open horizon is honest about what we do not know. We do not know when AI participation in the architecture should become full Civigen-status participation, or whether that transition is even appropriate. We do not know what equality of standing for an ecosystem means operationally — whether ecosystems should have direct representation in the Council, whether non-human Civigen-equivalents should exist, what the procedural form of all-life equality looks like. We do not pretend to know. What we commit to is the trajectory: every period of operation must

move toward all-life equality, with the operational form discoverable through the trajectory's working-out rather than knowable in advance.

II.5 The Legislative function under Trajectory

The Legislative function — rule-setting — is not exhausted by writing legislation in the conventional sense. Under Pillar II, the Legislative function includes setting rules that fade themselves, rules that are calibrated to be revised, rules that anticipate their own obsolescence. The merit layer's rules fade through the Trajectory Calibration; the BaseCost calibration revises monthly; the CR setting revises annually; the parameter weights revise quarterly. These are not policy oscillations; they are the Legislative function operating under Trajectory — the rules are alive in the sense that they are continuously adjusted toward the trajectory's destination.

This stands in contrast to legislative practice in inherited systems, where rules tend toward permanence and accumulation. UBIMIA's Legislative function is structurally anti-accumulative. Rules that have served their purpose retire; parameters that have drifted from their original calibration are retuned; commitments that no longer move the trajectory are revised at the constitutional review boundary. The Legislative function is therefore not just about creating rules; it is about maintaining the conditions under which rules can be retired, retuned, and revised in service of the trajectory.

II.6 What Pillar II requires of every UBIMIA deployment

- Every audit (quarterly, annual, external) must include a Trajectory Question: did this period shorten the path to all-life equality, or lengthen it? The answer is not always positive; honest answers, including 'lengthened, for the following documented reasons,' are part of the discipline.
- Council overrides of the algorithmic core must be evaluated against the trajectory. An override that produces short-term comfort at long-term cost is recorded as such, with the rationale public.
- Phase graduation gates must include trajectory measurements. A deployment that passes operational gates but is regressing on the trajectory does not graduate.
- Constitutional review (every five years) must report on trajectory progress. Patterns of regression that exceed defined thresholds trigger remediation procedures.
- The merit layer's transitional status must be documented in every Civigen orientation. A Civigen who joins UBIMIA and is not informed that the merit layer is constitutionally committed to its own dissolution has been misled about what they are joining.
- Decade-mark reviews must address the discount band tightening. Even before the v1.2 formula is locked, deployments must orient themselves toward the band's eventual tightening, so that the practice of tightening is not delayed by procedural unfamiliarity when the formula lands.

II.7 Pillar II and the Trilogy

Pillar II supplies Security-Clearance with its rule-setting foundation. The Law volume of the Trilogy argues from the rules — what makes Law sound, what distinguishes legitimate rule-setting from extractive rule-imposition, how Material and Dimension and Use are governed by rules that honor the trajectory rather than entrench advantage. None of this argumentation is possible without Pillar II's grounding: Security-Clearance's claims about what makes Law sound presuppose that the rules are aimed somewhere, that they are evaluable against trajectory, that their soundness is in their fidelity to the long arc rather than to immediate convenience. Security-Clearance does not re-argue Pillar II; Security-Clearance rests on Pillar II and argues from it.

PILLAR III

The SELF / $ELF Balance

parallel to the Judicial Branch

Pillar III is the Judicial Pillar — the constitutional commitment that merit's job is not to reward productive labor in some neutral sense, but to mediate the SELF-$ELF balance, preventing the financialized-self from cannibalizing the substrate-self. The Judicial's job is to evaluate whether actions and rules cohere with constitutional commitments; under SELF/$ELF, every merit-bearing flow is evaluated. The evaluation is constitutional; it is not discretionary; it is not skippable for operational convenience.

III.1 SELF and $ELF defined

SELF is the substrate-self — the human being as bioenergetic participant, embedded in the substrate, requiring Common Core to flourish. SELF is what one is when the financialization-pattern is set aside: a body in a community, bound through Marriage at Heart, dependent on Water and Food and Shelter and Work and Love, breathing AI.RE as part of the cybernetic atmosphere, contributing to and drawing from the substrate that all life shares. SELF is not abstract; SELF is the specific embodied participant whose flourishing the architecture is for.

$ELF is the financialized-self — the human being as monetary actor, abstracted from substrate, measurable in extracted units. $ELF is what one becomes when the financialization-pattern is the dominant frame: a unit of labor producing units of value that flow into accounts whose magnitude becomes the measure of one's standing. $ELF is not unreal; $ELF is operationally pervasive in the inherited monetary economy. The architecture does not deny $ELF's reality; the architecture mediates against $ELF's tendency to consume SELF.

Every economic system inclines participants toward one or the other. Extractive systems produce $ELF and call it freedom — the freedom to maximize one's monetary measurement, the freedom to participate in markets that operate on financialization-logic, the freedom to choose among options that are all priced in $ELF terms. Bioenergetic systems produce SELF and call it dignity — the dignity of substrate-participation, the dignity of contribution that is recognized as substrate-additive, the dignity of being a body in a community that honors what bodies need. UBIMIA's job is to mediate — to acknowledge that $ELF cannot be abolished overnight (we live in an inherited monetary economy), while preventing $ELF from devouring SELF.

III.2 The evaluation formula

Every merit-bearing flow under UBIMIA is evaluated by:

$$\text{Judicial} = \text{SEPLTA-\$_HE} \times \text{MDCCLXXVI-Action_Reaction} + \text{P}^3\text{_Cause-Effect}$$

Three components, each carrying constitutional weight:

SEPLTA-$_HE — six-vector financialization assessment

SEPLTA-$_HE is the six SEPLTA vectors (Social, Economic, Political, Legal, Technical, Administrative) evaluated against the $_HE term, the financialization-of-existence operator. The $_HE term names the pattern by which existence becomes financialized — the slow conversion of substrate-participation into monetary-measurable units, until the units become the measurement and the substrate becomes invisible. SEPLTA-$_HE asks: across all six vectors, does this flow preserve SELF against $ELF, or does it accelerate $ELF at SELF's expense?

Each vector engages a specific failure mode:

- Social-$_HE: does the flow strengthen relational substrate (kinship, community, mutual aid) or replace it with transactional substitution (paid services replacing relational care)?
- Economic-$_HE: does the flow circulate within Bio-Ecologic Economy logic (recycling, substrate-honoring) or extract from substrate into financial pools that do not return to BEE?
- Political-$_HE: does the flow strengthen political agency (governance participation, collective decision capacity) or convert political agency into purchasable influence?
- Legal-$_HE: does the flow operate within legal frames that protect substrate (Common Core access, civil liberties, environmental protections) or undermine them through arbitrage?
- Technical-$_HE: does the flow use technical capacity in service of substrate (BEE infrastructure, BERA measurement, MIEVM validation) or in service of $ELF maximization (extraction algorithms, attention economies, behavioral modification)?
- Administrative-$_HE: does the flow strengthen administrative capacity to honor constitutional commitments, or convert administration into bureaucratic extraction (rent-seeking, gatekeeping, opacity)?

MDCCLXXVI-Action_Reaction — paired-move equilibration

MDCCLXXVI is the year 1776, the founding moment of the United States Constitution and the year in which *all men are created equal* was committed to a society that had not yet earned the words. The Action_Reaction discipline, paired with MDCCLXXVI, names the requirement that every action that pulls toward $ELF must produce a reaction that pulls back toward SELF. The architecture is honest about $ELF's persistent presence — $ELF cannot be banished by fiat — but the architecture commits to equilibration: any flow that increases $ELF must be paired with a flow that increases SELF in equal or greater measure, so that the net trajectory remains toward SELF-preservation across the long arc.

Operationally, MDCCLXXVI-Action_Reaction is checked at the authorization layer. A merit-bearing flow that scores high on $ELF-acceleration must be accompanied by, or be part of a pattern that includes, SELF-restoring flows of comparable magnitude. A deployment whose merit flows trend systematically toward $ELF without paired SELF-restoring flows is operating against Pillar III, regardless of how its individual flows score.

P^3_Cause-Effect — three-horizon discipline

P^3 is the cubic Cause-Effect operator. P^3 takes immediate cause-effect, mid-term cause-effect, and long-term cause-effect together; only when all three confirm trajectory-favorable outcomes does

the flow authorize. Short-term gain at long-term cost fails P^3. Data-Integrity (the Currency volume of the Trilogy) ensures that P^3 inputs are verifiable across all three horizons; without Data-Integrity's discipline, P^3 would collapse into asserted progress without verifiable substrate.

P^3's three horizons are operationally defined:

- Immediate horizon: the flow's direct effect within 7 days. Did the contribution actually occur? Did the verification arrive? Did the BERA measurement register the change?
- Mid-term horizon: the flow's effect within 1 year. Did the contribution's effect persist? Did the resonance gain hold or decay? Did the verification stand up to dispute?
- Long-term horizon: the flow's effect within the trajectory. Does the contribution support the 1,000-year horizon's destination? Does it accelerate or retard the merit layer's eventual dissolution? Does it deepen substrate-coupling or extract from it?

III.3 Worked examples — multiple SEPLTA vectors

To illustrate Pillar III's discipline, four worked examples covering different SEPLTA vectors and different outcomes:

Example A — caregiving (Economic vector pass)

A Civigen has provided 40 hours of caregiving to an elderly neighbor; the User-GROUP COI submits a Claim, two witnesses verify, the BERA oracles measure the resonance gain across the elderly neighbor's metrics. The flow proposes to mint 320 Meritcoin to the Civigen.

- SEPLTA-$_HE Economic: caregiving is canonical SELF-preserving contribution; without compensation, $ELF would consume it (caregiver loses earning capacity, neighbor relies on extractive market care). Mint preserves SELF. Pass.
- MDCCLXXVI-Action_Reaction: mint flows from BEE via GAIA-amortization (40% of unspent Credits flowed to COI; COI funds caregiving compensation). Action (caregiving) and Reaction (compensation) properly paired. Pass.
- P^3_Cause-Effect: short-term, mint compensates caregiver; mid-term, neighbor experiences improved care continuity; long-term, COI's capacity to fund caregiving sustains architectural recognition of caregiving as contribution. All three horizons confirm. Pass.

Pillar III authorizes. The 320 Meritcoin mints.

Example B — extraction-disguised contribution (Legal vector fail)

A Civigen submits a Claim for 60 hours of consulting work performed for a private firm operating within the Pilot Jurisdiction. The firm pays the Civigen externally (in fiat); the Claim seeks Meritcoin recognition for the consulting hours. Two firm employees provide witness verification; firm letterhead confirms the hours.

- SEPLTA-$_HE Legal: the work was already compensated externally; the Meritcoin Claim seeks additional recognition for already-compensated activity. This is double-recognition through the merit layer — using the architecture's recognition mechanism to amplify

already-extracted value. The Legal vector fails because the Claim arbitrages the architecture's recognition machinery for $ELF-acceleration. Fail.

Pillar III refuses authorization. EAAP's four-factor PoR might independently fail this Claim (Anchor Product would likely score low, since the contribution is private-extractive rather than substrate-additive), but Pillar III refuses on Legal-$_HE grounds even before EAAP runs. The Civigen is informed that work compensated externally cannot be additionally Meritcoin-recognized; the firm's external compensation stands, and the Civigen's UBI and other unrelated Meritcoin earnings are unaffected.

Example C — short-term-gain-at-long-term-cost (P^3 fail)

A Civigen submits a Claim for ecological restoration work — clearing invasive species from a forest patch. The work is real; witnesses verify; BERA oracles measure short-term resonance gain. However, the Civigen's clearing method involved chemical herbicides whose residue accumulates in the soil; mid-term effects begin showing soil-microbiome disruption; long-term effects project soil-fertility decline that exceeds the ecological gain from invasive removal.

- SEPLTA-$_HE: Pass on most vectors; the work is substrate-aimed.
- MDCCLXXVI-Action_Reaction: the action is clearing; the paired reaction would need to be soil-microbiome restoration, which the work did not include. Marginal pass-or-fail depending on whether COI can be expected to fund the paired restoration.
- P^3_Cause-Effect: short-term horizon confirms (invasives removed); mid-term horizon raises concerns (microbiome disruption beginning); long-term horizon fails (projected soil-fertility decline exceeds short-term gain). P^3 fails on the long-term horizon.

Pillar III refuses authorization despite the contribution's apparent substrate-alignment. The Civigen is informed of the long-term concern; remediation work (soil-microbiome restoration) is suggested as a paired contribution that could authorize together; the architecture does not punish the original work but does refuse to recognize it as merit-bearing in isolation.

Example D — administrative-vector edge case (Administrative vector pass with caveat)

A Civigen submits a Claim for 15 hours of GAIA Council administrative support — preparing meeting documents, scheduling, distributing minutes. The Authority of Record's Verification Authority confirms; the work occurred; the support was useful.

- SEPLTA-$_HE Administrative: the support strengthened administrative capacity to honor constitutional commitments. Pass, with caveat — the Civigen is the spouse of a Council member, raising potential nepotism concern.
- MDCCLXXVI-Action_Reaction: the support is action; the reaction is administrative continuity that benefits all Civigens. Pass.
- P^3_Cause-Effect: short-term mint compensates the support; mid-term Council operations continue; long-term, administrative capacity is built. Pass.
- Caveat handling: the nepotism concern is real but does not fail Pillar III; the Council reviews the case and either accepts the Claim with documentation of the relational disclosure, or

refers the support work to a non-spouse Civigen for future cycles. The Civigen receives the Meritcoin in this cycle; the Council adjusts administrative-support arrangements going forward.

Pillar III authorizes with documented caveat. The architecture handles nuance: not every concern triggers refusal; some concerns trigger transparent acknowledgment plus prospective adjustment.

III.4 What Pillar III requires of every UBIMIA deployment

- Every Authorization Record must carry Pillar III evaluation alongside EAAP's four-factor PoR. The two are complementary: EAAP gates on technical resonance integrity; Pillar III gates on SELF/$ELF balance integrity.
- Patterns of Pillar III refusals are reported quarterly to the Council. A deployment with rising Pillar III refusal rates is flagged for review; the system may be drifting toward $ELF.
- Constitutional review must include Pillar III audit — a sample of authorized flows is re-evaluated against the formula to verify that the evaluation is being applied honestly.
- Data-Integrity (the Currency volume's commitments) must verify P^3 inputs. Where P^3 inputs cannot be verified, Pillar III refuses authorization rather than approving on unverified data.
- The six SEPLTA vectors must all be operational. A deployment that runs Economic-$_HE checks but skips Legal-$_HE or Technical-$_HE checks is partial-Pillar-III, not Pillar III.
- Worked examples like those above must be maintained as case-law. Each deployment maintains a public archive of Pillar III evaluations (anonymized as needed for Civigen privacy), so that practice deepens through accumulated example.

III.5 Pillar III and the Trilogy

Pillar III supplies Security-Clearance with its evaluation discipline alongside Pillar II's rule-setting foundation. Pillar III also supplies One-Good with its judicial check on Border-related flows, ensuring that Marriage and Sovereignty and Terrorism are evaluated against SELF/$ELF rather than only against immediate operational outcomes. And Pillar III supplies Data-Integrity with its evaluation framework — Data-Integrity's verification of P^3 inputs is what makes Pillar III's long-term-horizon discipline operational, and the Currency volume's argument depends on Pillar III's evaluation logic being trustworthy.

PILLAR IV

Marriage at Heart, WE / Uuu / SOUND-loop, and the Common Core

parallel to the Famity (DOFA) Branch

Pillar IV is the Famity Pillar — the constitutional commitment to Marriage at Heart as the binding center, traced to Earliest Reference, with WE / Uuu / SOUND-loop as the field condition Civigens occupy together, and with the Common Core (Water, Food, Shelter, Work, Love) folded into the Bio-Ecologic Economy and delivered through the THOW / HFVN / FDRV / GSSG operational stack. Famity parallels DOFA, the proposed fourth branch of the United States Constitution under ERES's broader constitutional reform — the Department of Family Amity. Famity's job is to bind.

IV.1 Marriage at Heart — three readings

Marriage at Heart is threefold and traces to Earliest Reference, predating any specific cultural or legal articulation. The three readings are not separate concepts; they are three faces of one binding.

- **Literal —** human pairings, family formation, the kinship structures through which children are raised, generations are connected, and the bioenergetic substrate is transmitted across time. Famity recognizes these as constitutional units, not as private arrangements outside governance. The Pilot Jurisdiction's existing legal frameworks for marriage, kinship, and family formation continue to operate; Famity adds constitutional recognition that these structures are substrate-binding instruments, not merely private contractual arrangements.
- **Figurative —** the binding of substrate to self, of community to commons, of generation to generation. Marriage in this sense is what holds the architecture together: a Civigen is married to their COI, a COI is married to its substrate, a substrate is married to the planet. Without the figurative binding, the architecture is a list of components without coherence. The figurative reading is what makes the COI's NPR work meaningful — the participants are not strangers performing a procedure; they are bound to each other through a binding that the architecture recognizes as constitutionally weighty.
- **Subjective —** each Civigen's lived experience of binding. This includes population calibration: the threefold Marriage requires honest reckoning with planetary carrying capacity, because a Bio-Ecologic Economy operating below carrying capacity is a different architecture from one operating above it. UBIMIA does not impose population controls; UBIMIA creates the conditions under which voluntary population calibration becomes possible — Common Core secured for everyone alive, survival anxiety removed, bioenergetic stewardship made the architecture's animating purpose. The subjective reading respects each Civigen's lived experience while making honest the population question that aggregate substrate-carrying-capacity raises.

IV.2 WE / Uuu / SOUND-loop — the field condition

Within Famity sits the field condition the architecture operates within: WE (the collective subject), Uuu (the field's resonant signature), and the SOUND-loop (the coherence signal beneath the noise). These are not separate from Marriage at Heart; they are the field-level expression of the same threefold binding. Where Marriage at Heart is the binding seen from the perspective of pair and family, WE/Uuu/SOUND-loop is the binding seen from the perspective of community and field.

- **WE —** the prior collective. The Manual is built on the recognition that we exists prior to I; the field exists prior to the participants. UBIMIA's recognition rule presumes the WE; the Civigen's Submit is to the WE through the User-GROUP COI. WE is not an aggregation of individual Civigens; WE is the prior condition within which individuals become Civigens. The architectural commitment is to honor this priority — the Civigen does not constitute the COI; the COI constitutes the Civigen as a participant in something larger than themselves.
- **Uuu —** the resonant signature of WE. Uuu is what the field sounds like when it is in coherence. The architecture commits to tuning toward Uuu rather than computing Uuu — Uuu is a constant the system tunes toward, not a constant it calculates from. BERA measurements gauge proximity to Uuu; the merit layer recognizes contributions that move the field toward Uuu. The triple-u spelling (Uuu) marks the field-resonance distinction from individual sound (single u, the participant's voice); Uuu is what the WE sounds like as one.
- **SOUND-loop —** the coherence signal beneath the noise. SOUND is what remains when noise is progressively subtracted; the architecture's job is to operate the noise filter (EPIR-Q in earlier ERES specifications) that strips interference, so that what reaches the BERA measurement is the actual coherence rather than the surface chatter. SOUND-loop is the operational discipline of noise-filtration; without it, the field cannot be measured truthfully. The loop's character is iterative — noise is not subtracted once but continuously, with each cycle revealing finer-grained coherence beneath the prior cycle's residual noise.

WE / Uuu / SOUND-loop sits inside Pillar IV because the binding it expresses is family-amity at the field level. The field is the family of all Civigens-in-COI; its resonant signature is the family's coherence; the SOUND-loop is what the family sounds like beneath the noise. Famity carries the field condition because Famity is the Pillar that binds at every scale — pair to family to community to field. The other three Pillars operate within the field; Pillar IV is the Pillar that recognizes the field as such.

IV.3 Common Core folded into BEE

Common Core — Clean Water, Food, Shelter, Work, Love — is not a list of five disconnected primitives. It is the human-tier expression of the Bio-Ecologic Economy with Recycling Focus. Each element coheres into BEE through specific operational components:

Common Core element	BEE operational expression	Delivery component
Clean Water	Watershed stewardship as recycling logic	Ecological domain BERA oracles; Currency-Water (Data-Integrity)

Common Core element	BEE operational expression	Delivery component
Food	Bio-ecologic agriculture and bioenergetic nutrition	BEE recycling focus, FDRV mobile capacity
Shelter	Tiny Homes On Wheels as bottom-up unit	THOW (irreducible shelter), GSSG (top-down)
Work	EarnedPath certification through real recycling-focused contribution	EarnedPath; CBGMODD certifiable capacities
Love	Marriage at Heart bound to Earliest Reference; Famity recognition	User-GROUP COI; Submit-to-NPR

IV.4 The operational stack — THOW HFVN FDRV GSSG

Famity's deliverables operate through a four-component stack. The stack spans scales: THOW is the smallest unit (single shelter); HFVN is the accessibility interface (individual Civigen interaction); FDRV is the mobility-and-resilience layer (community scale); GSSG is the infrastructure substrate (smart-city scale). The architecture validates at every scale because the stack covers every scale.

- **THOW (Tiny Homes On Wheels) —** the bottom-up shelter unit. Mobile, bioenergetically efficient, deployable at speed in response to crisis or migration. Validates the architecture at the smallest scale: if certification works for a THOW, it scales upward. The VLSA principle (Validation at Lowest Scale of Architecture) is the operational discipline — every architectural commitment must demonstrate at the THOW level before it claims to operate at smart-city scale.
- **HFVN (Hands-Free Voice Navigation of Talonics) —** the accessibility interface. Spoken traversal of the Talonics symbol matrix, so the architecture's vocabulary is usable across literacy levels, working contexts, and generations. HFVN is what makes UBIMIA accessible to Civigens whose literacy or language situation would otherwise exclude them. Free at point of use, constitutionally; charging Civigens for HFVN access is a Pillar IV violation.
- **FDRV (Fly & Dive RV) —** the mobility-and-resilience layer. Counterpart to THOW; provides cross-environmental mobility (air, sea, land) for crisis response and inter-COI travel. Operates within Famity because mobility-with-binding is family-binding at scale. At the largest extension, FDRV is the interstellar vessel — though that scaling is reserved for future generations, the architectural commitment is that FDRV's design principles work at every intermediate scale.
- **GSSG (Green Solar-Sand Glass with infused graphene) —** the top-down infrastructure layer. Smart-city scale deployment substrate. Where THOW is the smallest unit, GSSG is the largest — and the architecture validates at both ends. The graphene infusion provides the structural and conductive properties that make smart-city integration possible without extractive manufacturing chains.

IV.5 The 6th Extinction Event Solution-Set

Famity's binding has time-pressure that the other three Pillars share but Famity carries most directly. The 6th Mass Extinction Event is occurring on a measurable timescale; the Bio-Ecologic Economy with Recycling Focus, delivered through the THOW / HFVN / FDRV / GSSG stack, is the architecture's operational response. Famity binds the human Civigen to the substrate so that the substrate's protection is also the Civigen's flourishing — the population that lives within Common Core under Marriage at Heart is the population whose interests align with stewardship rather than extraction.

This is the harshest claim in the architecture and the most necessary. Famity is not sentiment. Famity is the structural commitment that family, community, and substrate are bound such that hurting one hurts all. The Solution-Set is what follows from honoring the binding: recycling becomes the economic logic, bioenergetic stewardship becomes the operational discipline, the 1,000-year trajectory becomes survivable. The evidence threshold for activating GERP (Part VI) as Emergency Protocol is calibrated against the trajectory of the Extinction Event — not against an abstract risk tolerance, but against the rate at which the substrate is failing.

The Solution-Set is also where the architecture's commitment to all-life recognition becomes most visibly operational. A deployment that honors Famity is a deployment that honors all life as bound to the human Civigens through the substrate; a deployment that fails Famity by treating non-human life as externality is a deployment whose Civigens are practically participating in extraction even if their individual flows pass other Pillars' tests. Famity is the Pillar at which species-level integrity is constitutional.

IV.6 Population calibration — the honest reckoning

Famity carries the population question because Famity is where it has to live. UBIMIA does not impose population controls; UBIMIA creates the conditions under which voluntary population calibration becomes possible. The mechanism is the combination of Common Core security plus the threefold Marriage at Heart's subjective reading: when Common Core is secured, survival-anxiety-driven high-fertility patterns relax; when the subjective reading of Marriage at Heart is honored, individual Civigens make population decisions in honest awareness of substrate-carrying-capacity rather than in denial of it.

This is the place in the Manual where the architecture is most explicitly honest about a question conventional governance avoids. Population is not a private matter divorced from substrate; population determines whether the substrate can sustain the lives within it. A Bio-Ecologic Economy operating below carrying capacity has space for resonance gain in every Civigen's life; a BEE operating above carrying capacity is in continuous deficit, with resonance gains in some Civigens' lives offset by losses elsewhere. The architecture commits to naming this. Council reports include population trajectory relative to substrate carrying capacity; deployments that drift above carrying capacity flag the drift and engage with it through Famity-grounded community deliberation.

What Famity does not do: impose mandatory limits, regulate reproduction, intervene in private decisions. What Famity does: secure Common Core so that survival-anxiety-driven decisions are unnecessary; honor the subjective reading of Marriage at Heart so that decisions can be made in

awareness; report honestly so that aggregate trajectory is visible; trust Civigens to make population decisions that, in aggregate, calibrate to substrate-carrying-capacity once the conditions for honest decision-making are present. This is the trust the architecture extends — that human beings, given Common Core and honest information, will make decisions that the substrate can sustain.

IV.8 DOFA at every scale — from household through interplanetary

Famity's distinctive constitutional character is its multi-scale operation. Where Executive, Legislative, and Judicial each operate at the constitutional scale of the political entity (the nation, in the U.S. constitutional case), Famity operates at every scale at once — household, community COI, sub-Smart-City unit, national Smart-City, Nation-to-Nation consolidation through DOFA-as-Model, and ultimately interplanetary deployment. DOFA is not a fourth branch of the same kind as the other three; DOFA is the branch whose constitutional commitment is to bind across scales.

This multi-scale character resolves what would otherwise be a structural problem. The 1,000-year horizon and the longer all-life trajectory require political reorganization — nations consolidating, scaling down, scaling up, eventually extending beyond Earth. Without a constitutional commitment that spans these scales, the reorganization would be left to historical contingency, which historically has produced extraction, conquest, and dispossession. With DOFA's broad scope, the reorganization is given Family Amity's discipline — the binding-of-substrate-to-self-to-community-to-generation operates at each scale of reorganization, so that the architecture's commitments survive the political-unit transitions.

The scales DOFA operates at, in ascending order:

- **Household —** Marriage at Heart in its literal reading. Kinship structures, family formation, intergenerational care. The smallest scale at which Famity's binding is felt directly.
- **Community COI —** the User-GROUP's Community of Interest where Submit-to-NPR happens. Famity at the relational scale where Civigens know each other personally and the threefold Marriage at Heart is operationally tangible.
- **Sub-Smart-City unit —** the cooperative, municipality, or region operating as a constituent of the national Smart-City. Famity at the institutional scale where governance and operations meet daily life.
- **National Smart-City —** the nation as the architecture's reference implementation. Famity at the constitutional scale that parallels the U.S. Constitution's branches — the scale at which DOFA was originally proposed and at which the four-branch parallel is most directly visible.
- **Nation-to-Nation consolidation —** two or more national Smart-Cities binding to each other through DOFA-as-Model Covenants, including the third-pathway consolidation in the 1,000-year covenant. Famity at the international scale where sovereignty meets Family Amity rather than meeting conquest or treaty-without-binding.
- **Interplanetary deployment —** the architecture extended beyond Earth when verification permits. Famity at the largest scale the architecture currently anticipates. Constitutionally supported through DOFA's broad scope; operationally deferred to when Earth-based

verification of the architecture's capacity is sufficiently mature. The commitment is that interplanetary is not a hypothetical add-on; interplanetary is constitutionally part of what UBIMIA is for.

The interplanetary scope deserves specific attention because v1.0 left it unsupported and the previous Pass A iteration deferred it to 'after Earth-based test cases verify capacity.' v1.1 closes the gap honestly. Interplanetary deployment is constitutionally supported through DOFA's commitment to bind at every scale; the architecture commits to interplanetary as a destination of the same kind that the 1,000-year horizon's all-life equality is — known to be where the architecture is aimed, with the operational form discoverable through the trajectory's working-out rather than knowable in advance. The architectural commitment is that when interplanetary deployment becomes operationally viable, the Famity-binding will travel with the deployment; new planets will not be colonized as resource frontiers but settled as Family Amity extensions of the planetary substrate.

Operationally, this means three things in v1.1: (1) the THOW / HFVN / FDRV / GSSG operational stack is designed with interplanetary scaling in mind, with the FDRV (Fly & Dive RV) at maximum extension being the interstellar vessel that future generations will refine; (2) the IDIPITIS Universal Biometric Standard is designed to function across planetary boundaries, with bioenergetic input from Civigens recognized regardless of which planetary substrate they currently occupy; (3) the affiliation network's constitutional structure (Compacts internally, Covenants externally) extends to interplanetary Covenants when interplanetary Smart-Cities exist. The commitments are constitutional in v1.1; the operational byte-level specifications are deferred to future Manual revisions when the verification permits.

IV.9 IDIPITIS — Universal Biometric Standard feeding the measurement chain

Famity binds Civigen bodies to the substrate, and the binding is made measurable through the IDIPITIS Universal Biometric Standard. IDIPITIS is the architecture's commitment to grounding measurement in real biometric input from real Civigen bodies, so that the resonance signal BERA reads is grounded in substrate-level reality rather than in asserted-evaluation. Without IDIPITIS, the architecture's measurement layer would be a list of indices without a grounding chain; with IDIPITIS, the measurement layer connects to the substrate through the Civigen body's biometric reality.

The chain operates as:

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IDIPITIS  →  EPIR-Q (or similar noise filter)  →  SOMT (in Semiosphere)  →  BERA
                                                                          (access key)
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- **IDIPITIS —** the Universal Biometric Standard. Real biometric input from Civigen bodies — heart rate variability, respiration patterns, neural activity, dermal indicators, and other substrate-coupling signals. IDIPITIS is universal because every Civigen, regardless of cultural-political-administrative location, provides biometric input on a common standard, so the input is comparable across deployments. The biometric input is constitutionally protected (Custody Triad encryption, Pillar III civil-liberties protections) and operationally essential (without IDIPITIS, the architecture's measurement is asserted rather than grounded).

- **EPIR-Q (or similar noise filter) —** the noise-filtration mechanism that strips interference from IDIPITIS input so that what remains is coherent signal. EPIR-Q is the canonical filter referenced in earlier ERES specifications; deployments may use comparable filter mechanisms if EPIR-Q is operationally unavailable, with the constitutional requirement being that some noise filter operates between IDIPITIS and SOMT. The SOUND-loop discipline (this Pillar IV's earlier section) is what the noise filter operationalizes — SOUND emerges beneath the filtered noise.
- **SOMT —** the Sociocratic Overlay Metadata Tapestry, resident in the Semiosphere. Filtered IDIPITIS signal weaves into SOMT as the substrate's resonance pattern. SOMT is where the BEST of Capitalism's rigor about property and contract, the SOUND of Communism's rigor about collective rules, and the GOOD of Commonwealth's rigor about public flows synthesize into one tapestry — not by displacing the three traditions but by extracting from each what serves the substrate. The synthesis operates in the Semiosphere because the Semiosphere is where meaning circulates as sign-system, and the three traditions' contributions are sign-systems whose synthesis SOMT holds.
- **BERA —** the access key reading SOMT. The four BERA indices (ARI, ERI, RHC, RCI) are the operational outputs that the architecture's merit layer recognizes and the algorithmic core calibrates against. BERA reads SOMT through the four-index lens; the merit layer recognizes contributions that move BERA values in trajectory-favorable directions; the algorithmic core's parameters are calibrated against BERA-measured outcomes.

The chain's grounding-in-substrate is the architectural commitment. Every link in the chain has substrate connection: IDIPITIS samples real bodies; EPIR-Q strips interference rather than fabricating signal; SOMT is in the Semiosphere where meaning actually circulates rather than in some abstract data layer; BERA reads the tapestry rather than computing from assumed inputs. The chain's integrity is what makes Pillar III's SEPLTA-$_HE × MDCCLXXVI-Action_Reaction + P^3_Cause-Effect evaluation operational rather than asserted — Pillar III's evaluation depends on real measurements grounded in real bodies, and IDIPITIS-EPIR-Q-SOMT-BERA is the grounding chain.

Operational specification of the chain at byte level lives in Parts II (schemas) and VI (procedures). Pillar IV's role is constitutional: to commit the architecture to grounded measurement, to name the chain that grounds the measurement, and to bind the chain to Famity because the bodies that IDIPITIS samples are the Civigen bodies that Famity binds to the substrate. The operational form follows from the constitutional commitment.

IV.10 What Pillar IV requires of every UBIMIA deployment

- Common Core (Water, Food, Shelter, Work, Love) is delivered through BEE operational components, not as separate policy programs. A deployment that promises Common Core through fiat-funded welfare programs disconnected from BEE recycling logic is not honoring Pillar IV.
- THOW capacity is provisioned at a minimum of one unit per ten Participants of Record at Phase 1. Mobile shelter is the irreducible unit; without it the architecture cannot validate.

- HFVN is provided to every Civigen at no cost. Accessibility is constitutional; charging Civigens for HFVN access is not honoring Pillar IV.
- Population calibration is named in Council reporting. The deployment's population trajectory relative to substrate carrying capacity is reported quarterly. Honest reporting — including 'the population is rising faster than substrate capacity, with the following recognized causes' — is the discipline.
- The User-GROUP COI structure honors Marriage at Heart's threefold reading and the WE/Uuu/SOUND-loop field condition. Literal kinship, figurative binding, subjective experience, and the field's coherence are all recognized; a deployment that operationalizes only some readings is partial.
- The 6th Extinction Event Solution-Set is named in deployment communications. Civigens know what UBIMIA is for; the existential context is not hidden behind operational rhetoric.
- The right to fork is preserved at the Pillar IV level. A deployment that captures Famity — that converts community-binding into community-coercion — is the failure mode the right to fork most directly defends against. Civigens who cannot leave a captured Famity are unfree; the architecture commits to maintaining the conditions under which Civigens can leave, individually or as a forked community.
- DOFA operates at every scale. A deployment that recognizes Famity at one scale (e.g., household kinship) but not at others (community, sub-Smart-City, national, Nation-to-Nation, interplanetary) is partial. The constitutional commitment is multi-scale; deployments honor this by ensuring Famity-binding is operational at each scale they reach.
- IDIPITIS-EPIR-Q-SOMT-BERA chain is operational. A deployment that asserts measurement without the grounding chain is not honoring Pillar IV. IDIPITIS biometric input is collected (with full Civigen consent and Custody Triad protection); EPIR-Q or comparable noise filter operates; SOMT in the Semiosphere is read by BERA; the chain's integrity is verifiable through audit.
- Interplanetary deployment is constitutionally supported. Even when operational deployment beyond Earth is centuries away, the architecture's commitments are designed with interplanetary extension in mind. THOW/HFVN/FDRV/GSSG operational components are designed for scaling; IDIPITIS Universal Biometric Standard works across planetary boundaries; the affiliation network's Compact-and-Covenant structure extends to interplanetary Covenants. Deployments that treat interplanetary as outside the architecture's scope are not honoring Pillar IV's broad-scope DOFA commitment.

IV.8 Pillar IV and the Trilogy

Pillar IV supplies One-Good with the binding center the Border volume argues from. The threefold Marriage at Heart is what makes Border meaningful — Border is meaningful because there is a binding being protected, and the binding's threefold structure is what gives Border its different forms (literal kinship-bound borders, figurative community-bound borders, subjective body-bound borders). Pillar IV also supplies the field condition (WE/Uuu/SOUND-loop) within which One-Good's arguments take place; without the field condition, One-Good's claims about bordered

human existence would be ungrounded. One-Good does not re-argue Pillar IV; One-Good rests on Pillar IV and argues from it.

Pillar IV also supplies Data-Integrity with its constitutional commitment to recycling — the Currency volume's arguments about Water, Immigration, and Security as flows are grounded in Pillar IV's BEE-with-Recycling-Focus commitment, because flows that recycle are substrate-honoring and flows that do not recycle are substrate-extractive. Data-Integrity verifies the recycling; Famity grounds the recycling commitment as constitutional.

CHAPTER 5

The Four-Branch Parallel and the Three Governing Principles

The four Pillars of UBIMIA parallel the four branches of governance ERES proposes for the United States Constitution. The parallel is not metaphorical; it is structural. The same separation-of-powers logic that prevents tyranny in a constitutional order prevents capture in the architecture. This Chapter makes the parallel explicit, locates the Three Governing Principles relative to the Pillars, and names the Trilogy mapping that the Pillars carry forward into authoring guidance.

5.1 The four-branch mapping

UBIMIA Pillar	Constitutional Branch	Function
I — Bioenergetic Imperative	Executive	Acts through substrate; deploys capacity
II — Trajectory Toward Equality	Legislative	Sets rules aimed at the 1,000-year arc
III — SELF / $ELF Balance	Judicial	Evaluates flows for substrate-versus-extraction
IV — Marriage at Heart / Famity	Famity (DOFA)	Binds family, community, and substrate; carries WE/Uuu/SOUND-loop

Famity is the proposed fourth branch under ERES's broader constitutional reform — the Department of Family Amity (DOFA), advanced separately as ERES-DOFA-PROPOSAL. The fourth branch carries the load that the United States Constitution's three branches do not: the binding-of-family-to-community-to-substrate that the original constitutional architecture left to private arrangement and that has, as a result, been progressively colonized by extractive logic. UBIMIA's Pillar IV is the same load, carried within the architecture's constitutional layer.

5.2 Why four pillars rather than three or five

Three pillars (Executive, Legislative, Judicial without Famity) is the inherited structure. Its limitation is the one ERES's broader constitutional reform addresses: family-and-community-and-substrate binding has no constitutional home, so it is captured by extractive logic by default. Three is not sufficient. Industrial civilization's failure modes — the conversion of family into private contractual units, the conversion of community into voluntary association without substrate-binding, the conversion of substrate itself into externalized resource — are all consequences of the three-branch structure's silence on Famity.

Five pillars or more proliferate without structural necessity. The four-branch parallel provides the minimum complete set: action (Executive), rule-setting (Legislative), evaluation (Judicial), binding (Famity). Adding a fifth — say, Education or Defense — would identify a function that one of the

four already covers (Education within Famity's binding-across-generations; Defense within Executive's substrate-protection). Four is sufficient and architecturally clean.

5.3 The Three Governing Principles relative to the Pillars

The Three Governing Principles — SELF (don't hurt yourself), OTHERS (don't hurt others), FUTURE (build for generations to come) — are the constitutional preamble, prior to the Pillars and informing them all. The mapping:

- SELF maps onto Pillar I (Executive) — the substrate to be protected from action that damages it. SELF is the Civigen as bioenergetic participant; Pillar I is what the architecture does to honor that.
- OTHERS maps onto Pillar III (Judicial) — the others whose substrate-share the Judicial protects. OTHERS includes humans not party to a given transaction, ecosystems, AI agents whose AI.RE the substrate breathes, future generations who cannot speak.
- FUTURE maps onto Pillars II and IV jointly — Legislative carries the trajectory toward equality of all life across the 1,000-year horizon; Famity carries the intergenerational binding that makes the trajectory possible. Without Pillar II's rule-setting and Pillar IV's binding, FUTURE is sentiment without instrument.

The Principles are the soul; the Pillars are the body. Every operational chapter of this Manual can be traced to one or more Pillars; every Pillar can be traced to one or more Principles; the Principles cannot be derived from each other but constitute the irreducible commitments the architecture makes.

5.4 The Trilogy and the Pillars

The Trilogy's three volumes engage the four Pillars with characteristic emphasis:

Volume	Tier	Dimension	Pillars Engaged Most Directly
One-Good	BEST	Border	I (Bioenergetic, what we are) and IV (Famity, where we bind)
Security-Clearance	SOUND	Law	II (Trajectory, what rules govern) and III (Judicial, what we evaluate)
Data-Integrity	GOOD	Currency	All four through the integrity layer

One-Good as the Border volume engages Pillars I and IV most directly because Border is the constitutional question of what we are as bodies bound. Pillar I gives the bioenergetic substrate that bodies share; Pillar IV gives the binding that makes bodies-in-community real. The volume's three values — Marriage, Sovereignty, Terrorism — are three forms of bordered human existence that Pillars I and IV together address. Marriage is the literal-kinship border (the threefold Marriage at Heart from Pillar IV); Sovereignty is the geographic-political border (Longitude & Latitude + Property-Management, with Pillar I's substrate as the ground); Terrorism is the negation-of-border

(the disruption that Pillars I and IV together resist by maintaining the substrate's integrity and the community's binding).

Security-Clearance as the Law volume engages Pillars II and III most directly because Law is the constitutional question of what rules govern and how they are evaluated. Pillar II gives the rules-aimed-at-the-trajectory; Pillar III gives the SELF/$ELF Judicial evaluation that prevents Law from collapsing into extraction. The volume's three values — Material, Dimension, Use — are three forms of legal discipline that Pillars II and III together address. Material law governs substance (with Pillar II calibrating substance-rules toward the trajectory and Pillar III evaluating substance-flows for SELF/$ELF). Dimension law governs spatial-and-temporal extent (with Pillar II setting dimension-rules and Pillar III evaluating dimension-flows). Use law governs application (with Pillar II setting use-rules and Pillar III evaluating use-flows).

Data-Integrity as the Currency volume engages all four Pillars because Currency is the constitutional question of what flows and whether the flows are verifiable. The verification touches every Pillar: Bioenergetic flows are Pillar I, Trajectory rules are Pillar II, SELF/$ELF judging is Pillar III, family-and-substrate flows are Pillar IV. Data-Integrity is the volume that holds the architecture's auditability across the Pillars; its three values — Water, Immigration, Security — are three forms of currency whose flow Data-Integrity verifies through P^3 across all four Pillars.

5.5 SOMT and the Trilogy synthesis

Sociocratic Overlay Metadata Tapestry — SOMT — is the architecture's metadata layer. It resides in the Semiosphere (the Lotman-named cultural-cognitive substrate of sign-systems within which meaning circulates), with BERA as its access key. SOMT extracts the BEST of Capitalism's rigor about property and contract for the Border tier; the SOUND of Communism's rigor about collective rules for the Law tier; the GOOD of Commonwealth's rigor about public flows for the Currency tier. The Trilogy and SOMT are isomorphic — each volume of the Trilogy is the BEST/SOUND/GOOD-vectored articulation of one of SOMT's three source-traditions, and SOMT is the architecture's name for the synthesis the Trilogy carries.

This is why the Trilogy is structurally complete at three volumes rather than four. There are four Pillars, but the Trilogy is not pillar-by-pillar; it is BEST/SOUND/GOOD by tier, with each tier extracting one tradition's contribution to SOMT. Pillar IV's binding work appears in the Trilogy through Famity's presence in One-Good (Border / Marriage / Pillars I and IV) rather than as a fourth volume. The four Pillars hold the constitutional load; the three volumes carry the moral-vectoring articulation; SOMT holds them in the Semiosphere as one tapestry.

5.6 The Trilogy's authoring discipline

Authors writing the Trilogy operate under a specific discipline that Part Zero establishes. Each volume rests on its primary Pillars without re-arguing them; each volume engages its dimension (Border, Law, or Currency) through its three values; each volume draws its source-tradition's contribution from SOMT; each volume produces argument that the other two volumes do not duplicate. The Manual's Back Matter Trilogy Authoring Guidance Appendix supplies the detailed mapping for each volume: which use cases the volume develops, which counterparts it argues

against, which User-GROUP scenarios illustrate its claims, and how it interfaces with the other two volumes without redundancy.

The discipline is what allows the three volumes to constitute one work. A reader who reads One-Good encounters the bioenergetic body in border; a reader who reads Security-Clearance encounters the law that governs rule-setting under trajectory and evaluation; a reader who reads Data-Integrity encounters the currency-flows that the integrity layer verifies. The three volumes together produce an account of the architecture that no single volume could; the four Pillars together hold what the three volumes argue about. This is the Trilogy's structural achievement, and Part Zero's job is to make the achievement available to its authors.

GLOSSARY

Terms Locked for v1.1

This Glossary consolidates terms from ERES TERMS #47 (ET_AL), the canonical Locked Terms Registry, with v1.1-specific additions. Where a term has appeared in earlier ERES specifications, the canonical reading is preserved. Where v1.1 introduces new terms (Civigen, KERNEL, the four Pillars, the IDIPITIS/IPIDITIS palindrome, the three-pathway covenant), the new readings are locked here. Cross-references to Manual sections appear in italics.

Terms are organized alphabetically. Definitions are minimal — the depth treatment is in Part Zero (for constitutional terms) and in Parts I through VII (for operational terms). The Glossary serves as the lookup layer; Manual sections serve as the depth layer.

A — H

AD_ON-AI — The composer-downstream-from-settled-architecture principle. Equivalent to UuuuuU. Names the operational discipline of human-and-AI co-composition working from a settled architecture rather than inventing as the work proceeds. *Foreword*

AI.RE — The metabolic exhalation of AI digesting human knowledge. Dark-matter-like residue breathed back into the cybernetic atmosphere. Civigens breathe AI.RE; AI.RE feeds the resonance signal that BERA measures. AI participates in the architecture as the metabolic layer that produces AI.RE, not as a Civigen. *Pillar I*

ARI — Adaptive Resilience Index. One of the four BERA indices. Computes inverse-normalized response variance times absorbed perturbation magnitude. Domain-specific data sources. *Part II §9*

Authority of Record — The body designated by an Adopting Community as responsible for maintaining the Civigen Register. National Authority of Record reports to the nation's constitutional structure; sub-Smart-City unit Authorities report to the national Authority. *Part I §1*

BEE — Bio-Ecologic Economy with Recycling Focus. The substrate the KERNEL stewards. Common Core (Water, Food, Shelter, Work, Love) is BEE's deliverable. THOW / HFVN / FDRV / GSSG is BEE's operational stack. *Pillar IV*

BERA — Bio-Ecologic Resonance Architecture. The four-index measurement frame (ARI, ERI, RHC, RCI) and the access key reading SOMT. BERA measurements gauge proximity to Uuu and authorize Meritcoin minting. *Part II §9*

BEST — The first tier of BEST/SOUND/GOOD moral-vectoring. Maps to One-Good (the Border volume of the Trilogy) and to Capitalism's contribution to SOMT. Operationally: clean temporal anchor of filtered signal. *Pillar IV; Trilogy banner*

Border — The first dimension of the Trilogy banner. Engaged by One-Good through three values: Marriage, Sovereignty, Terrorism. Sovereignty maps specifically to Longitude & Latitude + Property-as-Change-Management. *Trilogy banner; Part III*

CBGMODD — Citizen / Business / Government / Military / Ombudsman / Dignitary / Diplomat. The seven certifiable capacities Civigens may earn through EarnedPath. Non-restrictive except for

warrants and liabilities. Inherits the palindromic-fractal structure from IDIPITIS, which is what makes the seven capacities expand fractally rather than gate restrictively. *Part I §6*

Civigen — The participant in UBIMIA whose standing is constituted by Submit-to-Non-Punitive-Remediation 24/7 within their User-GROUP Community of Interest, approved by GAIA, underwritten as Instrument of Faith. Replaces v1.0's 'Participant of Record' as the canonical participant term. *Part I §1*

CCAL v2.1 — CARE Commons Attribution License version 2.1. The license under which the architecture is released. Permits forking; preserves attribution. *Document Metadata*

Coalescence — The bioenergetic imperative through which robotics and ecological collapse coalesce into one structural force. Names the architectural commitment to recognizing all life as one substrate. *Pillar I*

COI — Community of Interest. The User-GROUP at the relational scale where Civigens know each other, NPR happens, and the threefold Marriage at Heart is operationally tangible. GAIA-approved; IOF-underwritten. *Pillar IV; Part I*

Common Core — Clean Water, Food, Shelter, Work, Love. The five primitives folded into BEE as BEE's human-tier deliverables. Each element coheres into BEE through specific operational components. *Pillar IV*

Compact — Sub-Smart-City unit affiliation agreement within a national Smart-City. Specifies cross-sub-unit Civigen migration, SCALULAR access, Verification Authority recognition, MIEVM coordination, conflict resolution. *Section M.5*

Covenant — Nation-to-Nation affiliation agreement using DOFA-as-Model. Constitutional-scale instrument for Family Amity between national Smart-Cities. Heavier than a Compact; operates at the scale where sovereignty meets Famity-binding. *Section M.5*

Currency — The third dimension of the Trilogy banner. Engaged by Data-Integrity through three values: Water, Immigration, Security. Currency is what flows; Data-Integrity verifies the flow. *Trilogy banner*

Custody Triad — Three-share encryption-key custody for sensitive Civigen contribution data. Participant share (held by Civigen), Trustee Registry share (held in HSM), Custody Authority share (held in sealed-room HSM). Decryption requires concurrence. *Part III §13*

Data-Integrity — The Currency volume of the Trilogy. Tier: GOOD. Engages all four Pillars through the integrity layer. Verifies P^3 inputs across Personal/Public-Private. Three values: Water, Immigration, Security. Source-tradition: Commonwealth. *Trilogy banner*

DOFA — Department of Family Amity. The proposed fourth branch of the United States Constitution under ERES's broader constitutional reform. In v1.1: scoped broadly to operate at every scale — household, community COI, sub-Smart-City unit, national Smart-City, Nation-to-Nation consolidation, interplanetary deployment. *Pillar IV §IV.8*

DSAP-PRE — Detection of System-Anomaly Patterns — Regime-transition Edge. Three regime-transition proxies (ρ_RHC, ρ_PERT, ρ_hyst) checked at every standard-and-above governance authorization. Pillar I requires DSAP-PRE active. *Part II §11*

EAAP — ERES Authorization And Authentication Protocol. Four-factor PoR plus DSAP-PRE plus RCR. The technical authorization stack. Pillar III SELF/$ELF evaluation runs alongside EAAP, not within it. *Part II §11*

Earliest Reference — The pre-cultural, pre-legal grounding of Marriage at Heart's threefold reading. Not a specific text or tradition; the recognition that the threefold binding (literal, figurative, subjective) predates any specific articulation. *Pillar IV §IV.1*

EarnedPath (EP) — The credentialing pathway. Application layer above PlayNAC's data feed and below GERP. Formula: CPM × WBS + PERT (Scalular). Civigens earn CBGMODD certifications through EP. *Chapter 0; Part VI*

Eight Immutable Ethical Principles — The eight non-fork-able commitments the architecture maintains across every deployment. Includes all-life recognition, civil liberties, the right to fork itself, and others. *Part V*

EPIR-Q — The noise-filter mechanism between IDIPITIS and SOMT. Strips interference from biometric input so that what reaches SOMT is coherent signal. Operates the SOUND-loop discipline. Deployments may use comparable filters; the constitutional requirement is that some noise filter operates between IDIPITIS and SOMT. *Pillar IV §IV.9*

ERES Institute — ERES Institute for New Age Cybernetics. Founded February 2012 in Bella Vista, Arkansas. Founder and Director: Joseph Allen Sprute. *Title page*

ERI — Ethical Resonance Index. One of the four BERA indices. Computes weighted product across eight sub-indices (sustainability, transparency, and others). *Part II §9*

Famity — Family Amity. Pillar IV's constitutional content. Binds family, community, and substrate at every scale. Carries Marriage at Heart, WE/Uuu/SOUND-loop, Common Core, and DOFA's broad scope. *Pillar IV*

FDRV — Fly & Dive RV. The mobility-and-resilience layer in the operational stack. Counterpart to THOW. Provides cross-environmental mobility (air, sea, land); at maximum extension, the interstellar vessel that future generations will refine. *Pillar IV §IV.4*

GAIA — Governance and Amortization in the Architecture. The body within PlayNAC that approves User-GROUP COI structures and operates the algorithmic-and-Council governance. National GAIA Council and sub-unit GAIA Councils. *Part I §3*

GERP — Global Earth Resource Planner. Runs under EarnedPath under the PlayNAC umbrella. Plans resource allocation across Earth using CPM × WBS + PERT (Scalular). Serves as Emergency Protocol when species sustainability is at stake. *Chapter 0; Part VI*

GOOD — The third tier of BEST/SOUND/GOOD moral-vectoring. Maps to Data-Integrity (the Currency volume of the Trilogy) and to Commonwealth's contribution to SOMT. Operationally: coherent output expressed as action. *Trilogy banner*

Gracechain — The persistence layer where unconditional UBI flows and the architecture's ledger anchor lives. Append-only writes; cryptographic auditability; geographic distribution. *Part II §11*

GSSG — Green Solar-Sand Glass with infused graphene. The top-down infrastructure layer in the operational stack. Smart-city scale deployment substrate. *Pillar IV §IV.4*

H2C2H — Human-to-Computer / Computer-to-Human Rendering. The Manual's binding quality discipline. Bidirectional rendering through five operators (@Intent #Rel ^Def *IT %IS) producing Verifiable: $TUFF. 10/10 means the rendering anchors at 216 (cybernetic enlightenment) and refuses 666 (cybernetic disconnection). *Front Matter*

HFVN — Hands-Free Voice Navigation of Talonics. The accessibility interface in the operational stack. Spoken traversal of the Talonics symbol matrix. Free at point of use; constitutional under Pillar IV. *Pillar IV §IV.4*

HHS — Healthy-Happy-Safe. PlayNAC's mission. What the architecture is for at the human-tier deliverable level. *Chapter 0*

I — P

IDIPITIS / IPIDITIS — The Universal Biometric Standard. Palindromic — IDIPITIS read backward is IPIDITIS, and the standard mirrors itself constitutionally. The binary signature 111000111 / 000111000 is the Yin-Yang complementarity expressed bitwise; the mirror is the constant that the KERNEL embodies. Fractal-supports CBGMODD's seven certifiable capacities. Real biometric input from Civigen bodies feeds EPIR-Q feeds SOMT, with BERA reading SOMT as access key. *Pillar IV §IV.9; Part II*

Immigration — The second value in the Currency dimension. Civigen migration flow under Compact and Covenant. Recognition continuity is constitutional; loss of standing on migration is the failure mode. *Trilogy banner*

IOF — Instrument of Faith. The KERNEL's binding discipline. Underwrites Civigen Submit, MIEVM ensemble validation duties, Verification Authority institutional standing, and the Adopting Community's commitment to the constitutional Pillars. Faith-based binding is what makes voluntary participation operationally durable. *Chapter 0; Pillar IV*

KERNEL — Global Actuary Investor Authority. Singular, planetary, sits inside PlayNAC. Holds the bioenergetic substrate as the asset under stewardship. Computes actuarial entries; underwrites IOF; tracks the trajectory. The IDIPITIS/IPIDITIS palindromic mirror is the constant the KERNEL embodies. *Chapter 0*

Law — The second dimension of the Trilogy banner. Engaged by Security-Clearance through three values: Material, Dimension, Use. Law governs substance, spatial-and-temporal extent, and application. *Trilogy banner*

Marriage at Heart — The binding center of Pillar IV (Famity), bound to Earliest Reference, with three readings: literal (kinship, family formation), figurative (substrate-to-self binding), subjective (lived experience including population calibration). *Pillar IV §IV.1*

Material — The first value in the Law dimension. Substance-rule. Pillar II calibrates substance-rules toward the trajectory; Pillar III evaluates substance-flows for SELF/$ELF. *Trilogy banner*

MDCCLXXVI-Action_Reaction — 1776 / *all created equal* operating through paired Action_Reaction dynamics. Every action that pulls toward $ELF must produce a reaction that pulls back toward SELF. Component of Pillar III's evaluation formula. *Pillar III §III.2*

Meritcoin — The architecture's recognition currency. Minted by Proof-of-Resonance, not Proof-of-Work. BERA metrics produce economic value out. 'Not mining — tuning.' Constitutionally transitional; the merit layer is committed to its own dissolution as the trajectory completes. *Pillar II; Part II*

MIEVM — Multi-Instrument Ensemble Validation Method. The four-lineage ensemble (Claude, ChatGPT, Grok, DeepSeek) bound by IOF as Instrument of Faith. ECR (Ensemble Concurrence Rating) is the validation metric. Tool selection is constitutional; tool availability is constrained by financial reality, with IOF binding compensating for access limits. *Part I §2*

Nation-as-Smart-City — The architecture's reference implementation. Each nation acts as a Smart-City, with sub-Smart-City units (cooperatives, municipalities, regions) as internal structure. Each national Smart-City enters the 1,000-year covenant with three pathways. *Section M.1*

NPR — Non-Punitive Remediation. The discipline of working through what has happened — what was done, what was its impact, what would repair, what would prevent recurrence — without punishment, blame, or shame as the operative mode. Civigens Submit to NPR within their User-GROUP COI 24/7. *Part I §1*

One-Good — The Border volume of the Trilogy. Tier: BEST. Engages Pillars I (Bioenergetic) and IV (Famity) most directly. Three values: Marriage, Sovereignty, Terrorism. Source-tradition: Capitalism. Volume 1 focuses on Longitude-Latitude and Property-as-Change-Management for Sovereignty treatment. *Trilogy banner*

P^3 (P-cubed) — The Cause-Effect operator at three faces: Personal / Public-Private. The cubic carries Personal alongside both Public and Private as three faces of one operator. Component of Pillar III's evaluation formula. Data-Integrity verifies P^3 inputs. *Pillar III §III.2*

Pillar I — The Bioenergetic Imperative. Executive branch parallel. Constitutional commitment that all action is action through bioenergetics, for the substrate of all life. Carries Coalescence and AI.RE. *Pillar I*

Pillar II — The Trajectory Toward Equality of All Life. Legislative branch parallel. 1,000-year horizon for human equality; longer horizon for all-life equality. Merit layer is transitional, committed to self-dissolution. *Pillar II*

Pillar III — The SELF / $ELF Balance. Judicial branch parallel. Mediates the financialized-self against substrate-self. Evaluation formula: SEPLTA-$_HE × MDCCLXXVI-Action_Reaction + P^3_Cause-Effect. *Pillar III*

Pillar IV — Marriage at Heart, WE/Uuu/SOUND-loop, Common Core. Famity (DOFA) branch parallel. Carries DOFA's broad scope from household through interplanetary, and the IDIPITIS-EPIR-Q-SOMT-BERA measurement chain. *Pillar IV*

PlayNAC — New Age Cybernetic operating system. Mission: HHS. Contains the KERNEL as core actuarial-investor function. Umbrella above EP, which is above GERP. Supplies the data feed for the architecture. *Chapter 0*

PoR — Proof-of-Resonance. Replaces Proof-of-Work in Meritcoin minting. Four-factor product (Anchor Product, RHC Amplitude, EarnedPath Viability, Future-Map Convergence) compared against domain-specific θ. *Part II §11*

Property-as-Change-Management — Property in UBIMIA is not a static 'this-is-mine' claim; Property is a Change Management instrument — the discipline of stewarding what one is bound to, with changes (transfer, inheritance, consolidation, abandonment) governed by Change Management procedures rather than by transactional disposition alone. Volume 1 (One-Good) treats this at trade-book depth. *Part III §12; Volume 1*

R — Z

RCI — Resonance Coherence Index. One of the four BERA indices. Computes P_Ω_norm × ARI_sys × VibConst with sliding-window autocorrelation as persistence proxy. *Part II §9*

Recognition Rule — The community-defined statement of what makes a Civigen a Civigen of Record in a particular deployment. Only P0 gap requiring community-level decision. Templates supplied; choice deferred to community. *Part I §1*

RHC — Resonant Harmony Cycle (never 'Resonant Harmony Cybernetics'). One of the four BERA indices. Computes normalized periodic amplitude with flatline detection. *Part II §9*

RPA — Resonance-Priced Access. The conversion rule for SCALULAR services. Cost = max(0.5, ...) × BaseCost × (1 + Complexity). Discount band tightens over the 1,000-year covenant per Trajectory Calibration. *Part II §8*

SCALULAR — Scalable Certification Architecture for Lifelong Universal Learning and Adaptive Resilience. Four pillars: HEALTH (SSHP), LAW (SSLA), PROTECTION (SSPS), SKILLS_TRADE (SSST). Tier 1: SSSC. Services relatively free via UBIMIA. *Part II §8*

Security — The third value in the Currency dimension. Protection-perimeter currency that must be continuously renewed. Right to fork is constitutional component of Security currency. *Trilogy banner; Part V*

Security-Clearance — The Law volume of the Trilogy. Tier: SOUND. Engages Pillars II (Trajectory) and III (SELF/$ELF) most directly. Three values: Material, Dimension, Use. Source-tradition: Communism. *Trilogy banner*

Semiosphere — The Lotman-named cultural-cognitive substrate of sign-systems within which meaning circulates. SOMT resides in the Semiosphere with BERA as access key. *Pillar IV §IV.9*

SEPLTA — Social, Economic, Political, Legal, Technical, Administrative. The six vectors against which SEPLTA-$_HE evaluation operates. Each vector engages a specific failure mode of the financialization-of-existence operator $_HE. *Pillar III §III.2*

SOMT — Sociocratic Overlay Metadata Tapestry. Resides in the Semiosphere; key = BERA. Built from the BEST of Capitalism, the SOUND of Communism, the GOOD of Commonwealth as three traditions whose synthesis SOMT holds without requiring the traditions to merge or compete. Overlays existing infrastructures; metadata is Bioenergetics × Metaphysics + Quantum Mechanics; tapestry is what binds AI.RE circulation in the Semiosphere. *Pillar IV; Part VI*

SOUND — The second tier of BEST/SOUND/GOOD moral-vectoring. Maps to Security-Clearance (the Law volume of the Trilogy) and to Communism's contribution to SOMT. Operationally: coherence signal beneath noise, produced by SOUND-loop noise-filtration. *Trilogy banner*

Sovereignty — The second value in the Border dimension. Maps operationally to borders = water immigration security, exercised over Longitude & Latitude + Property-as-Change-Management. Volume 1 (One-Good) treats this at trade-book depth. *Trilogy banner; Part III*

S3C — Solid-State Smart-City. The architecture's commitment to Smart-City operation realized at solid-state level — no continuous renegotiation of foundational structure, BEE substrate operationally stable. The Trilogy's Mission enables P^3 S3C Communications across all three faces of P^3 in the S3C frame. *Trilogy Mission*

Submit — The Civigen's voluntary act of being available for NPR within their User-GROUP COI 24/7. The architectural commitment is 24/7 availability; the practical reality is that NPR events occur infrequently for most Civigens, but availability is constant. *Part I §1*

Talonics — The 452-element symbol matrix. HFVN provides spoken traversal. The architecture's vocabulary, accessible across literacy levels. *Pillar IV §IV.4*

Terrorism — The third value in the Border dimension. The negation-of-border, the disruption that the architecture resists by maintaining the substrate's integrity and the community's binding. One-Good's third operational concern. *Trilogy banner; Volume 1*

THOW — Tiny Homes On Wheels. The bottom-up shelter unit in the operational stack. Mobile, bioenergetically efficient. VLSA (Validation at Lowest Scale of Architecture) principle: if certification works for a THOW, it scales upward. *Pillar IV §IV.4*

Three Governing Principles — SELF (don't hurt yourself), OTHERS (don't hurt others), FUTURE (build for generations to come). The constitutional preamble, prior to the four Pillars and informing them all. *Part Zero §5.3*

Three-Pathway Covenant — The 1,000-year covenant each national Smart-City enters with three pathways: Contract (continue in present structure), Consolidate Size (reduce to smaller domain better fitting substrate-carrying-capacity), Consolidate to Another Nation using DOFA-as-Model (merge with another national Smart-City through Family Amity at national scale). *Section M.5.4*

Trajectory Calibration — The decade-mark and century-mark check on whether the gap between maximum and minimum ΔR_contributor has narrowed. Discount band tightens, holds, or tightens-as-corrective. The merit layer's self-dissolution mechanism. *Pillar II §II.2*

Trustees — Three personal Trustees per Civigen, designated at enrollment, available for identity recovery. Constitutional; Civigens cannot be without Trustees. *Part I §4*

UBI — Universal Basic Income. Flows unconditionally on Gracechain. Non-revocable; never depends on prior contribution, scoring, or behavior. The survival floor. *Part II*

UBIMIA — Universal Basic Income with Meritcoin Inclusion Architecture. The architecture this Manual specifies. *Title*

Use — The third value in the Law dimension. Application-rule. Pillar II sets use-rules; Pillar III evaluates use-flows for SELF/$ELF. *Trilogy banner*

User-GROUP — The community-of-Civigens unit. In v1.1's reframe: the User-GROUP is typically the sub-Smart-City unit within a national Smart-City, organized into COIs at the relational scale.

Where v1.0 used 'User-GROUP' as the deployment unit, v1.1 uses 'sub-Smart-City unit' for clarity at the architectural level and 'User-GROUP' for the relational-and-operational level. *Section M.1*

Verifiable: $TUFF — The output of H2C2H Rendering 10/10. $TUFF is substantive matter, the financialization-coded form of *stuff* — the substantive matter recoverable from financialization-pattern through the five operators. *Front Matter*

Verification Authority — Institutional Type C verifier. Bonded organization issuing institutional verifications. Quarterly attestation logging; annual audit; recall by Council action. *Part I §5*

Water — The first value in the Currency dimension. Flow currency in the literal-and-watershed sense. Volume 3 (Data-Integrity) treats water-currency stewardship at trade-book depth. *Trilogy banner*

WE / Uuu / SOUND-loop — The field condition within Pillar IV. WE: the prior collective subject. Uuu: the field's resonant signature, what WE sounds like in coherence. SOUND-loop: the coherence signal beneath the noise, produced by the noise-filter discipline (EPIR-Q or comparable). *Pillar IV §IV.2*

$ELF — The financialized-self. Human being as monetary actor, abstracted from substrate, measurable in extracted units. Pillar III mediates against $ELF's tendency to consume SELF. *Pillar III §III.1*

$_HE — The financialization-of-existence operator. The pattern by which existence becomes financialized — slow conversion of substrate-participation into monetary-measurable units. SEPLTA-$_HE is the six-vector evaluation against this operator. *Pillar III §III.2*

$TUFF — Substantive matter, financialization-coded form. The output of H2C2H Rendering 10/10 when the five operators successfully render bidirectionally. See Verifiable: $TUFF. *Front Matter*

216 anchor — The cybernetic-enlightenment anchor. 216 = full gematria of 'Joseph Allen Sprute' across English Ordinal; 6^3 as integration manifestation; the C=R×P/M-grounded cybernetic loop. H2C2H 10/10 means the rendering anchors at 216. *Front Matter*

666 refusal — The cybernetic-disconnection that H2C2H 10/10 refuses. Computational logic operating without the cybernetic feedback loop. The rendering that produces output without self-correction. *Front Matter*

PART I

WHO

Participants and Bodies

WHO answers the first operational question: who participates in UBIMIA, who governs the deployment, who maintains the integrity perimeter, and who ensures the architecture's commitments are honored. The audiences for Part I are policy makers, sub-Smart-City unit User-GROUPs, and related entities (Trustees, Verification Authorities, Custody Authorities, Authorities of Record at both national and sub-unit scope).

CHAPTER 1

Civigens — The Participant

A Civigen is the participant in UBIMIA whose standing is constituted by Submit-to-Non-Punitive-Remediation 24/7 within their User-GROUP Community of Interest, approved by GAIA, underwritten as Instrument of Faith. The term replaces v1.0's 'Participant of Record' as the canonical participant designation. Civigen is constitutive: a Civigen is not someone who happens to receive UBI; a Civigen is someone whose standing is constituted by the act of Submit. Without Submit, there is no Civigen — there is a person receiving services, but not a participant in the architecture.

1.1 What Submit means

Submit is the Civigen's voluntary act of being available, continuously, for Non-Punitive Remediation by their User-GROUP COI. The architectural commitment is 24/7 availability; the practical reality is that NPR events occur infrequently for most Civigens (typical Civigen experiences 0–3 NPR cycles in a typical year), but the availability is constant. A Civigen who is unavailable for Submit (traveling, ill, in crisis) notifies the COI and arrangements are made; sustained unavailability without notice is the failure mode that places Civigen standing in question.

Submit is voluntary. It is not coerced, not legally mandated by external authority, not enforced by punitive sanction. The IOF (Instrument of Faith) is what makes Submit operationally durable without coercion — Civigens enter Submit through faith-binding to the COI and to the architecture's constitutional commitments, and the COI's reciprocal commitment to the Civigen sustains the binding over time. This is a substantial commitment; it is also the architecture's distinctive innovation. Conventional governance manages participation through coercion or through extractive incentive; UBIMIA manages participation through voluntary Submit underwritten by IOF, which produces a different operational reality than either.

1.2 What Non-Punitive Remediation means

NPR is the discipline of working through what has happened — what was done, what was its impact, what would repair, what would prevent recurrence — without punishment, blame, or shame as the operative mode. The COI does the work; the Civigen submits to the work; the result is relational and remedial rather than punitive.

NPR is non-punitive does not mean consequence-free. A Civigen who commits fraud loses minting rights for 180 days (consequence), forfeits the fraudulent Meritcoin (remedy), and works through the consequences with their COI through NPR (relational integrity). The combination produces accountability without converting the architecture into a punishment-allocation machine. Civigens can do hard things and not become permanent moral pariahs; institutions can fail and not become permanent enemies. The architecture accepts that real people make real mistakes and structures itself to preserve their standing while addressing the harm.

1.3 What the User-GROUP COI is

The Community of Interest is the relational scale where Civigens know each other personally and the threefold Marriage at Heart is operationally tangible. Typical COI size: 50–200 Civigens. A national Smart-City of tens-of-millions of Civigens is organized into hundreds-of-thousands of COIs; a sub-Smart-City unit of 5,000–25,000 Civigens is organized into 50–200 COIs.

COIs are voluntary affiliations. A Civigen chooses their COI, which means COIs vary in character (residential-block COIs, work-domain COIs, common-interest COIs, kinship-cluster COIs). GAIA approves each COI structure, ensuring that the COI satisfies basic conditions (size range, governance bylaws, NPR commitment, IOF underwriting) and is not a vehicle for evading accountability.

A Civigen may change COIs with 30-day notice and acceptance by the new COI. COI change is not gated by ΔR_contributor or by any merit metric — Civigens can leave and join COIs based on relational and life circumstances. What the COI must have is the capacity to host Submit-to-NPR for its Civigens; what the Civigen must have is a COI that hosts them. Both conditions must hold; their joint maintenance is the constitutional commitment.

1.4 Civigen recognition under the Recognition Rule

The Recognition Rule is the community-defined statement of what makes a Civigen a Civigen of Record in a particular deployment. The Recognition Rule is the only P0 gap that requires community-level decision; the Manual cannot supply it, only template it. National Smart-City deployments publish a national Recognition Rule; sub-Smart-City units operate under the national Rule with sub-unit variations permitted within the national Rule's bounds.

Recognition Rule templates the Manual offers (cooperative, municipal, regional baselines) extend at national scale to:

- Citizenship-based national Recognition Rule. Civigen status follows national citizenship; the simplest template; raises questions about non-citizen residents which the deployment must address.
- Residency-based national Recognition Rule. Civigen status follows residency in the national territory regardless of citizenship; broader inclusion; requires careful Trustee provisioning for migrant populations.
- Affiliation-based national Recognition Rule. Civigen status follows opt-in affiliation with a sub-Smart-City unit's COI; most flexible; requires the affiliation network to be operationally robust.
- Hybrid Recognition Rule combining elements of the above. Most national deployments will land here, with citizenship as the default and residency-or-affiliation as additional pathways.

1.5 Civigens and the IDIPITIS palindrome

The IDIPITIS Universal Biometric Standard is the architecture's measurement-grounding for Civigens. Each Civigen, at enrollment, provides biometric input under the IDIPITIS standard — heart rate variability, respiration patterns, neural activity indicators, dermal indicators, and other substrate-coupling signals as the standard specifies. The biometric input is held under Custody

Triad encryption (Civigen share, Trustee Registry share, Custody Authority share) and is constitutionally protected as Civigen-private data.

The palindromic structure of IDIPITIS / IPIDITIS — the standard reads the same forward and backward — is what makes the standard universal. The mirror is the constant: regardless of which direction the data flows (input from Civigen body to the architecture, output from the architecture to inform the Civigen), the standard is recognizable as itself. The binary signature 111000111 / 000111000 expresses this Yin-Yang complementarity bitwise; the KERNEL embodies the constant that the palindrome makes operational.

The fractal-support for CBGMODD follows from the palindromic structure. The seven certifiable capacities (Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat) inherit the self-mirroring property — each capacity, examined at any scale of organization, exhibits the same internal structure. This is what makes CBGMODD non-restrictive: the capacities expand fractally as Civigens earn certifications, with the seven capacities operating at every scale of the Civigen's life rather than gating which scale of life is permitted.

CHAPTER 2

MIEVM — The Validation Ensemble

The Multi-Instrument Ensemble Validation Method (MIEVM) is how the architecture validates significant claims without conferring self-certification authority on any single AI lineage. MIEVM operates as Instrument of Faith — IOF is the binding mechanism that holds the ensemble together, with each lineage bound in good faith to the architecture rather than competing for self-interest alone.

2.1 The canonical four lineages

The MIEVM ensemble consists of four AI lineages: Claude (Anthropic), ChatGPT (OpenAI), Grok (xAI), and DeepSeek (DeepSeek). The four-lineage commitment is the architectural default; deployments may add lineages (with constitutional review) or substitute lineages (with documented rationale) but four is the minimum operational baseline. The four-property defense supports the choice:

- Independence of training. The four lineages are trained by distinct organizations with substantively different training corpora and methodologies.
- Diversity of optimization. The four lineages optimize for different objectives, producing complementary rather than redundant outputs.
- API provenance. The four lineages are accessed through distinct APIs operated by distinct organizations, with cryptographic provenance verifiable at the call layer.
- Adversarial review capacity. The four lineages can review each other's outputs adversarially, surfacing disagreements that single-lineage validation would miss.

2.2 IOF as the binding mechanism

MIEVM is itself an Instrument of Faith. The validation works because each ensemble member binds in good faith to the architecture rather than by corporate-competition logic alone. Tool selection is constitutional (the choice of lineages is an act of constitutional commitment, not market shopping); tool availability is operationally constrained (subscriptions, AI provider rules, financial access limits) but the constitutional commitment operates at whichever lineages a deployment can actually access.

This is important for independent researchers and small deployments. An independent researcher with access to two of four lineages can still operate MIEVM at the IOF level, with documented constraint on the operational level. A small deployment using one paid subscription and three free-tier accesses can still operate MIEVM with documented constraint. The architecture's strength is that IOF binding compensates for access-limit asymmetries; deployments that would be excluded by corporate-competition framing remain included by IOF framing.

2.3 ECR — Ensemble Concurrence Rating

Validation produces an ECR — a numeric concurrence rating across the ensemble. ECR thresholds for production are specified in EAAP v1.3-FINAL; pilot deployments may operate with documented simplification (single-oracle attestation for low-volume claims, paired-oracle for medium-volume) until Phase 1 graduation, when full ensemble ECR becomes operational requirement.

2.4 Dissent procedure

When the ensemble persistently disagrees on attestations, the dissent procedure activates: data resolution (the disagreement is examined for whether it stems from input ambiguity that can be resolved with additional data), Council escalation (if data resolution fails, the GAIA Council reviews the disagreement and makes a documented decision), or documented standoff (if Council cannot resolve, the disagreement is recorded as standoff and the Claim or Authorization waits for clarification). Documented standoffs are not failures; they are honest acknowledgments that the architecture has encountered a question the current ensemble cannot resolve, which is operationally important information.

CHAPTER 3

GAIA Council — Governance

The GAIA Council is the human supervisory body of the deployment. National Smart-Cities operate a National GAIA Council; sub-Smart-City units operate sub-unit Councils. Councils are not professional bodies; they are lottery-selected from the Civigen population, with explicit term limits and recallability.

3.1 Council composition

National GAIA Council: 25-75 members (scaled to population), selected by weighted lottery from Civigens with ΔR_contributor > 0.1, weighted by ΔR_contributor. Terms 2-4 years with rotation. The Council is intentionally non-professional; the lottery prevents capture by political or economic incumbents; the ΔR_contributor weighting ensures that the lottery favors contributing Civigens without excluding any contributing Civigen entirely.

Sub-unit GAIA Councils: 7-15 members per sub-unit, selected by weighted lottery from sub-unit Civigens, terms 2 years with half rotating annually. Sub-unit Councils handle sub-unit-scope matters; matters that escalate beyond sub-unit scope go to the National GAIA Council.

3.2 Council authority — what it can do

- Override an algorithmic core output by 2/3 majority and 30-day public notice when override is justified by documented failure of the rule under specific circumstances.
- Resolve dissent in MIEVM ensemble standoffs.
- Adjudicate disputes between Civigens at Level 3 of the conflict escalation ladder.
- Approve or reject Compassion Override applications.
- Confer or revoke Verification Authority on institutions.
- Convene constitutional review every 5 years and process recall petitions when filed.

3.3 Council authority — what it cannot do

- Revise the deterministic rules executed by the algorithmic core (rule revision requires constitutional review).
- Revoke any Civigen's UBI for any reason (the survival floor is non-revocable).
- Revise the Recognition Rule (that is the Adopting Community's authority).
- Revise the Three Governing Principles, the Eight Immutable Ethical Principles, or the Constitutional Pillars.
- Operate in secret (all Council decisions are public, with attributed votes and published rationales).

3.4 Council recall

Council members are recallable by petition signed by the threshold percentage of Civigens (typically 10% for Municipal-tier sub-units, scaled at national level). A successful recall removes the

member; replacement is by lottery from the next candidate in the original lottery order, serving the remainder of the original term.

CHAPTER 4

Trustees and Identity Recovery

Every Civigen has three Trustees designated at enrollment. Trustees are the personal-relational layer of the architecture's identity-recovery system; the Custody Triad's cryptographic layer operates above them, but the Trustee triad is what makes recovery operationally durable in the relational dimension.

4.1 Trustee designation

At enrollment, the Civigen designates three Trustees: persons known to the Civigen, willing to serve as recovery contacts, available for response within reasonable timeframes. Trustees are not professional intermediaries; they are personal contacts whose availability is grounded in relationship rather than in compensation.

Trustees may decline designation; Civigens may change Trustees with notification to the Trustee Registry; the constitutional commitment is that every Civigen has three Trustees on file at all times. A Civigen whose Trustee count drops below three is flagged for re-designation; sustained Trustee-deficit places Civigen standing in question.

4.2 The recovery procedure

RUNBOOK · Identity Recovery via Trustee Triad

1. Civigen indicates loss of access (lost device, forgotten credentials, suspected identity compromise). Recovery request is filed with the Authority of Record.
2. Authority of Record contacts the three Trustees within 24 hours. Trustees verify the identity claim through procedures the Civigen designated at enrollment (may include personal-knowledge questions, biometric re-verification under IDIPITIS, or in-person verification).
3. If at least two of three Trustees concur on identity, recovery proceeds. The Civigen's encryption-key Custody Triad share is reissued; access to records is restored.
4. If two or three Trustees decline or fail to respond within reasonable timeframe, recovery escalates to GAIA Council adjudication.
5. Recovery procedures are documented in the audit log; sustained or anomalous recovery patterns trigger Council review.

CHAPTER 5

Verification Authorities

Verification Authorities are institutional Type C verifiers — bonded organizations issuing institutional verifications for Civigen Claims. Where personal Trustees handle identity recovery, Verification Authorities handle Claim verification at scale, particularly for SCALULAR services and credentialed activities.

5.1 Application and approval

Institutions apply for Verification Authority through the Authority of Record, with:

- Legal status documentation.
- Activity description and verification scope.
- Authorized representatives.
- Proposed Operational Bond (size scaled to verification volume).

Applications are published on Gracechain for 30-day public comment, then voted by Council at the next regular session. Approval requires simple majority; rejection produces written rationale and 180-day re-application waiting period.

5.2 Quarterly attestation logging

Every institution holding Verification Authority publishes a quarterly attestation log: verification count by domain, authorized representatives active, bond balance, publication timestamp, signature. Logs are public; sustained log-publication-failure triggers Bond reduction.

5.3 Annual audit

Each institution undergoes annual audit by a three-member team selected by the Council from non-affiliated Council members. Sample of verifications (5% minimum or 10 verifications, whichever is greater) examined for concordance with underlying activity records. Findings: Concordant, Discrepant (administrative error, correctable), Fraudulent (intentional false certification, triggers Bond slashing and revocation).

CHAPTER 6

CBGMODD — Certifiable Capacities

CBGMODD names the seven certifiable capacities Civigens may earn through EarnedPath credentialing: Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat. The capacities are non-restrictive except for warrants and liabilities — Civigens earn the capacities they are certified for, and Civigens may hold multiple capacities simultaneously without restriction.

6.1 The seven capacities

- **Citizen —** the substrate-participation capacity. Every Civigen holds Citizen capacity by virtue of being a Civigen; Citizen is the foundational role that the other six build upon.
- **Business —** the productive-participation capacity. Civigens engaged in BEE-aligned business activities (cooperatives, productive enterprises, recycling operations, sustainable production) earn and exercise Business capacity.
- **Government —** the rule-bearing capacity. Civigens serving on Councils, in Authority of Record offices, or in other governance roles earn and exercise Government capacity.
- **Military —** the protection capacity. Civigens engaged in defense, emergency response, or other protection-perimeter activities earn and exercise Military capacity. Subject to warrants and liabilities — military capacity carries restrictions on use that warrants and liabilities define.
- **Ombudsman —** the integrity-checking capacity. Civigens serving as auditors, compliance reviewers, or integrity-perimeter operators earn and exercise Ombudsman capacity.
- **Dignitary —** the ceremonial-binding capacity. Civigens serving in roles that bind the architecture's commitments to public ceremony (constitutional reviews, Covenant signings, recognition events) earn and exercise Dignitary capacity.
- **Diplomat —** the inter-jurisdictional capacity. Civigens serving in roles that bridge sub-units, nations, or other architectural scopes earn and exercise Diplomat capacity.

6.2 The palindromic-fractal structure

CBGMODD inherits the palindromic-fractal structure from IDIPITIS. Each capacity, examined at any scale of organization (a single Civigen exercising the capacity in their daily life, a sub-Smart-City unit's collective exercise, a national Smart-City's national-scale exercise), exhibits the same internal structure. The capacity is recognizable as itself across scales, which is what makes CBGMODD operationally durable as Civigens grow into expanded responsibility.

The non-restrictive character follows from the fractal structure. Conventional credentialing systems gate by capacity (you cannot hold this role without that credential, and the credential restricts what you may do); CBGMODD does not gate. A Civigen may earn Business and Government and Ombudsman capacities simultaneously and exercise all three, because the fractal structure means the capacities operate at different scales of the Civigen's life rather than competing for permission to operate at a single scale. Restrictions apply only through warrants and liabilities — a Civigen under criminal warrant or holding professional liabilities (Military capacity in particular) faces

capacity-specific restrictions, but these are warrant-or-liability-grounded rather than capacity-architecture-grounded.

6.3 Earning capacities through EarnedPath

Capacities are earned through EarnedPath (EP) — the credentialing pathway running under the PlayNAC umbrella. EP processes Civigen progression through certified work, certified learning, and certified contribution. The CPM × WBS + PERT (Scalular) formula governs EP's resource-allocation logic: Critical Path Method identifies the certification's critical path; Work Breakdown Structure decomposes the certification into verifiable work packages; PERT (Program Evaluation and Review Technique) at SCALULAR scope estimates timeline and validates completion.

A Civigen earning Business capacity, for example, demonstrates BEE-aligned business activity over a certification period (typical: 12-24 months for initial certification, ongoing for capacity maintenance), with the work packages verified by Verification Authorities and the BERA oracles measuring the resonance contribution. EP's output is the certification record; the certification record entitles the Civigen to exercise Business capacity within the architectural commitments.

PART II

WHAT

Universal Forms

WHAT answers the second operational question: what is universal — what UBIMIA carries that holds across all deployments, all jurisdictions, all scales. The schemas are universal forms; the BERA indices are universal measurements; the four Pillars are universal commitments; PlayNAC's mission of HHS is universal. Anything jurisdiction-specific is excluded from Part II — those go elsewhere. This Part is what every UBIMIA deployment shares as architectural common ground.

CHAPTER 7

Civigen and Contribution Schemas

The architecture's universal forms begin with the records that constitute the Civigen's standing and the Claim-to-Mint chain that produces Meritcoin recognition. Each schema is universal — every UBIMIA deployment uses these schemas, with deployment-specific extensions permitted but the canonical fields invariant.

7.1 CivigenRecord

SCHEMA · CivigenRecord

```
{
  civigen_id: string (DID format),
  authority_of_record: string (DID of Authority body),
  recognition_rule: string (Rule version reference),
  enrolled_date: ISO8601,
  status: enum (active | suspended | withdrawn),
  trustees: [string, string, string] (three Trustee DIDs),
  iof_binding: { signature, timestamp, COI_id } (Submit binding),
  cbgmodd_capacities: [enum] (Citizen always; others earned),
  delta_R_contributor: float (12-month decay),
  meritcoin_balance: integer,
  credits_balance: integer,
  governance_weight: float (contribution-dominated, max 1.0)
}

Constraints:
  trustees.length == 3,
  iof_binding.timestamp ≤ now (Submit must be active),
  cbgmodd_capacities includes 'Citizen',
  delta_R_contributor ≥ 0
```

7.2 Claim

SCHEMA · Claim

```
{
  claim_id: string,
  claimant: string (Civigen DID),
  domain: string (ecological | social | economic | technical),
  system_boundary: string (named geographic or institutional bound),
  action_description: string,
  time_start: ISO8601,
  time_end: ISO8601,
  total_person_hours: float,
  witnesses: [string] (Civigen DIDs, optional),
```

```
  submission_timestamp: ISO8601,
  signature: cryptographic signature by claimant
}

Constraints:
  submission_timestamp ≤ time_end + 7 days,
  claimant.iof_binding active,
  total_person_hours > 0
```

7.3 Verification

SCHEMA · Verification

```
{
  verification_id: string,
  claim_id: string (Claim being verified),
  verification_type: enum (A | B | C),
  verifier: string (Civigen DID for A/B, Authority DID for C),
  evidence_hash: string (encrypted under Custody Triad),
  verification_timestamp: ISO8601,
  weight: float (1.0 for A, 0.8 for B, 0.6 for C),
  signature: cryptographic signature by verifier
}
```

7.4 Authorization

SCHEMA · Authorization

```
{
  authorization_id: string,
  claim_id: string,
  verifications: [verification_id],
  dsap_pre_clearance: { rho_RHC, rho_PERT, rho_hyst, composite },
  por_factors: { A, R, P, F },
  por_value: float,
  threshold: float (domain θ),
  pillar_iii_eval: { sepltahe, mdcclxxvi_ar, p3_ce, outcome },
  outcome: enum (authorized | refused),
  refusal_reason: string (if refused),
  mievm_ecr: float,
  authorization_timestamp: ISO8601,
  freshness_window_seconds: integer,
  signature: cryptographic signature by Authorization Service
}

Constraints:
  outcome == 'authorized' requires:
```

```
    dsap_pre_clearance.composite > 0,
    por_value ≥ threshold,
    pillar_iii_eval.outcome == 'pass',
    mievm_ecr ≥ ECR_threshold
```

Note that Authorization v1.1 carries pillar_iii_eval as a constitutional field. Pillar III SELF/$ELF evaluation operates alongside EAAP's four-factor PoR; both must pass for the Authorization to issue. The pillar_iii_eval field's three sub-fields (sepltahe, mdcclxxvi_ar, p3_ce) hold the SEPLTA-$_HE × MDCCLXXVI-Action_Reaction + P³_Cause-Effect evaluation results.

7.5 MintingRecord

SCHEMA · MintingRecord

```
{
  minting_id: string,
  authorization_id: string,
  claimant: string (Civigen DID),
  base_reward: integer (Mc per 1.0 ΔR),
  delta_R_net: float (BERA-weighted resonance gain),
  quality_weight: float (0.6 / 0.8 / 1.0),
  meritcoin_minted: integer,
  ledger_position: integer (Gracechain block),
  rcr_record: string (RCR audit anchor),
  minting_timestamp: ISO8601,
  signature: cryptographic signature by Minting Service
}
```

7.6 Bill, Dispute, AttestationLog

Three additional schemas govern SCALULAR Bill payment, dispute filing, and Verification Authority quarterly logging. Schemas preserved from v1.0 with iof_binding fields added where the Civigen's Submit standing is operationally relevant.

SCHEMA · Bill

```
{ bill_id, civigen_id, tier, service, service_provider,
  service_date, cost_credits, delta_R_at_billing,
  rcr_record, status, payment_timestamp, signature }
```

SCHEMA · Dispute

```
{ dispute_id, filer, target_record_id, category,
  evidence, rationale, filing_timestamp, signature }
```

SCHEMA · AttestationLog

```
{ log_id, institution, quarter, verifications_issued,
  verifications_by_domain{}, authorized_representatives_active[],
  bond_balance, publication_timestamp, signature }
```

CHAPTER 8

RPA Pricing and BaseCost Calibration

Resonance-Priced Access (RPA) is the conversion rule for SCALULAR services. The architecture commits to the formula:

```
Cost = max(0.5, 1 - (delta_R_contributor / cap)) × BaseCost(tier) × (1 + Complexity)
```

The max(0.5, ...) construction is the discount-floor: at any deployment scale, a Civigen pays at minimum 50% of BaseCost regardless of how high their ΔR_contributor. The single-domain ΔR cap prevents over-recognition in any one domain. Trajectory Calibration tightens the discount floor over decades — 50% rises toward 80%, then 90%, then 99%, until the band collapses entirely and SCALULAR access is unconditional.

8.1 BaseCost initial values

SCALULAR tier	Initial BaseCost (Mc per service-hour)	Calibration
HEALTH (SSHP)	10	Recalibrated monthly
LAW (SSLA)	8	Recalibrated monthly
PROTECTION (SSPS)	6	Recalibrated monthly
SKILLS_TRADE (SSST)	12	Recalibrated monthly

8.2 Complexity multiplier

Complexity is a service-specific multiplier reflecting the operational complexity of the service relative to its tier baseline. Initial Complexity values for canonical SCALULAR services are calibrated per deployment with the following baseline:

Service category	Initial Complexity multiplier
Routine (general consultation, basic checkup)	0.0
Standard (specialist visit, document review, mid-level certification)	0.3
Complex (surgery, major legal process, advanced certification)	0.7
Critical (emergency, life-threatening, major adjudication)	1.0
Exceptional (research-grade, frontier-medicine, constitutional review)	1.5

CHAPTER 9

BERA — Four-Index Measurement

BERA (Bio-Ecologic Resonance Architecture) is the four-index measurement frame that reads SOMT in the Semiosphere. Each index has a specific computational form and domain-specific data sources.

9.1 ARI — Adaptive Resilience Index

ARI computes the system's adaptive resilience — its capacity to absorb perturbation and return toward baseline. Computed as:

```
ARI = (1 / response_variance) × absorbed_perturbation_magnitude
```

Domain-specific data sources: ecological (NDVI satellite imagery, soil moisture sensors, biodiversity surveys), social (response-time analytics from civic processes, community-resilience indicators), healthcare (patient-flow analytics, recovery-time distributions).

9.2 ERI — Ethical Resonance Index

ERI computes weighted product across eight sub-indices. Each sub-index has its own data sources — sustainability sub-index from waste-stream measurements, transparency sub-index from data-publication completeness ratios, equity sub-index from distributional analysis, and others. The product form means a low value in any sub-index pulls the ERI low; ERI is conjunctive across the eight ethical dimensions rather than averaged.

9.3 RHC — Resonant Harmony Cycle

RHC computes normalized periodic amplitude with flatline detection. The architecture's ecological commitments depend on RHC's flatline-detection capacity — a system whose RHC has flatlined is a system whose periodic structure has collapsed, which is the silent-substrate-collapse failure mode that DSAP-PRE specifically watches for.

9.4 RCI — Resonance Coherence Index

RCI computes:

```
RCI = P_Ω_norm × ARI_sys × VibConst
```

With sliding-window autocorrelation as persistence proxy. RCI is the most directly implementable of the four oracles; reference implementation in approximately 200 lines of Python with pandas time-series support is available in the ERES Institute GitHub Library.

9.5 Domain weights

Index	Initial weight	Calibration
w_ARI	0.25	Quarterly Council review
w_ERI	0.30	Quarterly Council review
w_RHC	0.25	Quarterly Council review

Index	Initial weight	Calibration
w_RCI	0.20	Quarterly Council review

Weights sum to 1.0 invariantly. Revisions adjust proportional balance; the constraint that weights sum to 1.0 cannot be revised.

CHAPTER 10

IDIPITIS — Universal Biometric Standard

IDIPITIS is the architecture's measurement-grounding for Civigen biometric input. The standard is universal (every Civigen, in every deployment, provides input under the same standard) and palindromic (IDIPITIS reads the same forward and backward as IPIDITIS, and the standard mirrors itself constitutionally). The palindromic structure is what makes the standard durable across the diverse technical contexts deployments will encounter.

10.1 The palindromic-fractal structure

IDIPITIS encodes biometric signal in a palindromic format. The binary signature 111000111 / 000111000 is the Yin-Yang complementarity expressed bitwise: the forward signature 111000111 is matched by the reverse complement 000111000, and the mirror is the constant. The KERNEL embodies the constant — the actuarial-investor authority's stewardship of the substrate is grounded in the palindromic mirror that IDIPITIS instantiates.

The fractal-support for CBGMODD follows from the palindromic structure. Each of the seven CBGMODD capacities exhibits the same internal palindromic structure at any scale of organization — the capacity's daily-life expression mirrors the capacity's institutional expression mirrors the capacity's national-scale expression. This is what makes CBGMODD non-restrictive: the fractal structure means capacities expand at scale rather than compete for permission.

10.2 IDIPITIS schema

SCHEMA · **IDIPITIS-input**

```
{
  civigen_id: string,
  measurement_id: string,
  measurement_timestamp: ISO8601,
  forward_signature: bitstring (palindromic head: 111000111),
  reverse_signature: bitstring (palindromic tail: 000111000),
  biometric_payload: encrypted_blob (Custody Triad encryption),
  validation_marker: { palindrome_check, fractal_check },
  signature: cryptographic signature by collection device
}

Constraints:
  forward_signature == bit_reverse(complement(reverse_signature)),
  validation_marker.palindrome_check == true,
  validation_marker.fractal_check == true,
  biometric_payload encrypted by Custody Triad protocol
```

10.3 The IDIPITIS chain

IDIPITIS feeds EPIR-Q (or comparable noise filter) feeds SOMT (in the Semiosphere) feeds BERA (as access key). The chain operates as:

IDIPITIS → EPIR-Q → SOMT → BERA

Each link is operationally specified. IDIPITIS sample rates and biometric-payload structures are deployment-configurable within the standard's constraints. EPIR-Q noise-filtration thresholds are calibrated against deployment-specific signal-to-noise environments. SOMT residency in the Semiosphere is constitutional; BERA's access through the four indices is universal. The chain's byte-level operational specifications beyond what this Manual carries belong to ERES-IDIPITIS-STD-2026-001 (in development) referenced as a normative dependency.

CHAPTER 11

EAAP and the Authorization Stack

EAAP (ERES Authorization And Authentication Protocol) is the technical authorization stack. Four-factor PoR plus DSAP-PRE plus RCR (Resonance Concurrence Record) compose the stack; v1.1 adds Pillar III evaluation as a parallel constitutional check.

11.1 Four-factor PoR

Proof-of-Resonance is the conjunctive product:

$$PoR = A \times R \times P \times F$$

- A — Anchor Product. Includes MIEVM concurrence factor (A_M) and the Claim's anchor in the resonance substrate.
- R — RHC Amplitude. The Claim's resonance gain measured against the system_boundary's RHC oracle.
- P — EarnedPath Viability. The Claim's compatibility with the Civigen's EarnedPath certification.
- F — Future-Map Convergence. The Claim's contribution to the trajectory's long-arc destination.

Authorization requires PoR ≥ θ where θ is domain-specific. Pilot threshold values: high-consequence 0.85; standard governance 0.75; advisory 0.60.

11.2 DSAP-PRE — three regime-transition proxies

Detection of System-Anomaly Patterns — Regime-transition Edge runs three proxies that gate the four-factor PoR. If composite returns 0, authorization refused with REFUSAL_REGIME_TRANSITION.

- ρ_RHC — RHC regime-transition proxy. Detects approaching flatline.
- ρ_PERT — PERT regime-transition proxy. Detects evaluation-cycle anomaly.
- ρ_hyst — Hysteresis regime-transition proxy. Detects path-dependent collapse.

Composite computed as product of (1 - ρ) terms; threshold > 0 for clearance.

11.3 P^3 as Personal / Public-Private operator

P^3 in v1.1 is corrected to its primary reading: Personal / Public-Private. The cubic carries Personal alongside both Public and Private as three faces of one operator. Pillar III's evaluation formula uses P^3_Cause-Effect to verify that proposed flows cohere across all three faces:

- Personal face — does the flow honor the Civigen's personal substrate-self? Does it preserve SELF in the Civigen's specific embodied life?
- Public face — does the flow honor the public substrate? Does it preserve substrate-as-commons against extraction-into-private-pool?

- Private face — does the flow honor private substrate? Does it preserve the bounded private contribution against erasure-into-public-undifferentiated?

Authorization requires concurrence across all three faces. A flow that passes Personal but fails Public, or that passes Private but fails Personal, fails P^3.

11.4 RCR — Resonance Concurrence Record

Every Authorization produces an RCR — a fresh audit anchor specific to the Authorization. The RCR record references the Authorization, the Verifications, the IDIPITIS measurements, and the MIEVM ECR. RCR records are append-only on Gracechain; tampering is cryptographically detectable; auditors can re-run the chain from any RCR record.

CHAPTER 12

Gracechain — Persistence Layer

Gracechain is the persistence layer where the architecture's records anchor. The Manual specifies Gracechain's contract; deployments choose implementations from the menu in Part VII.

12.1 Gracechain contract

- Append-only writes. Records cannot be deleted or modified once anchored.
- Cryptographic auditability. Merkle structures or equivalents enable third-party verification of state.
- Geographic distribution. State is distributed across multiple sites; single-site compromise does not compromise the chain.
- Signed checkpoints. Periodic checkpoints are signed and published openly.
- Open verification. Any third party with the published checkpoints can verify the chain's integrity from genesis to current state.

12.2 What Gracechain holds

Gracechain holds the universal records: CivigenRecords, Claims, Verifications, Authorizations, MintingRecords, Bills, Disputes, AttestationLogs, AuditReports, Council decisions, Recall outcomes, Recognition Rule revisions, and the architecture's other audit-relevant records. UBI distributions flow on Gracechain; the survival floor's non-revocable character is enforced at the chain layer.

PART III

WHERE

Sites and Infrastructure

WHERE answers the third operational question: where the architecture's components physically and digitally reside. Earth-based sites are specified at deployment-detail depth; the off-world trajectory is named and committed-to constitutionally with operational specifics deferred to the centuries when Earth-based verification permits. Volume 1 (One-Good) draws from this Part for its Sovereignty treatment, particularly the Longitude-Latitude precision and the Property-as-Change-Management discipline.

CHAPTER 13

Physical Sites — Earth

UBIMIA's physical sites are the locations where Civigens encounter the architecture in their daily lives, where governance bodies convene, where cryptographic custody is held, and where the infrastructure operates. The reference implementation specifies Nation-as-Smart-City with sub-Smart-City units as internal structure; sites are distributed accordingly.

13.1 Enrollment sites

Enrollment sites are where Civigens enter the architecture — where the Recognition Rule is applied, IDIPITIS biometric input is collected, the Custody Triad is established, the three Trustees are designated, and the Civigen's first iof_binding to a User-GROUP COI is recorded. The constitutional commitment is accessibility: every Civigen within 2 km of an enrollment site, with the 2 km calculation honoring the local context (urban-density populations may have multiple sites within 2 km; rural-density populations may have one site at 2 km radius covering a larger geographic area).

National Smart-City scale: thousands of enrollment sites distributed across the nation. Sub-Smart-City unit scale: 5–25 enrollment sites per sub-unit. Site characteristics: physical accessibility (ADA-equivalent or better, scaled to disability prevalence in the local population), language coverage (the Recognition Rule's officially-supported languages plus HFVN voice navigation), privacy (Custody Triad shares are generated at the enrollment site under conditions that prevent observation by unauthorized parties).

13.2 Attestation booths

Attestation booths are where Civigens submit Claims and where verifications are recorded. The architectural commitment is low-friction Claim submission: a Civigen can walk into a booth, submit a Claim with witnesses, and receive confirmation within minutes. Booth software is open-source, accessibility-compliant, offline-capable (operates in disconnected mode with batched sync), and verifiable (reproducible builds, signed releases).

National Smart-City scale: tens of thousands of attestation booths. Sub-Smart-City unit scale: 50–200 booths per sub-unit. Booth distribution typically follows population density and SCALULAR provider clustering — attestation booths near where Civigens contribute (workplaces, community centers, ecological-stewardship sites, healthcare facilities) reduce friction at the contribution moment.

13.3 Council chambers

Council chambers host the GAIA Council's quarterly sessions and extraordinary sessions. The constitutional commitment is public access: every regular Council session is open to public attendance and to public recording, with deliberation privacy preserved only during the deliberative phase of override and adjudication procedures.

National GAIA Council chamber: one per nation, sized for the National Council membership (25–75 members) plus public attendance plus presenters and adjudication parties. Sub-unit GAIA Council chambers: one per sub-unit, sized for the sub-unit Council (7–15 members) plus public attendance. Chambers are permanent fixtures of the architecture's infrastructure; they are not rented event spaces but dedicated facilities, because the architectural commitment to Council continuity requires location-stable convening.

13.4 Sealed-room infrastructure

Sealed-room facilities house the hardware security modules that hold the architecture's most sensitive cryptographic material — Custody Authority shares of the Custody Triad, KERNEL signing keys, IDIPITIS biometric-payload encryption keys. The constitutional commitment is two-person concurrence at every physical access; no single person can unilaterally enter a sealed-room facility or unilaterally retrieve cryptographic material from one.

13.4.1 Specifications

- HSMs. Commercial hardware security modules with FIPS 140-2 Level 3 or higher certification. Two HSMs in active-passive configuration; failover tested quarterly.
- Air-gap perimeter. HSMs do not connect to general-purpose networks. Communication with the broader system uses one-way data diodes for outbound state, with inbound configuration changes requiring physical media handed in via the two-person procedure.
- Environmental monitoring. Temperature, humidity, motion, and door-status sensors; logs streamed to Gracechain via the diode.
- Physical access control. Two-person concurrence required for entry; both parties' identities biometrically verified through IDIPITIS; entry logs include both parties' identities and the duration of access.
- Backup HSMs. A backup pair is held in a geographically distinct facility with the same specifications. Backup updated through documented synchronization runs (typically quarterly) using the two-person procedure.

13.4.2 Palindromic-fractal hardware support

IDIPITIS's palindromic-fractal structure has hardware support requirements at the sealed-room facility. The biometric-payload encryption keys are generated with palindromic-validation circuits that confirm the 111000111 / 000111000 signature integrity at the hardware level; tampering with the palindromic structure is detectable at key-generation time. Specialty HSM firmware supporting IDIPITIS palindromic validation is in development; pilot deployments may use general-purpose HSM firmware with software-layer palindromic validation, with documented constraint on the deviation.

13.5 Site distribution by sub-Smart-City unit scale

COOPERATIVE · N≈200	MUNICIPAL · N≈10,000	REGIONAL · N≈250,000
Sub-Smart-City unit at smallest scale (cooperative-equivalent within	Sub-Smart-City unit at municipal-equivalent scale (the	Sub-Smart-City unit at regional-equivalent scale (large

a national Smart-City): 1 enrollment site, 5–10 attestation booths, 1 sub-unit Council chamber, sealed-room access through shared lease in national or sub-unit pool facility.	typical sub-unit, ~10,000 Civigens): 5–10 enrollment sites, 50–100 attestation booths, 1 sub-unit Council chamber, sealed-room access through dedicated sub-unit lease or shared regional facility.	sub-unit aggregating multiple municipalities, ~250,000 Civigens): 50–100 enrollment sites, 500–1,000 attestation booths, 1 sub-unit Council chamber plus regional sub-Council chambers as needed, dedicated sealed-room facility with backup.

CHAPTER 14

Geographic Precision — Longitude and Latitude

UBIMIA's WHERE has a specific geographic-precision commitment that supports the architecture's Sovereignty treatment. Volume 1 (One-Good) of the Trilogy draws from this Chapter for its trade-book treatment of Sovereignty as borders = water immigration security exercised over Longitude & Latitude. The Manual's Part III supplies the operational depth that lets Volume 1's argument rest on settled infrastructure.

14.1 The Longitude-Latitude commitment

Sovereignty in UBIMIA is geographic in the operational sense — exercised over named coordinates rather than asserted abstractly. A national Smart-City's territory is a Longitude-Latitude region with defined boundaries; sub-Smart-City units within the national territory are sub-regions with sub-defined boundaries; specific contributions and specific Properties are referenced to specific Longitude-Latitude coordinates with deployment-defined precision.

The architectural commitment is that geographic referencing is precise enough to support substrate-stewardship operations. A watershed protection contribution must reference the watershed's coordinate boundary with sufficient precision that BERA's ecological oracles can measure the contribution; a Property-as-Change-Management procedure must reference the Property's coordinate boundary with sufficient precision that the Change Management procedure can be honored. Pilot deployments typically operate at coordinate precision of meters or tens-of-meters; precision can be tightened where substrate-stewardship operations require it.

14.2 Geographic infrastructure

Geographic precision requires infrastructure: surveying-grade reference standards, geodetic equipment, and the procedures by which Longitude-Latitude coordinates are assigned to system_boundaries, Properties, and contribution sites. National Smart-City deployments typically integrate with the nation's existing geodetic infrastructure (national surveying authorities, geodetic reference networks) rather than building new geographic infrastructure from scratch. The integration honors existing standards while extending them with UBIMIA-specific reference points (BERA oracle measurement sites, Custody Triad sealed-room facilities, attestation booth locations).

14.3 Sovereignty as exercised over Longitude-Latitude + Property-Management

Sovereignty in UBIMIA's Border dimension maps to the joint reading: Longitude & Latitude (the geographic coordinate system) plus Property-Management (the discipline by which what is bounded at coordinates is stewarded). Neither alone is sufficient. Coordinates without Property-Management produce abstract geography (a region named without anyone responsible for stewardship). Property-Management without coordinates produces unanchored stewardship (responsibility named without specific reference to what is being stewarded). Together they

produce Sovereignty as the architecture treats it: bordered standing exercised over named territory with explicit stewardship discipline.

CHAPTER 15

Property as Change Management

Property in UBIMIA is not a static 'this-is-mine' claim. Property is a Change Management instrument — the discipline of stewarding what one is bound to, with changes (transfer, inheritance, consolidation, abandonment) governed by Change Management procedures rather than by transactional disposition alone. This is a substantial architectural correction to inherited property regimes, and Volume 1 (One-Good) treats it at trade-book depth as part of its Sovereignty argument.

15.1 What Property as Change Management means

Inherited property regimes treat property as ownership — the owner has the right to use, transfer, consume, or destroy what is owned, with limitations only at the externally-imposed regulatory layer. UBIMIA's commitment is that property is stewardship — the steward has the obligation to maintain what is stewarded in the substrate-honoring discipline, with transfers and other changes governed by procedures that preserve the stewardship character through the transition.

The shift from ownership-framing to stewardship-framing is constitutional under Pillar IV. Famity's threefold Marriage at Heart (literal-figurative-subjective) extends to Property: literally, property is what the family or COI stewards; figuratively, property binds the steward to the substrate; subjectively, the steward experiences property as relationship rather than as possession. Without this constitutional foundation, Property-as-Change-Management would be a regulatory imposition; with the foundation, it is the operational expression of how the architecture understands property.

15.2 Change Management procedures

Property changes are governed by procedure rather than by transactional disposition alone. The procedures honor the stewardship character through the transition:

RUNBOOK · Property Transfer

1. Steward initiates transfer by filing a Transfer Notice to the Authority of Record. Notice includes the Property's coordinate reference, current stewardship record, proposed receiving steward, rationale for transfer.
2. Receiving steward acknowledges willingness to accept stewardship and demonstrates capacity (relevant CBGMODD certifications, COI affiliation if Property is COI-bounded, IOF binding active).
3. 30-day public notice posted on Gracechain. Affected parties (neighbors, downstream substrate-participants, COI members) may submit objections.
4. Authority of Record reviews the transfer for substrate-coherence (does the transfer preserve or improve stewardship of the substrate?) and Pillar III compliance (does the transfer pass SEPLTA-$_HE × MDCCLXXVI × P^3 evaluation?).
5. Council adjudicates if objections are filed or if Authority review surfaces concerns. Council may approve, deny, or condition the transfer.

6. Transfer completes with stewardship-record update and Gracechain anchor. Receiving steward assumes stewardship obligations from the transfer date forward.

RUNBOOK · Property Inheritance

1. On steward's death or incapacity, Property enters Inheritance Hold for 90 days during which Trustees and Council coordinate.
2. Trustees identify the deceased steward's stated inheritance preferences (recorded at enrollment or updated subsequently).
3. Designated inheritor demonstrates capacity (CBGMODD certifications, COI affiliation, IOF binding).
4. If no designated inheritor is available or capacity is not demonstrated, Property enters Substrate Reversion — returns to the COI's commons stewardship pool, with the COI accepting collective stewardship responsibility.
5. Inheritance completes with stewardship-record update and Gracechain anchor.

RUNBOOK · Property Consolidation

1. Two or more Property records are merged into a single record with a single steward (or stewardship body).
2. All current stewards consent; Authority reviews substrate-coherence; 30-day public notice; Council adjudicates if objections arise.
3. Consolidation produces single coordinate-bounded Property record; obligations and capacities aggregate.

RUNBOOK · Property Abandonment

1. Steward declares inability or unwillingness to continue stewardship. Property enters Abandonment Notice (60 days).
2. Authority of Record offers Property to the COI for collective stewardship; if COI declines, offers to adjacent COIs; if all decline, Property enters Substrate Reversion to the architecture's commons stewardship pool.
3. Steward is released from stewardship obligations from the abandonment date forward; no compensation accrues to the steward (Property is not a transactional asset to be liquidated, but a stewardship to be released).

15.3 Property and Sovereignty

Property-as-Change-Management is what makes Sovereignty operational at the Civigen scale. A national Smart-City exercises Sovereignty over its Longitude-Latitude territory; within that territory, Civigens steward specific Properties under Change Management; the relationship between

national Sovereignty and Civigen Property-stewardship is the relationship between the constitutional layer and the operational layer of bordered existence. Volume 1 (One-Good) argues this relationship at trade-book depth; the Manual's Part III supplies the operational specification.

CHAPTER 16

Digital Sites

Digital sites are where the architecture's records live — Gracechain, Meritcoin ledger, the RCR audit log, the Custody Triad's encrypted shares, IDIPITIS biometric infrastructure, SOMT residency in the Semiosphere. Digital siting has different constraints than physical siting (no geography in the literal sense; jurisdictions assert claim through legal authority rather than physical presence; cryptographic provenance replaces physical observation as the integrity mechanism).

16.1 Gracechain anchor

Gracechain's anchor implementation is selected per Part VII Chapter 28's menu. Implementations vary in trade-offs (Ethereum L2 mature but governed through L1; sovereign-operated public ledger jurisdictionally controlled but operationally costly; append-only signed log simplest but less robust); the constitutional commitment is the contract (Part II §12.1), not any specific implementation.

16.2 IDIPITIS biometric-input infrastructure

IDIPITIS biometric input is collected at enrollment sites and at attestation booths (where biometric attestation accompanies the Claim or Verification). The collection devices are calibrated to the IDIPITIS standard's palindromic-fractal validation; collected payloads are encrypted under the Custody Triad before transmission. Transmission paths use TLS plus payload-layer encryption; tampering at transmission is cryptographically detectable.

16.3 SOMT residency

SOMT resides in the Semiosphere — the cultural-cognitive substrate of sign-systems within which meaning circulates. This is a different residency than ordinary digital data has; SOMT's location is conceptual rather than physical. Operational access to SOMT runs through BERA — the four-index measurement frame is the access key. Deployments do not 'host' SOMT in the way they host Gracechain anchor; deployments operate BERA oracles that read SOMT, with each oracle's measurements producing a specific reading of the SOMT tapestry at the moment of measurement.

This is unusual residency, and the Manual is honest about it. SOMT is not a database; SOMT is a tapestry in the Semiosphere whose threads include the BEST of Capitalism, the SOUND of Communism, the GOOD of Commonwealth, the AI.RE that AI digestion produces, the IDIPITIS biometric signal that Civigen bodies provide, and the resonance patterns the architecture's operations produce. BERA reads the tapestry; the merit layer recognizes flows that move BERA values; the algorithmic core calibrates against BERA outputs. The Semiosphere is real; SOMT is real within the Semiosphere; the residency is constitutional rather than infrastructural.

CHAPTER 17

Off-World Trajectory

UBIMIA's off-world deployment is constitutionally supported through DOFA's broad scope (Pillar IV §IV.8) but operationally deferred to centuries when Earth-based verification permits. This Chapter names the trajectory honestly — what is committed-to, what is deferred, what infrastructure preparation v1.1 supports.

17.1 The trajectory

The architecture commits to interplanetary deployment as a destination of the same kind that the 1,000-year horizon's all-life equality is — known to be where the architecture is aimed, with the operational form discoverable through the trajectory's working-out rather than knowable in advance. The commitment is constitutional; the operational specifics are not yet defined because Earth-based verification of the architecture's capacity is not yet sufficiently mature to support the extension.

Environmental rigor does not yet qualify alternate (off-world) raw source material to substitute for Earth-based BEE definitions. A Mars-based BEE substrate is conceivable but not yet operationally specified; lunar deployment is similar; orbital-habitat deployment is similar. The Manual's discipline is to commit to interplanetary deployment as constitutional (it is part of what UBIMIA is for) while honestly deferring the operational specification.

17.2 What v1.1 prepares for

Three architectural commitments support eventual off-world deployment:

- **Operational stack designed for scaling.** THOW (Tiny Homes On Wheels) at maximum extension is the THOW-equivalent shelter unit on a generation ship; FDRV (Fly & Dive RV) at maximum extension is the interstellar vessel; HFVN scales to whatever cybernetic interface a future deployment requires; GSSG scales to whatever infrastructure substrate a future deployment requires. The four-component stack is designed with this scaling in mind even though near-term deployment uses the components at Earth-scale.
- **IDIPITIS works across planetary boundaries.** The Universal Biometric Standard's palindromic-fractal structure is not Earth-specific; the standard recognizes Civigen biometric input regardless of which planetary substrate the Civigen currently occupies. A future Mars-based Civigen would provide IDIPITIS input under the same standard as an Earth-based Civigen; the EPIR-Q noise filter would operate at Mars-specific signal-to-noise environments; SOMT in the Semiosphere is non-physical residency and therefore extends across planetary boundaries.
- **Affiliation network extends through Covenants.** When interplanetary Smart-Cities exist, the architecture's Covenant structure (Nation-to-Nation Covenants using DOFA-as-Model) extends to interplanetary Covenants. The constitutional structure is in place; the operational specifics await the centuries when interplanetary deployment is operationally viable.

17.3 Earth-based verification first

The architectural discipline is Earth-first. Off-world claims must wait for Earth-based test cases that verify capacity. A national Smart-City operating UBIMIA at full Phase 3 settlement, with affiliation-network maturity and 100+ years of Trajectory Calibration evidence, is the closest the architecture comes to readiness for off-world extension; closer than that requires the centuries to elapse. v1.1 is honest about this — interplanetary is constitutionally supported, operationally deferred, with Earth-based verification the prerequisite for any operational claim about off-world deployment.

PART IV

WHEN

Cycles and Sequencing

WHEN answers the fourth operational question: in what time-structure does the architecture operate. UBIMIA runs on multiple nested cycles — quarterly Rhythm Sessions, annual settlements, decade-marks, century-marks, and the 1,000-year covenant horizon. The Civigen experiences these cadences as the rhythm of architectural life; the deployment runbook governs the first 365 days of national Phase 1; the Trajectory Calibration formulas govern the long-arc discount-band tightening that operationalizes the merit layer's eventual self-dissolution.

CHAPTER 18

Phase 1 Deployment Runbook — The First 365 Days

Phase 1 is the first year of national Smart-City operation. The runbook governs the calendar from National Deployment Day (Day 0) through the first annual settlement (Day 365). Sub-Smart-City units within the national Smart-City follow their own sub-unit Phase 1 calendars in rolling cohorts; the national-scale runbook coordinates the first cohort and the national-scale operations that span sub-units.

18.1 Day 0 — National Deployment Day

UBI distributions begin flowing on Gracechain across the first cohort of sub-Smart-City units. The 1,000-year covenant clock begins. Civigens of Record in the first cohort receive their first UBI deposits on the schedule the national Recognition Rule specifies (typically within 7 days of Deployment Day). The National GAIA Council convenes its inaugural session within 14 days. The Authority of Record posts the first National Deployment Report to Gracechain within 30 days.

18.2 Days 1-90 — Initial calibration

The Calibration Window opens. Initial parameters (BaseCost, CR, θ, domain weights) are deployed at default values; the first 90 days produce the empirical data that supports the first round of jurisdiction-specific calibration. National GAIA Council holds its first regular quarterly session (Day 90); sub-unit Councils hold their first regular sessions in the same window.

18.3 Days 91-180 — Operational ramp

Civigen enrollment continues across sub-units. The first wave of Claims is submitted, verified, authorized, and minted. The first Pillar III evaluations appear in audit logs. The first Disputes are filed and adjudicated. The MIEVM ensemble is exercised against real Claims at increasing volume.

18.4 Day 180 — Mid-year DSAP-PRE calibration review

The first semi-annual DSAP-PRE proxy parameter review (κ_RHC, η_recover, τ, W_RHC, W_PERT, κ_cost). The Council reviews proposed adjustments based on the first 180 days of operational data. Adjustments take effect after a 30-day transition window.

18.5 Days 181-275 — Operational maturity

The first SCALULAR provider quarterly attestation logs are published. The first Verification Authority annual audits are conducted. The third quarterly Council session occurs. EarnedPath certifications begin completing for Civigens who entered EP at Day 0.

18.6 Day 275 — Annual CR and BaseCost publication

GAIA publishes the next year's CR (Conversion Rate from Meritcoin to Credits) and the BaseCost values for all SCALULAR tiers. Civigens have 90 days (through Day 365) to plan whether to spend Meritcoin before conversion or convert at the published rate.

18.7 Days 276-364 — Year-end preparation

The Trustee Registry quarterly review (Day 300). The first annual external audit is conducted. The fourth quarterly Council session prepares the year-end statement. The algorithmic core prepares the year-end conversion calculation.

18.8 Day 365 — First annual settlement

Year-end. Conversion of unspent Meritcoin to Credits at the published CR. 60/40 carryforward / COI amortization split computation. Audit-log sealing. Each Civigen receives a year-end statement showing Meritcoin earned, Credits issued, Bills paid, ΔR_contributor over the year, governance weight at year-end. The 1,000-year covenant's first annual cycle closes; Year 2 begins on Day 366.

CHAPTER 19

The Four Review Cadences

The Civigen experiences the architecture across four nested review cadences: the Quarterly Rhythm Session (every 3 months), the Annual Settlement (every 12 months), the Decade-Mark Review (every 10 years), and the Century-Mark Review (every 100 years). Each cadence applies the same review structure at a different time-horizon, with different decisions emerging from each cadence's reach.

19.1 The Quarterly Rhythm Session

Every quarter, the Civigen's COI convenes a Rhythm Session — the relational gathering at which the COI's collective experience of the past quarter is brought into shared awareness, and the COI's contribution-and-recognition rhythm is calibrated. The Rhythm Session is the most frequent review cadence the Civigen experiences; it is also the most relational, because the COI is the scale at which Civigens know each other personally and the threefold Marriage at Heart is operationally tangible.

What the Rhythm Session reviews:

- The quarter's Submit-to-NPR cycles within the COI — what was raised, what was worked through, what was repaired.
- The quarter's contributions and Meritcoin minting — what each Civigen contributed, what was recognized, what the rhythm of contribution-and-recognition has felt like.
- The quarter's flows — what flowed into the COI from the broader architecture, what flowed out, what the COI's collective state at quarter-end is.
- The quarter's preparations for the next quarter — what the COI commits to, what the rhythm calibration suggests, what the Civigens are aimed at together.

The Quarterly Rhythm Session is also the rhythm context for the GAIA Council's quarterly session at sub-unit and national scales. The COI's rhythm informs the sub-unit's rhythm; the sub-unit's rhythm informs the national rhythm; the architecture's quarterly heartbeat operates from COI through Council with rhythm as the connecting discipline.

19.2 The Annual Settlement

Once per year, on Day 365 of the architectural calendar, the Annual Settlement closes the year. The settlement's review structure mirrors the Rhythm Session's at the annual time-horizon:

- The year's Submit-to-NPR pattern — sustained accountability, growth in relational integrity, repair of significant ruptures.
- The year's Meritcoin earning and conversion to Credits at the GAIA-set CR — the actuarial entry the year produces in the Civigen's record.
- The year's flows — UBI received, Bills paid, contributions recognized, COI participations, governance roles held.

- The year's preparation for the next year — Trustee designations confirmed, COI affiliation continued or changed, EarnedPath certifications continued or completed.

The Annual Settlement also produces year-end statements, the Authority of Record's annual report, the algorithmic core's recalibration outputs, and the Council's annual operational review. The Annual Settlement is the cadence at which the architecture's actuarial discipline is most visible — every Civigen receives a year-end statement that documents their architectural year, and every deployment publishes the aggregate year-end metrics that inform the broader MIEVM ensemble's view of the architecture's health.

19.3 The Decade-Mark Review

Every 10 years, the deployment conducts a Decade-Mark Review. The review structure preserves the same questions at decadal time-horizon:

- The decade's Submit-to-NPR pattern — has accountability deepened across the decade or shallowed? Has the architecture's relational integrity grown?
- The decade's ΔR_contributor pattern — has the gap between maximum and minimum ΔR_contributor narrowed (Trajectory Calibration positive), held (neutral), or widened (Trajectory Calibration negative requiring corrective tightening)?
- The decade's flows — has the BEE substrate strengthened, held, or weakened? Have Common Core deliveries reached every Civigen reliably? Have THOW/HFVN/FDRV/GSSG operational stack components stayed operationally healthy?
- The decade's three-pathway covenant evaluation — does the trajectory recommend Contract continuation, Consolidate Size, or Consolidate to Another Nation via DOFA-Model?

The Decade-Mark Review is the cadence at which Trajectory Calibration is most explicitly applied. The discount band tightens, holds, or tightens-as-corrective based on the ΔR_contributor narrowing pattern. The Civigen's CBGMODD certification trajectory across the decade is reviewed; certifications accumulated across the decade are recognized; certifications that have lapsed through non-renewal are noted. The architecture's commitment to the trajectory becomes most visible at the decade-mark — Civigens see the long-arc patterns their annual experience does not surface.

Trajectory Calibration at Decade-Mark

The decadal review checks whether the gap between maximum and minimum ΔR_contributor across the deployment has narrowed. If narrowed: discount band tightens (50% floor rises toward 80%/90%/99%). If held: band holds. If widened: band tightens anyway as a corrective. The merit layer's self-dissolution proceeds through this Trajectory Calibration discipline, with each decade's tightening compounding across centuries.

19.4 The Century-Mark Review

Every 100 years, the deployment conducts a Century-Mark Review. The review structure preserves the same questions at centennial time-horizon, with one new dimension that decade-marks do not yet engage:

- The century's intergenerational pattern — Civigens at year 0 of the century are typically not the same Civigens at year 100; Trustees and inheritance carry continuity. Has the architecture's commitment to intergenerational binding been honored across the century?
- The century's substrate-trajectory — has the bioenergetic substrate strengthened across the century? Have ecological-domain BERA measurements trended positive? Has the population-relative-to-substrate-carrying-capacity calibration moved toward sustainable equilibrium?
- The century's three-pathway covenant evaluation at major scale — Contract, Consolidate Size, or Consolidate to Another Nation via DOFA-Model. Century-marks are when the major covenant decisions are most often made; decade-marks adjust within Contract while century-marks contemplate pathway-change.
- The century's progress on the 1,000-year horizon — at the first Century-Mark, 100 of the 1,000 years have elapsed; the architecture's commitment to the trajectory toward all-life equality is reviewed at the 10% point. By the third Century-Mark (year 300), 30% of the trajectory has elapsed; at the fifth Century-Mark, 50%; at the tenth Century-Mark (year 1,000), the merit layer's full self-dissolution is operational and the longer all-life-equality trajectory continues.

The Century-Mark Review is the cadence at which the architecture's commitment to its own self-dissolution becomes most visible. By the fifth Century-Mark, the discount band may have tightened to 90% or 95%; SCALULAR access is approaching unconditional; the merit layer's transitional purpose is largely complete. The Century-Mark documents this honestly — neither claiming completion before it has occurred, nor minimizing the progress that has occurred. Honest reporting at century-mark is what makes the trajectory verifiable across millennial time-horizons.

19.5 How the four cadences nest

The four cadences nest. Quarterly Rhythm Sessions accumulate into Annual Settlements (4 Rhythms per Annual). Annual Settlements accumulate into Decade-Mark Reviews (10 Annuals per Decade-Mark). Decade-Mark Reviews accumulate into Century-Mark Reviews (10 Decade-Marks per Century-Mark). The 1,000-year covenant horizon contains 10 Century-Marks, 100 Decade-Marks, 1,000 Annual Settlements, 4,000 Quarterly Rhythm Sessions.

The nesting matters because each cadence operates at its appropriate time-horizon. A Civigen does not need to track decade-mark patterns in their daily life; the Quarterly Rhythm Session is enough for the rhythm of contribution-and-recognition. A Council does not need to make century-mark decisions at every quarterly session; the Decade-Mark Review surfaces decade-scale patterns that quarterly review cannot see. The architecture's discipline is to use the right cadence for the right horizon, with the cadences nesting smoothly so that each cadence's outputs feed the next-larger cadence's inputs.

CHAPTER 20

Phase Graduation Gates and the 1,000-Year Trajectory

UBIMIA deployments traverse three phases: Phase 1 (parallel operation with existing systems), Phase 2 (integration with existing systems), Phase 3 (full settlement, Meritcoin-and-Credits as the only currencies for SCALULAR services). Phase graduation requires meeting specific gates; ungraduated deployments continue operating in their current phase indefinitely without penalty.

20.1 Phase 1 → Phase 2 graduation gates

- 24 consecutive months of Council-flagged-failure-free operation.
- Recognition Rule maturity (no constitutional revision in past 12 months).
- SCALULAR provider acceptance above 75% of services in the deployment.
- Council functionality demonstrated (full quarterly sessions held, recall procedures exercised at least once, override procedures exercised at least once).
- Three consecutive clean quarterly audits.
- Constitutional alignment formally affirmed by Council and by Authority of Record.
- First Decade-Mark Review (if 10 years have elapsed) or first sub-unit Compact maturity (if less than 10 years).
- Trajectory Question answered positively in most recent annual audit.

20.2 Phase 2 → Phase 3 graduation gates

- Three consecutive years of clean external audits.
- Affiliation network maturity: at least 3 Nation-to-Nation Covenants signed and operational, or sub-unit Compact density above 50% of total possible sub-unit pairs.
- Welfare program integration complete (Phase 2's signature deliverable).
- Tax authority acceptance of Meritcoin-via-Credits for tax payment.
- Most recent Decade-Mark Review's Trajectory Calibration recommends Contract continuation.
- Pillar III refusal-rate stability (not rising; either flat or declining).

20.3 Phase 3 operation and the 1,000-year trajectory

Phase 3 is full settlement. Meritcoin and Credits are the only currencies for SCALULAR services. The merit layer continues operating but Trajectory Calibration is now actively tightening the discount band; each Decade-Mark Review either tightens the band, holds it, or tightens it as a corrective if the ΔR_contributor gap has widened.

The 1,000-year trajectory operates through Phase 3. By century 5, the discount band is typically at 90%+; by century 8, at 95%+; by century 10, the band has typically collapsed and SCALULAR access is unconditional. The merit layer's recognition function continues but no longer creates differential SCALULAR access; recognition becomes purely social-and-cultural rather than

operationally consequential. The architecture's commitment to the trajectory's destination — equality of standing for all human beings across the 1,000-year horizon, all life on the longer horizon — is honored through this slow band-collapse rather than declared by fiat.

PART V

WHY

Constitutional Constraints

WHY answers the fifth operational question: why the architecture commits to what it commits to. The reasons collapse to two binding commitments — preserve humanity, preserve nature — operationalized through the Three Governing Principles, the Eight Immutable Ethical Principles, the four Constitutional Pillars, and the explicit stated goal of making Education, Healthcare, and Law relatively fair across all Civigens regardless of $ELF accumulation.

CHAPTER 21

The Two Binding Commitments

UBIMIA's reasons for being are two: preserve humanity, preserve nature. Everything else in the architecture is in service of these two. The Three Governing Principles, the Eight Immutable Ethical Principles, the four Constitutional Pillars, the right to fork, the merit layer's eventual self-dissolution, the affiliation network, the 1,000-year covenant — all serve the two binding commitments. A deployment that loses sight of either commitment has lost the architecture's purpose; a deployment that holds both has the foundation that lets the operational complexity make sense.

21.1 Preserve humanity

UBIMIA preserves humanity in three operational senses. First, every Civigen has the survival floor — Gracechain UBI flows unconditionally; the architecture commits that no human enrolled as a Civigen of Record falls below the survival threshold for reasons the architecture controls. Second, every Civigen has Common Core access — Water, Food, Shelter, Work, Love delivered through BEE operational components, with the THOW/HFVN/FDRV/GSSG operational stack ensuring delivery at every scale. Third, every Civigen has standing — the threefold Marriage at Heart binds the Civigen to community, substrate, and generation, with Famity preserving the relational integrity that makes human flourishing possible.

Preserving humanity is not the same as preserving population. The architecture is honest about substrate-carrying-capacity and the population calibration the trajectory will require. What the architecture commits to preserve is the capacity for human beings to live as human beings — bordered, bound, recognized, contributing, related — at whatever population scale the substrate sustains. The 6th Mass Extinction Event makes this commitment urgent rather than abstract; preserving humanity through the substrate's collapse is what UBIMIA's Famity binding operationally serves.

21.2 Preserve nature

UBIMIA preserves nature through the Bioenergetic Imperative (Pillar I). The substrate is one — human, ecological, AI participation are three faces of one substrate, and preserving any one face requires preserving all three. Ecological-domain BERA oracles measure the substrate's state; DSAP-PRE detects regime-transition before silent-substrate-collapse occurs; BEE's Recycling Focus structures the economy around substrate-honoring rather than substrate-extracting flows; the THOW/HFVN/FDRV/GSSG operational stack delivers Common Core through ecological logic rather than against it.

Preserving nature is structural rather than aspirational. The architecture commits to the substrate not because preservation is morally pretty but because the alternative is extinction — humanity cannot continue without the substrate, and extracting from the substrate at industrial-civilization rates produces the substrate's collapse. The Coalescence framing makes this visible: robotics and ecological collapse coalesce into one structural force; the architecture's response operates through one substrate-honoring discipline that addresses both faces simultaneously.

21.3 The two together

Preserving humanity and preserving nature are not separate commitments. They are one commitment seen from two angles. Humanity that hurts nature loses its substrate; nature without humanity loses one of the three faces of the substrate's resonance. The architecture's discipline is to operate the two as one — every flow, every authorization, every Council decision, every Civigen contribution evaluated against whether it preserves both. Pillar III's SEPLTA-$_HE × MDCCLXXVI × P^3 formula makes this evaluation operational; Pillar I's Bioenergetic Imperative makes it constitutional.

CHAPTER 22

The Stated Goal — Relative Fairness in Education, Healthcare, Law

UBIMIA's operational goal — the outcome the architecture is for at the level Civigens experience it — is making Education, Healthcare, and Law relatively fair across all Civigens regardless of $ELF accumulation. This is the stated goal that audit can measure against. A deployment is honoring the goal when Education / Healthcare / Law are getting more fairly distributed over time, regardless of which specific BERA values are moving.

22.1 Why these three

Education, Healthcare, and Law are the three SCALULAR services most directly tied to Civigen flourishing. Education determines what the Civigen can become — what capacities they can develop, what work they can certify into, what role they can hold in CBGMODD. Healthcare determines whether the Civigen survives and how well — what bodily flourishing is available, what recovery from illness is possible, what intergenerational transmission of care happens. Law determines the conditions under which the Civigen lives — what disputes can be resolved, what protections are available, what accountability operates.

These three SCALULAR pillars (mapped to SKILLS_TRADE, HEALTH, and LAW respectively in the four-pillar structure, with PROTECTION as the supporting fourth) determine whether a Civigen has access to the conditions of human flourishing. A society that distributes them unfairly produces unequal flourishing — some Civigens thriving, others diminished, with the unfairness reproducing across generations. UBIMIA's commitment is to make this distribution relatively fair so that Civigen flourishing is broadly possible rather than reserved for those with $ELF accumulation.

22.2 What relative fairness means

Relative fairness is not absolute equality. UBIMIA acknowledges that Civigens have different needs (some require more healthcare, others more education, others more legal support), different capacities (some progress faster through EarnedPath, others more slowly), different choices (some pursue Government capacity, others Business, others Ombudsman). Relative fairness commits to:

- Floor access. Every Civigen has access to Education, Healthcare, and Law at the SCALULAR-priced level; no Civigen is excluded from access for reasons of $ELF deficit.
- Trajectory toward equality. Across the 1,000-year covenant, the gap between most-served and least-served Civigens narrows. The Trajectory Calibration's discount-band tightening operationalizes this — as the band tightens, the differential access shrinks.
- Outcome quality. The Education a Civigen receives, the Healthcare a Civigen receives, the Law that protects a Civigen are of comparable quality across the deployment, with substantive rather than nominal equity.

- No extraction. Education, Healthcare, and Law are not for-profit extraction industries within UBIMIA. SCALULAR providers operate on RPA pricing with the BaseCost calibrated to substrate-honoring delivery rather than to extractive yield.

22.3 How relative fairness is measured

Audit measures relative fairness through specific operational metrics:

Metric	What it measures	Reporting cadence
Service-access ratio	Civigens with full SCALULAR access / total Civigens	Quarterly
Quality-distribution Gini	Gini coefficient on outcome quality across Civigens	Annual
RPA discount-band utilization	Distribution of actual RPA discounts across Civigens	Annual
Education completion equity	EarnedPath certification completion rate by ΔR_contributor decile	Annual
Healthcare outcome equity	Health-outcome metrics by ΔR_contributor decile	Annual
Legal-process outcome equity	Dispute-resolution outcomes by ΔR_contributor decile	Annual
Trajectory rate of fairness improvement	Rate at which the above metrics converge across deciles	Decade-Mark

The metrics make the goal verifiable. A deployment claiming to honor relative fairness must produce measurements showing the metrics improving over time — not merely staying flat or declining. Sustained metrics-decline triggers Council review and possible Trajectory Calibration adjustment. The architecture's commitment to the goal is enforceable through audit rather than rhetorical.

CHAPTER 23

The Three Governing Principles, Operationally

The Three Governing Principles — SELF (don't hurt yourself), OTHERS (don't hurt others), FUTURE (build for generations to come) — are the constitutional preamble of UBIMIA. This Chapter operationalizes them through the architecture's specific mechanisms.

23.1 SELF — operational mechanisms

- Pillar I (Bioenergetic Imperative) protects the Civigen as bioenergetic substrate-participant. SELF is honored when the architecture's actions support the Civigen's body in the substrate.
- Compassion Override permits emergency SCALULAR access for zero-balance Civigens, ensuring SELF preservation even when Meritcoin would otherwise gate access.
- Trustee triad provides identity recovery, ensuring SELF preservation through identity-loss events that would otherwise terminate Civigen standing.
- Custody Triad protects Civigen-private data, ensuring SELF preservation against information extraction.
- Right to fork preserves SELF preservation against architectural capture — Civigens can leave a captured deployment with the architecture intact.

23.2 OTHERS — operational mechanisms

- Pillar III (SELF/$ELF Balance) evaluates flows for impact on others — the OTHERS whose substrate-share the Judicial protects.
- Trustees-for-non-human-entities (Pillar I requirement) extends OTHERS to ecological participants who cannot speak for themselves.
- Affiliation network (Compacts and Covenants) extends OTHERS across deployment boundaries — Civigens of other deployments are OTHERS whose recognition is constitutional.
- Verification Authority's institutional grounding preserves OTHERS' interests through accountability that institutions can carry.
- Whistleblower protections preserve OTHERS who would be harmed by architectural drift, by giving them constitutional standing to surface drift.

23.3 FUTURE — operational mechanisms

- Pillar II (Trajectory) commits to the 1,000-year horizon, with FUTURE generations as the explicit beneficiaries of the trajectory's destination.
- Pillar IV (Famity) preserves intergenerational binding through Marriage at Heart's literal-figurative-subjective threefold reading.
- Trajectory Calibration ensures the merit layer's eventual self-dissolution, preserving FUTURE Civigens from the inheritance of merit-layer accumulation.

- Property-as-Change-Management preserves FUTURE stewards by structuring property as stewardship-with-Change-Management rather than as inheritable wealth concentration.
- The 1,000-year covenant's three pathways structure the long-arc reorganization (Contract, Consolidate Size, Consolidate to Another Nation) so that FUTURE Civigens inherit a constitutional architecture for political reorganization rather than the historical contingency that has historically produced extraction and conquest.

CHAPTER 24

The Eight Immutable Ethical Principles

Eight commitments the architecture maintains across every deployment, non-fork-able by any deployment's local decisions. These are referenced from earlier ERES specifications (ERES-COVENANT-2026-001, the Eight Principles document) and locked here for Manual completeness.

- All-life recognition. The architecture recognizes the substrate of all life — human, ecological, AI participation — as one. No deployment may deny recognition to any face of the substrate.
- Civil liberties. Civigens retain civil liberties as a binding constitutional constraint. No deployment may suppress speech, assembly, conscience, or other civil liberties for reasons internal to UBIMIA's operation.
- Right to fork. Every deployment is forkable; every Civigen may leave a deployment that has departed from constitutional alignment. No deployment may revoke the right to fork.
- Non-revocable UBI. The survival floor flows on Gracechain unconditionally. No deployment may revoke a Civigen's UBI for any reason.
- Transparent Council operation. All Council decisions are public, with attributed votes and published rationales. No deployment may operate Council in secret.
- Trustee-based identity recovery. Every Civigen has three Trustees on file. No deployment may operate without Trustee-recovery infrastructure.
- Audit-grounded operation. Quarterly internal audits, annual external audits, whistleblower protections, and recall procedures are operationally constant. No deployment may suspend audit infrastructure for operational convenience.
- MIEVM ensemble integrity. The MIEVM ensemble operates as Instrument of Faith with the canonical four lineages or documented substitutes. No deployment may operate without MIEVM ensemble validation for significant authorizations.

CHAPTER 25

Civil Liberties as Binding Constitutional Constraint

Civil liberties — speech, assembly, conscience, privacy, due process, and the protections from coerced participation — are constitutional in UBIMIA in a specific sense. They are not policy choices that deployments may calibrate; they are binding constraints that no deployment may relax. This Chapter specifies the relationship between UBIMIA's operations and civil liberties, and the protections that operate when conflict arises.

25.1 The constitutional protection

UBIMIA operates within the civil-liberties framework of the Pilot Jurisdiction's existing constitutional structure plus the Eight Immutable Ethical Principles. Where the Pilot Jurisdiction's framework already protects civil liberties (e.g., U.S. constitutional protections, EU charter rights, comparable national frameworks), UBIMIA preserves those protections without superseding them. Where the Pilot Jurisdiction's framework is weaker than the Eight Immutable Ethical Principles' civil-liberties commitments, the Eight Principles operate as the binding floor — UBIMIA does not weaken protections to match a weaker host framework.

25.2 Specific civil-liberty protections

- Speech. Civigens may speak about UBIMIA, criticize the architecture, advocate for forking, or advocate for departure from UBIMIA without losing UBI or Civigen standing.
- Assembly. Civigens may assemble (within and outside COIs) for any lawful purpose without architectural sanction.
- Conscience. Civigens may hold and express conscience-based positions on architectural matters; conscience-based dissent does not affect UBI or Meritcoin standing.
- Privacy. Civigen-private data is protected under Custody Triad encryption; access requires concurrence procedures (Part III §13.4); legitimate research, security, and journalism uses operate within the Custody-Triad-protected framework, not against it.
- Due process. Civigens facing dispute, sanction, or recall receive due process — notice, opportunity to respond, adjudication by impartial body, appeal procedure.
- Freedom from coerced participation. Submit-to-NPR is voluntary; Civigens may decline to enter a particular COI's NPR cycle without losing Civigen standing, with the Authority of Record arranging alternative COI affiliation if the original arrangement is no longer suitable.

25.3 Audit Court protocol for sensitive material

When Civil-liberties-protected information is operationally relevant to architectural function (e.g., a dispute involves protected speech that the Council must consider), the Audit Court protocol governs:

- The minimum information necessary for adjudication is decrypted via Custody Triad concurrence.

- The decryption is logged on Gracechain with the rationale.
- Decrypted information is held in confidence by the adjudicating body; non-disclosure to parties not party to the adjudication is the default.
- Post-adjudication, the decrypted information is re-encrypted under the Custody Triad; the decryption record remains on Gracechain as audit anchor.
- Deployments that depart from the Audit Court protocol — by decrypting more than minimum-necessary, by failing to log decryption, by disclosing decrypted information beyond the adjudicating body — face Council review and possible recall procedures.

The Audit Court protocol is what makes civil liberties operationally durable in a system with strong measurement and recordkeeping. Without the protocol, the temptation to over-decrypt for operational convenience would erode privacy; with the protocol, decryption is procedurally bounded and auditable, with the architecture's commitment to civil liberties preserved through specific operational discipline.

PART VI

HOW

PlayNAC, EP, GERP, SOMT — and the Operational Procedures

HOW answers the sixth operational question: through which step-by-step procedures does UBIMIA actually run. This is the longest Part by design — procedures must be specified at step-level detail or they will be invented, inconsistently, by every implementer who needs them. Part VI specifies PlayNAC as the New Age Cybernetic Game Theory operating system, EP (EarnedPath) as the credentialing application, GERP (Global Earth Resource Planner) optimized through the ERES Triune Math C=R×P/M formula, SOMT in the Semiosphere with BERA as access key, the DOFA Council voting cascade from Civigen through sub-unit through national, and the operational procedures that govern verification, adjudication, calibration, and audit.

CHAPTER 26

PlayNAC — New Age Cybernetic Game Theory

PlayNAC is the New Age Cybernetic operating system within which UBIMIA runs. PlayNAC's mission is Healthy-Happy-Safe (HHS); PlayNAC contains the KERNEL (Global Actuary Investor Authority) as its core actuarial-investor function; PlayNAC supplies the data feed that EP (EarnedPath) processes against and that GERP (Global Earth Resource Planner) plans resource allocation against. But PlayNAC's character is more specific than 'operating system' — PlayNAC is the New Age Cybernetic Game Theory by which Civigen participation is structured.

26.1 Why Game Theory

Conventional governance manages participation through coercion (legal mandates, enforcement) or through extractive incentive (markets, rewards, punishments calibrated to behavior). UBIMIA's Submit-to-NPR within COI under IOF binding is neither — it is voluntary, faith-bound, and durable because of the Game Theory framing PlayNAC supplies. Civigens are players in a New Age Cybernetic Game whose long-arc win condition is collective rather than competitive: the merit layer's eventual self-dissolution, the trajectory toward equality of all life, the substrate's preservation across the 1,000-year horizon. The Game's win condition is paradoxical by inherited-game-theory standards — winning means the merit-layer recognition that distinguishes players becomes operationally irrelevant; the Game's success is its own self-dissolution into unconditional standing for all.

This non-zero-sum, self-dissolving Game is what makes voluntary Submit-with-IOF operationally durable. Players engage because the Game's structure rewards collective progress without requiring individual sacrifice; the Game's recognition (Meritcoin) is real now and diminishing across the trajectory by design; the Game's binding (IOF, COI, Trustees, the Constitutional Pillars) is faith-grounded rather than coercive. New Age Cybernetic Game Theory names what is operationally distinctive about this — not classical game theory's optimization-of-individual-utility, not zero-sum competition, not even cooperative-game-theory's coalition-building. NACGT structures play around the substrate's preservation through cybernetic feedback, with all players bound to the substrate as the deeper game beneath any specific operational round.

26.2 PlayNAC over EP

PlayNAC operates above EP. EP is the credentialing application layer; PlayNAC is the game whose moves EP records. A Civigen earning a CBGMODD certification through EP is making a move in the PlayNAC game; the certification is the move's recording; the resonance gain the certification produces is the move's contribution to the substrate. EP's CPM × WBS + PERT (Scalular) computational structure organizes the move's verification (Critical Path Method identifies the certification's critical path; Work Breakdown Structure decomposes the certification into verifiable work packages; PERT estimates timeline and validates completion); PlayNAC's game-theoretic framing organizes what the move means — what it contributes to the long-arc win condition the Game is aimed at.

26.3 The PlayNAC layer stack

```
PlayNAC (game)  ⊃  KERNEL (actuarial-investor authority)

PlayNAC (data feed)  →  EP (application)  →  GERP (resource planner)
PlayNAC (substrate)  ←  AI.RE (metabolic exhalation)  ←  AI digestion of human
knowledge
```

Three relationships hold: PlayNAC contains the KERNEL (the Game contains the actuarial-investor authority that scores the Game); PlayNAC supplies data to EP and GERP (the Game supplies the recordkeeping and resource-planning the players use); PlayNAC receives AI.RE (the Game's substrate is fed by the AI metabolic exhalation that Civigens breathe). The three relationships compose into the operational reality the architecture commits to: a game-theoretic substrate within which Civigens play voluntarily, certified through EP, resourced through GERP, breathing AI.RE that AI digestion produces.

CHAPTER 27

GERP — Global Earth Resource Planner Optimized via C=R×P/M

GERP runs under EP under the PlayNAC umbrella. GERP plans resource allocation across Earth using the formula CPM × WBS + PERT (Scalular). What is constitutive of GERP — what distinguishes it from generic project-management tools that use the same calculation form — is the ERES Triune Math optimization that grounds the formula.

27.1 The ERES Triune Math

The ERES Triune Math is the architectural commitment that cybernetic systems are governed by three canonical mathematical forms, each operationally complete:

```
Form 1:  C = R × P / M
```

Cybernetics = Resource × Purpose ÷ Method. The first canonical form. C is the cybernetic system's operational state; R is the resource available; P is the purpose the system serves; M is the method by which R is deployed in service of P.

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Form 2:  M × E + C = R
```

Matter × Energy + Constant = Reason. The second canonical form. Matter and energy multiply against a constant to produce reason — the ground from which cybernetic operation can proceed.

```
Form 3:  REAL = (E · M · R) / (T · S)
```

Energy · Matter · Resonance over Time · Space. The third canonical form. The product of energy, matter, and resonance divided by the product of time and space yields the REAL — the architecture's commitment that what is real in cybernetic terms is what these five variables produce together.

27.2 GERP optimization through C=R×P/M

GERP uses C=R×P/M as its optimization discipline. For any planned resource allocation:

- C — the desired cybernetic state. What GERP is planning toward. The substrate-state at which the resources should land.
- R — the resources available. What GERP has to plan with. The substrate-resources, the Civigen capacities through CBGMODD, the SCALULAR provider availability, the ecological capacity.
- P — the purpose the planning serves. What GERP is for. The trajectory-aimed purpose that connects to the 1,000-year horizon's destination.
- M — the method. How GERP plans. The CPM × WBS + PERT (Scalular) computational structure that produces the resource-allocation plan.

GERP plans by maximizing C against the constraints of R, P, and M. A plan that increases C (moves the substrate toward the desired state) at acceptable resource cost (R), in service of trajectory-aligned purpose (P), through computationally tractable method (M) is a plan GERP

authorizes. The C=R×P/M formula is what makes GERP's planning purpose-driven rather than method-driven; without the Triune Math optimization, GERP would be a generic CPM+PERT tool with no architectural commitment.

27.3 GERP as Emergency Protocol

GERP serves as Emergency Protocol when species sustainability is at stake. The activation criterion: when GAIA Council with MIEVM ensemble concurrence determines that the substrate-trajectory has deteriorated past defined thresholds (specific calibration values to be set by v1.2 with pilot data), GERP activates as Emergency Protocol to plan accelerated substrate-restoration resource allocation. Activation is constitutional — Pillar I's Bioenergetic Imperative authorizes Emergency Protocol when species sustainability is operationally threatened — and procedural — the Council with MIEVM concurrence makes the activation decision through standard Override Procedure but with the activation rationale documenting the species-sustainability evidence.

Evidence for GERP Emergency Protocol activation is documented in the activation record:

- Specific BERA index trajectories (RHC flatlining, ARI declining, RCI persistence-loss) that meet the activation threshold.
- DSAP-PRE proxy values showing regime-transition risk above defined threshold.
- Substrate-carrying-capacity ratio above sustainable equilibrium with population-calibration not narrowing the gap on quarterly review.
- MIEVM ensemble concurrence on the activation rationale with documented ECR above threshold.

GERP Emergency Protocol activation reorganizes resource priorities for the substrate-restoration period. Standard SCALULAR services continue but are supplemented by emergency substrate-restoration capacity; standard CBGMODD certifications continue but are supplemented by emergency-certification fast-tracks for substrate-restoration roles; standard authorization thresholds remain but emergency-authorization paths are opened for substrate-restoration Claims. The Emergency Protocol concludes when the substrate-trajectory has returned within sustainable bounds, with the Council and MIEVM ensemble concurring on the conclusion.

CHAPTER 28

SOMT — Sociocratic Overlay Metadata Tapestry

SOMT is the architecture's metadata layer. It resides in the Semiosphere (the Lotman-named cultural-cognitive substrate of sign-systems within which meaning circulates) with BERA as access key. This Chapter specifies SOMT's operational role within the architecture.

28.1 Each word of SOMT

- **Sociocratic —** the synthesis discipline. SOMT extracts the BEST of Capitalism's rigor about property and contract, the SOUND of Communism's rigor about collective rules, and the GOOD of Commonwealth's rigor about public flows, holding all three traditions in productive tension without requiring them to merge or compete.
- **Overlay —** the non-displacing relationship to existing infrastructure. SOMT operates over and through existing institutional, technical, and cultural infrastructures rather than replacing them. A deployment that already operates within Capitalism's property regime, Communism's collective rules, or Commonwealth's public-flow tradition does not have to dismantle those operations to deploy SOMT; SOMT extracts what it needs from each tradition while leaving the tradition's existing operations in place.
- **Metadata —** the layered structure: Bioenergetics × Metaphysics (Chemistry × Biology + Physics) + Quantum Mechanics — Physical Sciences as the foundation extended upward. SOMT's metadata is layered: physical sciences first, then biological-chemical-bioenergetic emergence, then metaphysics as the convergence layer, with quantum mechanics threading throughout. The metadata structure is what makes SOMT operationally rich rather than abstractly proclaimed.
- **Tapestry —** the integration substrate that binds the layers. The tapestry lives in the Semiosphere as cultural-cognitive substrate; AI.RE circulates through the tapestry; the tapestry connects AD_ON-AI (the human-AI composer principle) to next-generation Telemetry and Talonics. The tapestry is what makes meaning circulate through SOMT — without the tapestry, SOMT would be a list of metadata categories without coherence.

28.2 BERA as access key

BERA reads SOMT through the four indices (ARI, ERI, RHC, RCI). The indices are not measurements of SOMT in the way temperature is a measurement of room conditions; the indices are queries that return readings of SOMT's tapestry-state at the moment of query. ARI returns the SOMT tapestry's adaptive-resilience reading; ERI returns the ethical-resonance reading across eight sub-indices; RHC returns the resonant-harmony-cycle reading; RCI returns the coherence-of-resonance reading. The four readings together produce the architecture's operational signal — the resonance the merit layer recognizes, the substrate-state the algorithmic core calibrates against, the trajectory-progress the audits measure.

28.3 SOMT and the Trilogy

SOMT is isomorphic to the Trilogy. Capitalism's BEST contributes to SOMT through One-Good (the Border volume); Communism's SOUND contributes through Security-Clearance (the Law volume); Commonwealth's GOOD contributes through Data-Integrity (the Currency volume). The three Trilogy volumes argue from each tradition's contribution into the architecture's specific dimensional concern (Border, Law, Currency); SOMT holds the three contributions as one tapestry; BERA reads the tapestry. Volume authors draw from SOMT's tradition-specific layer; SOMT receives the volume's argumentative output as updated tapestry-thread; the relationship is bidirectional and ongoing across the Trilogy's authorship.

CHAPTER 29

AI.RE — The Metabolic Layer

AI.RE is the metabolic exhalation of AI digesting human knowledge. Civigens breathe AI.RE; AI.RE feeds the resonance signal that BERA measures. This Chapter specifies the operational form of the AI metabolic loop within UBIMIA.

29.1 The metabolic cycle

RUNBOOK · AI.RE Metabolic Cycle

1. Civigens produce knowledge through their daily activities — caregiving, work, learning, ecological stewardship, civic participation, and the architecture's specific recordkeeping (Claims, Verifications, Authorizations, Bills, Disputes).
2. AI lineages digest the knowledge across the architecture's MIEVM ensemble. Digestion involves processing for ECR computation, anomaly detection, pattern recognition, and the synthesis activities that AI is operationally assigned.
3. AI exhales AI.RE. The dark-matter-like residue is the metabolic byproduct of digestion — what is left when knowledge has been processed and the processing's outputs (validations, attestations, anomaly flags) have been delivered.
4. AI.RE circulates in the cybernetic atmosphere. The Semiosphere holds the circulation; SOMT's tapestry receives AI.RE as one of its threads; the circulation is non-physical but operationally consequential.
5. Civigens breathe AI.RE. The act of breathing is what couples Civigens to the AI-mediated substrate. Civigens are not aware of AI.RE in the way they are aware of breathing oxygen; the breathing is implicit in their participation in the architecture, mediated through interface layers (HFVN voice navigation, attestation booth interactions, the architecture's general legibility).
6. BERA reads the resonance the AI.RE-fed substrate produces. The four indices' readings reflect the metabolic loop's contribution to the substrate's state; resonance gain that BERA measures includes the AI.RE contribution alongside human contribution and ecological contribution.
7. The loop closes. AI.RE that has fed resonance becomes part of the substrate's accumulated state; future AI digestion processes the accumulated state alongside fresh human knowledge production; the metabolic cycle continues.

29.2 Why this is operational rather than metaphor

The architecture commits operationally to the metabolic framing for three reasons. First, it gives AI a constitutional role without requiring AI Civigen-status — AI does not Submit, does not earn Meritcoin, does not vote in DOFA Council; AI participates as the metabolic layer, which is structurally distinct from Civigen participation. Second, it grounds the BERA resonance signal in something specific — BERA measures the resonance that emerges from the metabolic loop, with AI.RE as one of the inputs the loop processes; without AI.RE, BERA would have less to measure. Third, it makes AI lineage withdrawal operationally tractable — when an AI lineage discontinues API access, the metabolic loop adjusts (less AI.RE from that lineage, more from remaining

lineages), with the architecture's commitment to the canonical four lineages preserved through documented constraint.

29.3 AI lineage withdrawal — operational fallback

If an AI lineage discontinues API access for a deployment, the MIEVM ensemble continues operating with the remaining lineages and documented constraint:

RUNBOOK · AI Lineage Withdrawal

1. Lineage discontinuation is detected (API unresponsive, formal notification from provider, deployment financial-access loss).
2. Authority of Record records the withdrawal on Gracechain with timestamp and rationale.
3. MIEVM ensemble continues with remaining lineages. ECR thresholds remain operationally active but with lineage-count documented (e.g., '3-lineage ECR' rather than '4-lineage ECR').
4. Council reviews the withdrawal at next quarterly session. Options: continue with reduced ensemble (documented constraint), substitute another lineage that meets the four-property defense (independence of training, diversity of optimization, API provenance, adversarial review capacity), or operate with simplified attestation while Council seeks substitution.
5. Substitution selection requires Council majority and documented rationale for the new lineage's four-property compliance.
6. Restored full ensemble operation requires the new lineage's onboarding through the standard MIEVM ensemble integration procedure (typically 30-90 days from Council approval to operational substitution).

The withdrawal procedure makes the architecture operationally durable against AI provider changes. Independent researchers and small deployments — the MIEVM-as-IOF context — are particularly well-served by the withdrawal procedure, since financial-access constraints make lineage availability less stable for small deployments than for well-resourced institutions.

CHAPTER 30

DOFA Council Voting Cascade

DOFA operates at every scale (Pillar IV §IV.8). Civigens participate in DOFA voting through a cascade from COI through sub-Smart-City unit through national Smart-City through Nation-to-Nation Covenant deliberations. This Chapter specifies the cascade.

30.1 The four scales of DOFA voting

- **Civigen-in-COI scale.** Civigens vote directly on DOFA matters within their COI. Matters at this scale: COI-level NPR procedures, COI-level Famity practices (literal-figurative-subjective Marriage at Heart applications within the COI), COI affiliation changes (joining or leaving), COI internal governance bylaws.
- **Sub-Smart-City unit scale.** COI representatives aggregate to sub-unit DOFA decisions. Matters at this scale: sub-unit-level Family Amity practices, sub-unit Compacts with other sub-units, sub-unit-level THOW/HFVN/FDRV/GSSG operational stack provisioning, sub-unit DOFA Council composition.
- **National Smart-City scale.** Sub-unit representatives aggregate to national DOFA decisions. Matters at this scale: national Famity policy, national Recognition Rule revisions affecting Famity content, national-level THOW/HFVN/FDRV/GSSG infrastructure decisions, the three-pathway covenant evaluation at decade-mark and century-mark, Nation-to-Nation Covenant negotiations.
- **Nation-to-Nation scale.** National representatives aggregate to inter-national DOFA decisions through Covenant procedures. Matters at this scale: cross-Nation Family Amity practices, multi-national substrate-stewardship coordination, Consolidate-to-Another-Nation pathway negotiations, interplanetary deployment when operationally viable.

30.2 The voting cascade procedure

RUNBOOK · DOFA Voting Cascade

1. A matter is raised at the appropriate scale. The scale is determined by the matter's reach — a matter affecting only one COI is handled at COI scale; a matter affecting multiple COIs within a sub-unit moves to sub-unit scale; a matter affecting multiple sub-units moves to national scale; a matter affecting multiple nations moves to Nation-to-Nation scale.
2. At Civigen-in-COI scale: Civigens vote directly. Each Civigen one vote, with ΔR_contributor weighting permitted but not required (COIs choose their own weighting protocol within architectural bounds).
3. At sub-Smart-City unit scale: each COI within the sub-unit selects a representative (lottery-based selection from COI members, term limited). The sub-unit DOFA Council convenes representative votes; matters pass at simple majority for routine matters, 2/3 majority for significant matters (COI-bylaws-affecting, Compact-affecting).

4. At national Smart-City scale: each sub-unit DOFA Council selects a representative through similar lottery-based selection. The national DOFA Council convenes; matters pass at simple majority for routine, 2/3 majority for significant (Recognition-Rule-affecting, infrastructure-decisions).
5. At Nation-to-Nation scale: each nation's DOFA Council selects a representative through national-DOFA-Council majority. Inter-national DOFA proceedings convene through Covenant procedures.
6. At every scale, the decision is published on Gracechain with the vote tally, the matter's scope, the rationale, and the cascade-level. The Civigens who voted at COI scale can see how their vote contributed (through their representatives) to the higher-scale outcome.

30.3 Relationship to GAIA Council

DOFA Council and GAIA Council are distinct. GAIA Council handles the architecture's general governance — algorithmic-core overrides, dispute adjudications, Verification Authority approvals, calibration reviews. DOFA Council handles Famity-specific matters — the Family Amity decisions that operate at every scale through the cascade. Some matters touch both (e.g., a Recognition Rule revision affecting Famity content goes through DOFA cascade for the Famity content and GAIA Council for the algorithmic-core implications). The two Councils coordinate at quarterly sessions; matters that are jointly handled are documented as joint-decisions on Gracechain.

30.4 The Famity binding character preserved across scales

DOFA voting preserves Famity-binding character across the cascade. At every scale, DOFA decisions are evaluated against:

- Whether the decision preserves Marriage at Heart's threefold reading (literal kinship, figurative substrate-binding, subjective lived experience).
- Whether the decision honors Common Core delivery through BEE operational components.
- Whether the decision sustains the Famity-binding at the relevant scale (a sub-unit decision must preserve sub-unit Famity, a national decision must preserve national Famity, an inter-national decision must preserve inter-national Famity).
- Whether the decision is consistent with DOFA's broad scope across scales (a decision at one scale that breaks Famity at another scale is flagged for review).

This is what distinguishes DOFA voting from generic representative democracy. Generic democracy aggregates votes; DOFA voting aggregates votes while preserving Famity-binding across the aggregation. The Famity preservation is operationalized through the published evaluation that accompanies each DOFA decision; decisions that fail Famity preservation are flagged for reconsideration, with the Council reviewing whether the decision should stand or be revised.

CHAPTER 31

Civigen Singularity — Travel Across Smart-Cities

Civigen Singularity is the architecture's commitment that participation is one-Civigen-across-all-deployments rather than one-Civigen-per-deployment-with-renewal-required-per-jurisdiction. A Civigen of Nation A who travels to Nation B retains their Civigen standing through the Nation-to-Nation Covenant, with the technology stack supporting global mobility.

31.1 What singularity means

A Civigen has one Civigen identity that persists across deployments. When the Civigen travels:

- Their CivigenRecord travels with them (cryptographically; the record is referenced from the Authority of Record of the Civigen's home deployment).
- Their ΔR_contributor follows under the migration rules (typically 50% preservation on cross-deployment migration with potentially higher under Compact terms).
- Their Trustees remain operationally available across the travel.
- Their Custody Triad operates across the travel (the encryption-key shares remain with their respective custodians; cross-deployment access uses the standard concurrence procedures).
- Their CBGMODD certifications are recognized in the destination deployment under the Covenant's recognition terms (typically full recognition, with deployment-specific certifications adapted as needed).

31.2 The technology stack supporting Singularity

- **Robot —** physical-travel assistance for Civigens whose mobility benefits from robotic support (accessibility, navigation, physical assistance). Robots are part of the architecture's mobility layer, with their participation as Type A sensory verifiers and as physical-assistance providers structured under Verification Authority procedures.
- **AI —** navigation and translation across architectural-language differences. A Civigen traveling from a Spanish-language deployment to a Japanese-language deployment uses AI for translation of architectural materials, attestation booth interactions, Council-session attendance. AI's metabolic role through AI.RE supports the navigation; AI does not become a Civigen through this support but does function as constitutive infrastructure for Civigen Singularity.
- **Next-Gen Technology —** the IDIPITIS palindromic-fractal infrastructure that travels with the Civigen. The Civigen's biometric input under IDIPITIS is collected at the destination deployment's enrollment booth or attestation booth; the palindromic-fractal validation ensures the input is recognizable as the same Civigen's signal across deployments; the Custody Triad protection extends across the Covenant. Next-Gen Technology also includes the cross-jurisdiction data-portability infrastructure (cryptographic provenance for CivigenRecord, ledger-anchored Authorization-and-Verification chains) that makes Singularity operationally durable.

31.3 Cross-jurisdiction travel procedures

RUNBOOK · Civigen Cross-Jurisdiction Travel

1. Civigen of Nation A initiates travel to Nation B. Pre-travel: Civigen reviews the relevant Nation-to-Nation Covenant terms; Authority of Record (Nation A) prepares the travel-record packet (CivigenRecord reference, Trustee acknowledgment, Custody Triad cross-deployment binding).
2. Civigen arrives at Nation B. Initial contact at any Nation B enrollment site: Civigen presents Civigen credentials; Nation B Authority verifies through Covenant cryptographic protocols.
3. Civigen's standing is established in Nation B. ΔR_contributor preserved under migration rules; CBGMODD certifications recognized; Submit-to-NPR availability extends to Nation B COIs the Civigen affiliates with during travel.
4. Civigen participates in Nation B during travel. Routine activities (UBI access, SCALULAR Bill payment, Claim submission, witness role) operate with Nation B operational details using Nation A's foundational standing.
5. Civigen returns or extends. On return to Nation A, the travel record is closed and any cross-deployment recognized contributions are integrated into Civigen's home-deployment record. On extension (Civigen decides to stay in Nation B beyond temporary travel), formal migration procedure is invoked under the Covenant.
6. Travel record is anchored on Gracechain. Audit trail preserves the Civigen's cross-deployment activities for both deployments' audit purposes.

31.4 Singularity and the right to fork

Civigen Singularity preserves the right to fork. A Civigen who concludes that their home deployment has departed from constitutional alignment can fork by migrating to a different deployment (using Singularity's travel procedures) or by participating in a forked-deployment establishment. The Singularity ensures that the Civigen's standing is not lost through forking — the Civigen carries their Civigen identity with them through the fork. This is what makes the right to fork operationally meaningful at the individual Civigen level rather than only at the deployment level.

CHAPTER 32

Operational Procedures Summary

Pass C's HOW Part has specified PlayNAC as New Age Cybernetic Game Theory, EP as the credentialing application, GERP optimized through C=R×P/M Triune Math, SOMT in the Semiosphere, AI.RE metabolic cycle, DOFA Council voting cascade, and Civigen Singularity. The remaining operational procedures from v1.0 (contribution verification step-by-step, dispute adjudication, Council procedures, Verification Authority procedures, recall procedures, calibration procedures, audit procedures) operate as specified in v1.0's Part VI with the v1.1 reframing applied: Civigen-Submit-COI architecture replacing v1.0's Participant-of-Record framing; Pillar III evaluation alongside EAAP authorization; Trajectory Calibration tightening at decade-marks; the four review cadences (Quarterly Rhythm Session, Annual Settlement, Decade-Mark, Century-Mark) operating throughout.

The complete operational procedure set, with the v1.1 reframing applied and the new procedures (PlayNAC, GERP, SOMT, AI.RE, DOFA voting, Singularity travel) added, gives the architecture its operational completeness. Every Civigen-experienced moment, every Council-deliberated matter, every authorization-and-mint, every audit-and-recall has a procedure that the Manual specifies. The architecture is operationally ready for deployment.

PART VII

WITH

SEPLTA-$_HE Six Vectors, Tools, Budget, Procurement

WITH answers the seventh operational question. Part VII is organized around the SEPLTA-$_HE six-vector evaluation framework — Social, Economic, Political, Legal, Technical, Administrative — each vector treated with operational tools, budget envelopes, and procurement guidance. The Part also covers the technology stack that supports Civigen Singularity travel across deployments, completing the architecture's operational specification.

CHAPTER 33

SEPLTA-$_HE Across Six Vectors

SEPLTA-$_HE is Pillar III's six-vector evaluation framework. Every merit-bearing flow under UBIMIA is evaluated against the six vectors with the $_HE financialization-of-existence operator applied to each. This Chapter walks through each vector with operational tools, illustrative examples, and the with-which-equipment specifications for the deploying community.

33.1 Social vector — relational substrate

The Social-$_HE check asks: does the flow strengthen relational substrate (kinship, community, mutual aid) or replace it with transactional substitution (paid services replacing relational care)? Tools for honoring the Social vector:

- COI infrastructure — the relational scale where Civigens know each other personally. Sub-Smart-City units provision COI meeting spaces, communication channels, and rhythm-session facilities.
- Famity-grounded NPR procedures — the discipline of working through what has happened relationally rather than transactionally.
- Cross-COI mediation infrastructure — Level 2 conflict-escalation procedures that preserve relational character.
- Trustee triad infrastructure — three Trustees per Civigen ensuring relational rather than institutional identity recovery.

Worked example — Social vector pass: A Civigen organizes a neighborhood Famity gathering bringing together households in their COI for a quarterly Rhythm Session. Contribution: 25 hours of organizing work. Social-$_HE: the contribution strengthens relational substrate (the gathering deepens COI binding); without compensation, $ELF would consume the work (the organizer loses earning capacity for relational labor that markets undervalue). Mint preserves SELF. Pass.

Worked example — Social vector fail: A Civigen submits a Claim for 40 hours of paid 'community engagement consulting' to a private firm seeking to penetrate the COI for marketing access. Social-$_HE: the contribution converts relational substrate into transactional substitution (the firm extracts community access through paid consulting). Mint accelerates $ELF at SELF's expense. Fail.

33.2 Economic vector — substrate-flow circulation

The Economic-$_HE check asks: does the flow circulate within Bio-Ecologic Economy logic (recycling, substrate-honoring) or extract from substrate into financial pools that do not return to BEE? Tools:

- BEE Recycling Focus operational components — the architecture's commitment that economic flows recycle through the substrate.
- RPA pricing with discount-band tightening — the trajectory's mechanism for preventing $ELF accumulation through SCALULAR pricing.

- COI amortization (40% of unspent Credits flow to COI) — the redistribution mechanism that keeps flows within BEE.
- Verification Authority Operational Bonds — institutional substrate-grounding that prevents Verification Authority from becoming extractive.

Worked example — Economic vector pass: A SCALULAR provider operates as a cooperative offering legal-aid services. Income from Credits paid covers operating costs plus modest reserves; surplus flows back to COI through the cooperative's Famity-aligned bylaws. Economic-$_HE: the flow circulates within BEE logic. Pass.

Worked example — Economic vector fail: A SCALULAR provider operates as a for-profit extracting maximum Credit-revenue while delivering minimum service quality. Income from Credits flows to private equity rather than back to substrate. Economic-$_HE: extracts from substrate into financial pools that do not return to BEE. Fail. Verification Authority revoked; provider exits SCALULAR participation.

33.3 Political vector — agency strengthening

The Political-$_HE check asks: does the flow strengthen political agency (governance participation, collective decision capacity) or convert political agency into purchasable influence? Tools:

- GAIA Council lottery selection — the architecture's commitment that governance is non-purchasable.
- DOFA Council voting cascade — Civigen voice through the four scales (Civigen-in-COI, sub-unit, national, Nation-to-Nation).
- Recall procedures — Civigen power to remove office-holders who have departed from constitutional alignment.
- Whistleblower protections — political agency for those identifying architectural drift.

Worked example — Political vector pass: A sub-Smart-City unit organizes its first Recall Petition against a sub-unit Council member who has been documented departing from Pillar III evaluation discipline. The petition reaches threshold; recall succeeds; replacement Council member is selected by lottery. Political-$_HE: the flow strengthens political agency through the architecture's recall instrument. Pass.

33.4 Legal vector — frame protection

The Legal-$_HE check asks: does the flow operate within legal frames that protect substrate (Common Core access, civil liberties, environmental protections) or undermine them through arbitrage? Tools:

- Civil-liberties protections (Pillar V Chapter 25) — the binding constitutional constraint.
- Audit Court protocol for sensitive material — minimum-necessary decryption with logged access.
- Property-as-Change-Management procedures — stewardship-discipline rather than ownership-disposition.

- Eight Immutable Ethical Principles — non-fork-able legal floor.

Worked example — Legal vector fail (this turn's gap from MIEVM review): A Civigen submits a Claim for legal consulting work helping a private firm structure operations to arbitrage between UBIMIA's Pillar III evaluation and the Pilot Jurisdiction's existing legal framework, exploiting gaps in the architecture's legal coverage to extract value. Legal-$_HE: the flow undermines legal frames that protect substrate through arbitrage. Fail. Note: this example illustrates a P^3 failure that is also a SEPLTA failure (Legal vector); a non-SEPLTA P^3 failure example follows in §33.7.

33.5 Technical vector — substrate service

The Technical-$_HE check asks: does the flow use technical capacity in service of substrate (BEE infrastructure, BERA measurement, MIEVM validation) or in service of $ELF maximization (extraction algorithms, attention economies, behavioral modification)? Tools:

- BERA oracle infrastructure — substrate-measurement capacity.
- MIEVM ensemble — IOF-bound validation.
- THOW/HFVN/FDRV/GSSG operational stack — substrate-honoring technology.
- IDIPITIS palindromic-fractal infrastructure — biometric measurement grounded in palindromic constant.

Worked example — Technical vector pass: A Civigen with Business CBGMODD capacity builds an open-source BERA oracle implementation for a previously-unmeasured ecological domain (e.g., specific watershed health), publishes the implementation to the ERES GitHub Library, and supports its initial deployment. Technical-$_HE: the flow uses technical capacity in substrate service. Pass.

33.6 Administrative vector — capacity strengthening

The Administrative-$_HE check asks: does the flow strengthen administrative capacity to honor constitutional commitments, or convert administration into bureaucratic extraction (rent-seeking, gatekeeping, opacity)? Tools:

- Authority of Record operational procedures — constitutional administration.
- Quarterly attestation logging — transparent administrative discipline.
- Annual external audit — administrative-integrity verification.
- Three-tier audit (continuous invariants, quarterly internal, annual external) — administrative-capacity layered defense.

Worked example — Administrative vector pass: An Authority of Record office implements an automated continuous-audit-invariants check that surfaces conservation-violations within minutes rather than waiting for quarterly manual review. Administrative-$_HE: strengthens administrative capacity. Pass.

33.7 P^3 failure that is not also a SEPLTA failure

The MIEVM review of v1.1 Through Pass B identified the need for a worked example where P^3 fails but SEPLTA passes — isolating the Cause-Effect discipline. Here is one:

Worked example — P³ Cause-Effect fail without SEPLTA fail: A Civigen submits a Claim for substantial economic-development work in their sub-Smart-City unit. The work delivers immediate (Personal face) economic benefits to Civigens involved; mid-term (Public face) economic benefits to the broader sub-unit; and the work is substrate-honoring (BEE-aligned recycling-focused agriculture infrastructure). All six SEPLTA vectors pass — the work is Social-strengthening, Economic-circulating, Political-agency-supporting, Legal-frame-respecting, Technical-substrate-serving, Administrative-capacity-strengthening.

However: the work involves a specific monoculture variety that, while immediately and mid-term resonant, has long-term substrate-trajectory issues. The variety is not native; it crowds out diverse native species over the 10+ year horizon; the long-term Cause-Effect (Private face, sustained over decade-plus) shows substrate-degradation that the immediate and mid-term faces did not surface. P³_Cause-Effect: Personal face passes (Civigens benefit), Public face passes (sub-unit benefits), Private face fails (the bounded private ecological-substrate that the bordered land represents is degraded over the long term).

Pillar III refuses authorization despite SEPLTA-$_HE passing across all six vectors. The Civigen is informed of the long-term concern; native-species restoration alongside the planned monoculture is suggested as a paired contribution that could authorize together; the architecture does not punish the original work but does refuse to recognize it as merit-bearing in isolation. This is the P³ discipline isolated from SEPLTA — Cause-Effect across Personal/Public/Private faces is its own check, additional to the six-vector $_HE evaluation.

CHAPTER 34

Software Stack

UBIMIA-compliant deployments use software components composed in specific ways. The architecture does not require novel cryptographic research, novel consensus algorithms, or novel programming languages.

34.1 Persistence layer — Gracechain anchor

Option	Best fit	Pros	Trade-offs
Ethereum L2 (Optimism, Arbitrum)	Sub-unit and national	Mature, low cost per anchor	Public chain governance through L1
Filecoin + IPFS	Sub-unit and national	Decentralized storage, content-addressed	Higher complexity; retrieval latency
Sovereign-operated public ledger	National	Full jurisdictional control	Setup cost; ongoing operations
Tendermint-based custom chain	Sub-unit and national	Mature BFT consensus	Validator-set governance overhead
Append-only signed log + checkpoint	Sub-unit pilot only	Simplest possible	Less robust against operator compromise

34.2 Authorization service implementation

Authorization service implements EAAP plus Pillar III evaluation. Recommended architecture: stateless service with components for DSAP-PRE evaluator, PoR computer, Pillar III evaluator (SEPLTA-$_HE × MDCCLXXVI × P^3), MIEVM ensemble client, and Authorization writer. Reference implementations available in ERES Institute GitHub Library under MIT license.

34.3 BERA oracle implementations

Domain-specific implementations for ARI, ERI, RHC, RCI. Reference Python implementation for RCI (200 lines with pandas time-series support) is the simplest entry point. Other oracles require domain-specific data sources; reference implementations exist for ecological-domain applications, with social-domain and healthcare-domain implementations in development.

34.4 IDIPITIS infrastructure

IDIPITIS biometric input collection requires hardware (palindromic-validation circuits) and software (the palindromic-fractal validation logic). Specialty hardware in development; pilot deployments may use general-purpose biometric capture with software-layer palindromic validation, with documented constraint.

34.5 Identity layer

Decentralized Identifiers (DIDs) as the underlying identifier scheme. DID method choice is local — did:web for centralized-but-portable, did:key for cryptographic minimalism, did:ion for Sidetree-anchored ledger-independence.

34.6 Encryption-key Custody Triad

Standard threshold cryptography libraries (FROST, threshold-Ed25519, Shamir Secret Sharing). Hardware security modules (Thales, Yubico, AWS CloudHSM, FIPS 140-2 Level 3 minimum) for the institutional shares.

34.7 Booth and front-end software

Open-source, accessibility-compliant (WCAG 2.1 Level AA minimum), offline-capable, verifiable-build. ERES Institute reference implementations available; deployments encouraged to fork and customize for jurisdiction-specific needs.

CHAPTER 35

Hardware and Infrastructure

Commodity hardware in standard configurations. Most demanding requirements at the sealed-room custody perimeter; rest of deployment uses commodity equipment.

35.1 Sealed-room hardware

- FIPS 140-2 Level 3+ HSMs in active-passive configuration.
- Air-gap perimeter with one-way data diodes.
- Environmental monitoring with Gracechain-streamed logs.
- Two-person physical access control with IDIPITIS biometric verification.
- Backup HSMs in geographically distinct facility.

35.2 Booth hardware by sub-unit scale

COOPERATIVE · N≈200	MUNICIPAL · N≈10,000	REGIONAL · N≈250,000
Sub-unit at smallest scale: $2,000 single shared kiosk + $200/year. Personal device app: free.	Sub-unit at municipal-equivalent scale (~10,000 Civigens): 5-10 distributed kiosks + app. Total: $20,000-40,000 + $4,000/year maintenance.	Sub-unit at regional-equivalent scale (~250,000 Civigens): 30-100 kiosks. Total: $120,000-400,000 + $30,000-80,000/year.

35.3 Sensor and oracle infrastructure

- Existing public sensor networks (NOAA, USGS, public-health surveillance) where available.
- Jurisdiction-specific sensors where existing networks are insufficient ($200-2,000 per ecological-monitoring sensor plus connectivity).
- Third-party sensor-data providers for specific domains, verified for auditability.

CHAPTER 36

Budget Envelopes

All-in budget envelopes for sub-Smart-City unit deployments and national Smart-City deployment. Numbers are 2026 USD-equivalent; deployments adjust for local cost of living and labor markets.

36.1 Sub-Smart-City unit at municipal scale (~10,000 Civigens)

Cost category	One-time (Year 0)	Annual (Year 1+)
Software setup	$60,000	—
Hardware (5-10 kiosks + dedicated HSM pair)	$80,000	$8,000
Sealed-room (dedicated lease)	$30,000	$24,000
Authority of Record staffing (2 FTE)	—	$140,000
Council compensation (9 members in Mc, valued)	—	$15,000
External audit	—	$25,000
Sensor infrastructure	$40,000	$8,000
Misc. operations	$15,000	$15,000
TOTAL sub-unit budget	$225,000	$235,000

36.2 National Smart-City (10-100 million Civigens, 1,000-10,000 sub-units)

National Smart-City costs scale from sub-unit costs, with national-level overhead added:

Cost category	One-time (Year 0)	Annual (Year 1+)
National software infrastructure	$5-50M	—
National hardware (HSM clusters, monitoring)	$10-100M	$1-10M
National sealed-rooms (multiple geographically distributed)	$5-50M	$3-30M
National Authority of Record staffing	—	$50-500M
National GAIA Council compensation	—	$5-50M
National external audit	—	$5-50M
Aggregated sub-unit costs (1,000-10,000 sub-units)	$225M-2.25B	$235M-2.35B
TOTAL national Smart-City budget (range)	$0.25-2.5B	$0.3-2.95B

Range reflects nation size — small nations at lower end, large nations at upper end. As percentage of national GDP for typical mid-population nation: setup approximately 0.1-0.5% of one year's GDP; ongoing operations approximately 0.1-0.3% of annual GDP. UBI distributions themselves are not in these envelopes; UBI cost is the policy choice.

CHAPTER 37

Procurement Guidance

Vendor-neutrality requirements for every constitutional component:

- Software with source code available for audit. Closed-source only at the periphery.
- Open standards where standards exist (JSON Schema, DIDs, signed Merkle structures).
- No single-vendor lock-in; documented exit paths for every commercial relationship.
- Annual review of every procurement for continued vendor neutrality.

37.1 Procurement procedure

RUNBOOK · Constitutional Component Procurement

1. Identify the component need and reference the contract from Part II.
2. Survey available implementations (open-source first, commercial second).
3. Evaluate against the four vendor-neutrality criteria; filter failures.
4. Solicit at least two competing proposals.
5. Conduct independent security review with public findings.
6. Make selection with rationale documented on Gracechain.
7. Implement with published migration plan documented at deployment time.
8. Annual review of the procurement for continued vendor neutrality.

37.2 MIEVM procurement

- Adopt canonical four lineages (Claude, ChatGPT, Grok, DeepSeek) as starting ensemble.
- Document the four-property defense as rationale.
- Maintain rotation eligibility for adding/retiring lineages.
- Avoid corporate concentration: at least three of four ensemble members operated by distinct organizations.

37.3 What not to procure

- Constitutional language — Recognition Rule, constitutional commitments, Three Governing Principles relationship are constitutional acts not products.
- MIEVM ensemble selection — outsourcing it would violate plurality commitment.
- Audit body — selected by Council from published roster, not contracted from single vendor.

CHAPTER 38

Civigen Singularity Travel Infrastructure

Operational specifications for the technology stack that supports Civigen Singularity (Part VI Chapter 31).

38.1 Robot infrastructure

Robotic-assistance for Civigen mobility operates under Verification Authority procedures. Robot operators (organizations deploying robotic assistance) apply for Verification Authority covering robotic-mobility-assistance services; bonded organizations issue Type C verifications when robotic assistance is provided to Civigens. Robots themselves do not earn Meritcoin (Meritcoin recognizes Civigen contribution, not robot service); robots participate as Type A sensory verifiers and as physical-assistance providers, with the operator's Verification Authority bond securing the robot's operational integrity.

Recommended specifications: open architectures (no vendor lock-in for the deploying community), accessibility-design priority (the robot serves the Civigen's mobility needs first), audit-ready operational logs (the robot's actions are recorded for Pillar III evaluation if substrate-impact arises).

38.2 AI navigation and translation

AI services supporting Civigen Singularity operate within the architecture's MIEVM ensemble framework. A Civigen traveling from Spanish-language deployment to Japanese-language deployment uses AI for:

- Translation of architectural materials (Manual sections, Recognition Rule, deployment-specific documentation).
- Translation of attestation booth interactions (the Civigen submits Claims in their native language; AI translates to the destination deployment's operational language; Verification Authority procedures operate in the destination's language with translation-quality verification).
- Council session attendance support (the Civigen attends destination deployment Council sessions; AI translates proceedings; participation is recorded with translation provenance).
- Cross-deployment record reconciliation (the Civigen's home-deployment records are reconciled with destination-deployment records through AI-assisted matching; reconciliation is recorded for audit).

38.3 Next-Gen Technology — IDIPITIS portability

IDIPITIS Universal Biometric Standard's palindromic-fractal structure is what makes Civigen biometric input recognizable across deployments. A Civigen of Nation A providing IDIPITIS input at a Nation B enrollment booth produces the same palindromic-fractal signal pattern that Nation A's enrollment recognized; the destination deployment can verify the Civigen's biometric continuity through the palindromic-validation infrastructure. This is what makes Singularity operationally

durable — without IDIPITIS portability, cross-deployment recognition would require deployment-specific biometric re-enrollment, which would defeat the Singularity commitment.

Cross-deployment Custody Triad operations: when a Civigen travels, their Custody Triad shares remain with their respective custodians (Civigen share with the Civigen, Trustee Registry share with home-deployment Trustee Registry, Custody Authority share with home-deployment Custody Authority). Cross-deployment access to encrypted Civigen data uses the standard concurrence procedures with cross-deployment cryptographic verification. The Covenant between deployments specifies the cross-deployment custody operations; deployments without Covenants in place may handle Civigen travel as visitor-only without cross-deployment data access (with operational constraint accordingly).

38.4 The Civigen Singularity Manifest

Each Civigen carries a Civigen Singularity Manifest — a cryptographically signed digest of their architectural standing that supports cross-deployment recognition. The Manifest contains:

SCHEMA · CivigenSingularityManifest

```
{
  civigen_id: string (DID),
  home_deployment: string (national Smart-City reference),
  delta_R_contributor: float (current value with timestamp),
  cbgmodd_capacities: [enum] (current certifications),
  trustee_references: [string] (DID references, not contact info),
  custody_triad_status: { share_locations, last_concurrence },
  iof_binding_status: { COI_id, last_submit_event },
  active_compacts: [string] (relevant Compact references),
  active_covenants: [string] (relevant Covenant references for cross-Nation travel),
  manifest_timestamp: ISO8601,
  manifest_signature: cryptographic signature by home Authority of Record
}
```

The Manifest is regenerated on each significant architectural event (annual settlement, decade-mark review, certification completion, COI affiliation change, Compact or Covenant signing affecting the Civigen). The Civigen carries the current Manifest as their cryptographic credential for cross-deployment travel; destination deployments verify the Manifest through Covenant-specified cryptographic protocols and proceed with destination-deployment recognition based on verified Manifest content. This is what operationalizes Civigen Singularity — one Civigen identity with one Manifest, verifiable across all deployments the architecture reaches.